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Development Authority**

FIRE REGULATIONS

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MINISTRY OF HOUSING & CONSTRUCTION

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FOREWORD TO THE THIRD EDITION

Recent umpteen number of gutting of shops and factories in Sri Lanka and disturbing fires in world arena opened eyes in global construction circles while the necessity of implementing stringent regulations, revising of standards to suit current trends and continuous monitoring of such adaptations highlights a great significance.

In Sri Lankan context Urban Development Authority or Local Authority is taken as the Authority Having Jurisdiction (AHJ). Such AHJs always depend on Fire Services Department recommendation and certification. Fire Department needs to abide by the Regulations published by the Construction Industry Development Authority (CIDA) and implementation to be strictly complied by the local Fire Brigades CID Act No 33 of 2014 specifies the use of Standard documents published by CIDA in any "Identified Construction Works" (ICW) by its clauses 28 & 46.

There are over 100 regulations in this document revised by a competent set of persons attached to the Industry. Initiated in the year of 1997, it had been subjected to the 1st revision in the year 2006. The introduction of a Refuge Floor for every 10 floors in all Super High-Rise buildings is a latest addition to this edition. Another salient inclusion is Fire requirements for building under construction. This is a proactive measure to avoid property and life safety in sites of many constructions taking place all over the country.

The group of persons who dedicated their service to revise and prepare this national standard publication and their representing organizations are as follows.

- (1) Chief of Fire Services - Fire Dept CMC - Mr. Rohitha Fernando
- (2) Institute of Fire Engineers (IFE) SL Branch- Gr.Capt. Kolitha Sri Nissanka
- (3) Building Services Engineering Sectional committee of IESL : Eng. L.C.R.De Silva
- (4) Lanka Association of Building Service Engineers (LABSE) ; Eng. Wijitha Perera
- (5) Sri Lanka Institute of Architects (SLIA) ; Archt. Dilshan De Mel
- (6) Association of Consulting Engineers of Sri Lanka(ACESL) : Eng. Ruzzel De Ziiva
- (7) Institute of Quantity Surveyors — SL : Ch. Q.S. Mahinda Gunawardana
- (8) Former Chief of Fire Dept : Mr. Jayampathi Kannangara
- (9) IESL 2nd nominee : Eng. Chandimal Jayakody
- (10) Eminent Fire Engineer/ Expert/ Compiler of Publication - Eng. Gamini Nanayakkara

I wish to take this opportunity to express my sincere gratitude to the members of the Fire committee & the Steering Committee on CIDA publications for the valuable contributions made for the successful completion of this revision.

I wish to present this second edition to the Construction Industry with confidence considering the careful thinking and hard work put in by those involved.

ENG. M R JEYACHANDRAN
CHAIRMAN
CIDA

FOREWORD TO THE SECOND EDITION

The first edition of the Fire Regulations was prepared by the following committee and published in 1997.

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|-----|------------------------|---|---|
| 1. | Mr. N D Dickson | - | Director (Planning) / UDA |
| 2. | Mr. S W Molligoda | | Managing Partner / Molligoda Associates |
| 3. | Mr. K A H Ranaweera | | Addl. Director / My, of LGH&C |
| 4. | Mr. K M I de Silva | | Chief Fire officer / CMC |
| 5. | Mr. K D M Udugama | | Deputy CFO / CMC |
| 6. | Mr. Somapala de Silva | | Chief Planning Officer / CMC |
| 7. | Dr. Milroy Perera | | Partner / M/s. Selvaratnam & Perera |
| 8. | Mr. E I Munasinghe | | Adviser / ICTAD |
| 9. | Mr. Mervyn Gunasekara | | Deputy Director / ICTAD |
| 10. | Mr. W V K M Abeysekara | | Deputy Director / ICTAD |
| 11. | Wg. Cdr. C A 0 Dirckze | | Fire Engineering Consultant |

After using this document for several years, a necessity of a revision was felt by ICTAD hence a sub committee was formulated and the committee consisted of the following;

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|-----|-------------------------|--|
| 1. | Mr. J Kannangara | Fire Chief / Fire Service Dept. |
| 2. | Mr. P Mervyn Gunasekara | MD / LAN Management Development Service |
| 3. | Wg. Cdr. C A 0 Dirckze | Fire Engineering Consultant |
| 4. | Arch. B E T Fernando | Deputy Director (Enforcement) /UDA |
| 5. | Mr. Russel de Silva | Executive Secretary / IESL |
| 6. | Ms. S D H de Silva | Director (Works)/ Ministry of Housing and Construction |
| 7. | Mr. R Gurusinghe | Electrical Consultant / Dept. of Buildings |
| 8. | Arch. R Nadesapillai | Secretary / SLIA |
| 9. | Mr. Dayaratne Perera | Planning Officer / CMC |
| 10. | Mr. K P W Rajasiri | Director (Development) / ICTAD |
| 11. | Mr. C Amarakoon | Deputy Director (TD) / ICTAD |

The resource person Wg. Cdr. C A 0 Dirckze revised this edition under the guidance of the sub committee to suit the current development in the construction industry.

I wish to acknowledge with grateful thanks the services of Wg. Cdr. C A 0 Dirckze and all the members of the steering committee and the sub committee. Meanwhile my special thanks are also due to Mr. J Kannangara, Mr. J K Lankatilake, Mr. K P W Rajasiri, Mrs. Vajira Ranasinghe and Mr. C Amarakoon who are contributed in numerous ways in the publication of this document. I also appreciate the assistance rendered by Mrs. W L Lalitha Ranjani for the secretarial work.

I Hope that this publication will be a useful guide for Engineers, Designers, Architects, Consultants, Property Developers and others to provide their clients with the required level of fire protection for life and property

VIDYAJYOTHI PROF. LAKSHMAN ALWIS
CHAIRMAN
ICTAD

FOREWORD TO THE FIRST EDITION

The necessity for a Code of Practice for Fire Precautions in buildings had been a long felt need for our country. ICTAD realize this fact and had been working on the preparation of a such code for quite some time.

Now it has become a reality due to the tremendous effort made by the committee appointed for the formulation of fire regulations for Sri Lanka. Hence I would like to thank all the members of the committee Mr. N D Dickson, Mr. S W Molligoda, Mr. K A H Ranaweera, Mr. K M I de Silva, Mr. K D M Udugama, Mr. Somapala de Silva, Dr. Milroy Perera, Mr. E I Munasinghe, Mr. Mervyn Gunasekera, Mr. W V K M Abeysekera and Wg. Cdr. C A 0 Dirckze.

I believe that this document will be a useful guide for Engineers, Designers, Architects and others to provide their clients with the required level of fire protection for life and property. Finally, I would like to thank Wg. Cdr. C A 0 Dirckze, Technical Editor of this document and Mr. G Gaffoor, Mr. L P C Liyanarachchy, Former Director (Development) and the Development Division Staff for their contributions to make this event a success.

PROF. D S WIJEYESEKERA
CHAIRMAN
ICTAD

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DEFINITIONS

In these Regulations, unless the context otherwise requires, the following expressions have the meanings hereby respectively assigned to them and other expressions have the meanings assigned to them by the Urban Development Authority Planning and Building Regulations.

Expression	Definition
Accepted codes of practice	- Codes, standards or manuals acceptable to CIDA.
Aggregate	Any material other than cement and water used in the making of concrete and complying with the requirements of the respective Sri Lanka Standards or in the absence of such a standard, the respective British standard.
AHJ	Authority Having Jurisdiction; Local Authority Fire Brigade/ Chief Fire Officer/ Fire Prevention Officers under a statutory organization.
Air Conditioning	- The process of treating air so as to control simultaneously its temperature, humidity, purity, distribution and movement and other properties to meet the requirements of the air conditioned space.
Air Change	- Is a measure of the air volume added to or removed from a space divided by the volume of the space.
Apartment Complexes	- A building consisting of individual residential units in a multi storey building sharing common services and access facilities.
Approved	- Approved by the AHJ.
Approved Plan	- A plan of a building or any building work or any land subdivision approved by the Urban Development Authority or the relevant Local Authority in accordance with the law and the Regulations.
Atrium	An atrium within a building is a large open space created by an opening, or a series of openings, in floor assemblies, thus connecting two or more storeys Atrium may be covered or open at the top and is used for purposes other than those associated with small shafts, such as for stairs, elevators and various services. The sides of the atrium maybe open to all floors, to some of the floors, or closed to all or some floors by unrated or rated fire-resistant construction.
B.S.S	- British Standard Specifications
Balcony	- Any stage, platform oriel window or other similar structure projecting outwards from the wall of a building beyond the outer face of an external wall of the building and supported by brackets, corbels or cantilevered.

Base	In relation to a wall or pier; (a) The underside of the course immediately above the footings, if any, or in the case of a wall carried by a beam, above the beam; and (b) In any other case the bottom of such wall or pier.
Basement	A storey which is located below the corresponding natural ground level in its entirety or a storey of building which is below the ground floor and the floor of which is situated as such a level that more than half of the height of such storey is below the level of the ground adjoining its perimeter walls for more than half the length of such perimeter walls.
Bressumer	A beam or girder which carries a wall.
British Standard Code of Practice	The current Code of Practice issued for Fire Systems by the British Standards Institution.
Business Premises	- A building or part thereof designed, adapted or used for the carrying on business transactions.
Car Parks	(a) Covered car park means a parking area which is housed in a building or a part of a building which is roofed over as distinct from a car parking area in the open air. (a) Multi Storey car park means a covered car park as defined above or below ground level. (b). Automated mechanized car parks means vehicles are directed to parking slots by use of mechanical automated means methods. <i>Note:</i> <i>Car park designed to accommodate only cars, motorcycles & passenger or light good vehicles weighing not more than 2500 gross kg.</i>
Ceiling	- The covering to the underside of a floor joist, ceiling joists or floor slab excluding in every case any supporting beam, and where no such covering exists, means the underside of a floor joist, floor slab, roof collar or tie excluding any supporting beam
Code of Practice	Code of practice is the document specifying standard of practice acceptable to the Relevant Authority. The Relevant Authority may adopt requirements stipulated in the stated year of publication of any referred Code of Practice or at its discretion adopt those specified in a later version.
Column	- A part of a construction which by its resistance to compression in the direction of its length and to bending actions induced by such compression, supports and transmits a load.
Compartment	A portion of a building which is separated from adjoining portions by walls and floors of the required standard of fire resistance. A roof space above the top storey of a compartment is included in that compartment.

Concealed Space	- A space enclosed by elements of a building (including a suspended Ceiling or raised floor or space between curtain walling and the floor slab or spandrel wall) or contained within an element but not a room, cupboard, circulation space, protected shaft or space within a flue, chute, duct, pipe or conduit.
Cavity Barrier	- A construction provided to close a concealed space against penetration of smoke or flame, or provided to restrict the movement of smoke or flame within such space.
Corridor	A passage providing a means of access from rooms or spaces to an exit.
Construct	Build, erect and place in position and includes reconstruct, rebuild, re-erect and replace in position.
Dead-end travel Distance	- A dead-end refers to a situation within a common area, normally a corridor or lift lobby spaces, where exit is only possible from one end, with no possible escape from the other end.
Dead Load or Dead Loading	The weight of all walls, floors, roofs, partitions and other like permanent structures.
Depth	The measured distance at ground level between the front line of the building and the back line of the rear main wall which separates the main building from the open space.
Detached Building	A building which is used or is intended, adapted, or designed to be used for living purposes and is a self-contained unit.
Direct Distance	The shortest distance from a point in a room or space, measured within the external enclosure walls of the room or space to the relevant exits, ignoring internal walls, partitions and fittings other than the enclosure walls of exit passageways or exit staircases
Duct	- A passageway for conveying all building services.
Dwelling House or Dwelling Unit	A building or part of a building consisting of a room or group of rooms fanning a self-contained unit with independent living, cooking and sanitary facilities used exclusively as a domestic residence.
Element of Structure	- Element of Structure (a) A member forming part of the structural frame of a building or any other beam or column but not a member forming part of a roof structure only, (b) A load-bearing wall or load-bearing part of a wall, (c) A floor, including a compartment floor, other than the lowest floor (in contact with the ground) of a building, (d) An external wall, (e) A separating wall, (f) A compartment wall, and (g) A structure enclosing a protected shaft (protecting structure).

Emergency Lighting and exit lighting	- Emergency Lighting and Exit Lighting (a) .Emergency lighting means lighting provided with a secondary source of power supply with automatic changeover from a battery back-up with re-charging facility. (b) Exit lighting means that part of emergency lighting which is provided to illuminate the exit routes provided with a secondary source of power supply, with automatic changeover from a battery back-up with re-charging facility.
Evacuation Lift	A lift used as part of the evacuation sequence for people requiring assistance, which has appropriate structural, electrical and fire protection and is capable of being taken under control by an authorized person.
Exit	A means of escape from the interior of the building to an exterior space which is provided by the use of the following either singly or in combination: exterior door openings, exit staircases, exit ramps or exit passageways. In the case of an exit leading to a separate building, exits also include link ways, walkways, bridges and balconies. Exit shall not include access stairs, aisles, corridor doors or corridors and access doors leading to rooms or spaces in occupancy areas.
Exit passageway	A horizontal extension to a vertical exit staircase or a passage leading to an open exterior space.
Exit width	Clear width of the exit pathway.
External Cladding	- Material fixed to the outside face of an external wall for weather protection or decorative purpose
Factory	A building or part thereof designed, adapted or used for- (a) The making of any article, commodity or product or part thereof; or (b) The altering, repairing, ornamenting, finishing, cleaning, washing or the breaking up or demolition of any article, commodity or product or part thereof or The adapting for sale or assembly of any article, commodity or product or part thereof.
Fire Doors, Fire Windows and Fire Shutters	A door or shutter provided for the passage of persons, air or objects which, together with its frame and furniture as installed in a building, is intended, when closed, to resist the passage of fire and/or ingress of gaseous products of combustion and is capable of meeting specified performance criteria.
Fire Damper	- A device installed in an air distribution system, designed to close automatically upon detection of heat to restrict the passage of flame
Firefighting lift	- A lift with fire protection measures, including controls that enable it to be used under the direct control of the fire and rescue service in fighting a fire.

Firefighting lobby	A smoke-stop lobby which is adjacent to a fire lift and exit staircase designated For use by the firefighting team during an emergency. The lobby shall not be used for any other purposes during such emergencies.
Firefighting shaft	- A protected enclosure containing a fire-fighting stair, fire-fighting lobbies, a fire main and, if provided, a fire-fighting lift together with any machinery space.
Fire Resistance Rating	The time, in minutes or hours, that materials or assemblies have withstood a fire exposure as established in accordance with the relevant test procedures of NFPA or BS or any other test procedure acceptable to AHJ.
Fire stops	A seal provided to close an imperfection of fit or any joint between elements, components or construction in a building so as to prevent and restrict penetration of smoke and flame through that imperfection or joint.
Flat	A building consisting of individual residential units in a multi storey building sharing common services and access facilities.
Floor	Includes any horizontal platform forming the surface of a storey and any joist board, timber, stone concrete, steel or other substance connected with or forming part of such platform.
Floor Area	- The horizontal area of a floor of a building measured from the exterior faces of exterior walls or in the case of a common wall separating two buildings from the centre line of such common wall and shall include all roof projections and balconies exceeding 1 m in width and all areas having a roof and capable of being enclosed.
Floor area, Gross	The floor area within the inside perimeter of the outside walls of the building under consideration with no deduction for stairs, closets, columns or other features.
Floor area. Net	- The floor area within the inside perimeter of the outside walls of the building under consideration excluding stairs, closets, columns or other features.
Habitable Floor	- A floor or part thereof, including roof level, regardless whether it is opened to sky or not, designated to be used for any purpose/ activity other than housing lift motors, fire pumps, water supply pumps, cooling towers, water other such equipment. Such purpose / activity shall include terrace, garden and playground and other M & E plants.
Habitable height	The height measured from the average level of the ground adjoining the outside of the external walls of the building to the finished floor level of the highest habitable floor.
Hardwood Timber	- Varieties are as given in the CIDA specifications for Building Works SCA-4/1 Yolumc I.
Headroom	- The clear vertical distance between landings, floors and ceilings, soffits etc., where people are expected to move;

Heliports (elevated)	- A facility designed to accommodate operation of helicopters, including landing area and all related functions above ground level.
Height	In relation to- (a) A room means the vertical distance measured between the finished floor level and the underside of the ceiling; (b) Any storey means the vertical distance measured between the upper surface of the floor immediately above it; (c) A wall means the vertical distance measured from the base of the wall to its highest part or, in the case of a gable to half the height of the gable. (d) A building, means the vertical distance measured from the corresponding location to its highest floor level.
High-Rise Building	Any building greater than 30 m in height and less than 60 m in height, where building height is measured from the lowest level of fire department vehicle access to the floor of the highest habitable floor.
Health care Facilities	Means a building or part thereof used for medical or other treatment or care for persons suffering from mental or physical illness, disease of infirmity and for the care of infants, convalescents or infirm aged persons.
Holding Area	Externally ventilated area located adjacent to firefighting shaft at each 10 th floor from the top most habitable floor of buildings having a habitable floor above 60 meters.
Hotel	- Building or group of buildings in which there are more than 10 guest rooms for hire, primarily used by transients who are lodged with or without meals.
Housing Complex	- A group of dwelling units on a site which is permanently in "common enjoyment" and may include a block of flats.
Industrial Buildings	- Where industrial processes are carried out.
Low-rise Building	Any building not greater than 18 m in height, where building height is measured from the lowest level of fire department vehicle access to the floor of the highest habitable floor. Low Rise Category 1 - Total area up to 800 m ² Low Rise Category 2 - Total area more than 800 m ²
"Listed"	Equipment, materials, or services included in a list published by an organization acceptable to the AHJ and concerned with evaluation of products or services and that maintains periodic inspection of production of listed equipment, materials and periodic evaluation of services and confirm that the equipment, material, or service meets appropriate and designated standards or has been tested and found suitable for a specified purpose.
Maintained	- Maintained in an efficient state and in good and proper working order.
Mechanical Ventilation	- The process of supplying or removing air to or from a building or part thereof by mechanical means or devices.
Medium-Rise Building	Any building greater than 18m in height and less than 30m in height, where building height is measured from the lowest level of fire department vehicle access to the floor of the highest habitable floor.

Mezzanine Floor	- Means any subsidiary storey interposed between two main storeys of a building and where the floor area of the intermediate floor is not more than 1/2 of the floor area.
Natural Ventilation	Means the supply of outside air to a building or removal of inside air from a building by means of windows and other openings due to wind outside and convection effects arising from temperature or vapor pressure differences (or both) between inside and outside of the building.
NFPA	National Fire Protection Association
Non-Combustible Material	Means any material which neither burns nor gives off an inflammable vapour in sufficient quantity to ignite when subjected to the test for combustibility prescribed in relevant BS, NFPA or any other test procedure acceptable to AHJ
Occupant load	The 'occupant load' of a building or part thereof means the total number of persons that may occupy such building or part thereof at any given time.
Office	Means a building or part thereof used for office purposes or for the purposes of administration, clerical work, book keeping, accounting, drawing, editorial work or banking;
Partition	Means a temporary or easily removable vertical structure made of panel work, covered with metal, wood or plastic sheets or other similar material used for the sub-division of spaces within a building;
Party Wall	A wall constructed to separate longitudinal sections of a building to prevent the horizontal spread of fire through the attic or the roof space from one section to the other.
Person with Disability	Any person who, as a result of any deficiency in his physical or mental capabilities, whether congenital or not, is unable by himself to ensure for himself, wholly or partly, the necessities of life.
Pitched Roof	A roof having an inclination of more than seven and one-half degrees with the horizontal.
Positive Suction	- Where the suction head to the pump has a water column above the centre line of the suction inlet.
Places of Public Assembly	- Include all buildings or portions thereof used for gathering of people exceeding 50.
Qualified Person	Sri Lankan national who has obtained his professional qualification in the specified field as listed below as recognised by the relevant professional institute. (a) Chartered Architect, competent in the discipline of Fire Protection and Detection, registered with Architectural Registration Board (b) Competent person having formal qualifications developed through education, training and assessed via examination being a corporate member of Institution of Fire Engineers (SL) (c) Corporate Member of Institution of Engineers Sri Lanka listed in the Directory of Building Services Engineers under the discipline of Fire protection and detection.

- (d) Corporate Member of any other recognized institution, competent in the discipline of Fire Protection and Detection, as accepted by CIDA.

Area of Refuge	In the building under consideration, an area of refuge is an area adequately separated from the rest of the building by fire resisting construction (see regulation Clause 2(52) for details), and evacuees from the rest of the building enter the area of refuge using an external corridor/ staircase that links this area to the rest of the building. An area of refuge may also be an area in an adjoining building under consideration by fire resisting construction and evacuees similarly enter this area of refuge using an external corridor.
Refuge Floor	- A protected floor that serves as a place of temporary refuge for the occupants of the building in case of fire for buildings having a habitable floor above 60 meters.
Repair Garage	- Means any premises or part thereof where repairing, painting automobile body and fender work or any process connected therewith is carried on;
Residential Building	- Means a building designed, adapted or used for residential dwelling purposes.
Restaurant	- Means a building or part thereof to which the public has access and used for the carrying on of any business where the primary purpose is the sale of food stuffs with or without cooking facility.
Roof	- Space above the top storey of a compartment is included in that compartment.
Educational Establishments	- Means a building or part thereof designed, adapted or used for the dissemination of knowledge;
Shop or Shopping Centre	- Means a building or part thereof to which the public has access and used for the carrying on of a trade or business for the purpose of display and sale of
Smoke dampers	- A device installed in an air distribution system, designed to close automatically upon detection of smoke, to interrupt migratory air flow, and to restrict the passage of smoke.
Sprinkler system	- An automatic sprinkler installation conforming to the requirements of BS/ NFPA or an equivalent standard approved by AHJ and comply with Reg. 1(13) of this regulation.
Storey	The space between the upper surface of every floor and the surface of the floor next above it, or if there is no such floor then the underside of the tie or collar beam of the roof or other covering or if there is neither a tie nor a collar beam then the level of half the vertical height of the underside of the rafters or other support of the roof.
Smoke check doors	- Door or set of doors placed in an internal corridor to restrict the spread of smoke by reducing draft.
Super High Rise Building	- Any building greater than 60 m in height, where building height is measured from the lowest level of fire department vehicle access to the floor level of the highest occupiable storey.

Superimposed Load or Superimposed Loading	- All loading other than dead load.
Verandah Way	- A covered footway at the side of a street;
Walls	(a) "Blind wall" means a solid wall constructed using solid material having no openings. (a) "Boundary wall" means any wall, enclosure or screen built on or along a boundary line of a parcel of land for the purpose of separating such land from another adjoining parcel of land; (b) "Cross wall" means an internal wall dividing a party wall or an external wall into distinct lengths; (c) "External wall" means an outer wall or vertical enclosure of a building not being a party wall even though it may adjoin a wall of another building when such wall is constructed at the boundary it shall be considered as a blind wall; (d) "Load bearing wall" means a wall which supports any load in addition to its own weight; (e) "Non-load bearing wall" means a wall which supports no load other than its own weight; (t) "Panel wall" means a non-load bearing wall in frame construction built between columns or piers and wholly supported at each storey; (g) "Partition wall" means an internal wall used for the purpose of subdividing a storey of a building into sections and which supports no load other than its own weight; (h) "Party walls" means a wall forming part of a building and used or constructed to be used along any part of its height or length for the separation of adjoining buildings, lands, or parts of the building that belong to different owners or are intended to be occupied by different persons; (i)* "Retaining wall" means a wall used to resist the lateral displacement of any material; (j) "Fire rated glass walls" means a wall constructed by fire 0 glass that resist spread of fire within or sometimes between structures to provide passive fire protection
Warehouse	Storage occupancies including all buildings or structures utilized primarily for the storage or sheltering of goods, merchandise, products, vehicles etc.
Abbreviations and Symbols	- The abbreviations and symbols listed in the Table I are used in these regulations
Designation of Purpose Groups	- For the purpose of this regulation, every building or compartment shall be regarded according to its use or intended use as falling within one of the purpose groups set out in Table 2.

For designation of purpose group, where a building is divided into compartments used or intended to be used for different purposes, the purpose group of each compartment shall be determined separately, provided that where the whole or part of a building or compartment (as the case may be) is used or intended to be used for more than one purpose, only the main purpose of use of that building or compartment shall be taken into account in determining into which purpose group it falls.

CHAPTER 1**ADMINISTRATION AND APPLICATION**

Purpose of this Chapter is to provide minimum design regulations to safe guard life and property by regulating and controlling approving design, construction, quality of materials, use and occupancy, locations and maintenance of all buildings and structures within the area of jurisdiction.

Application	Reg. 1 (1)	The provisions of the regulation shall apply to the construction, alteration, repair, equipment, use and occupancy, maintenance, relocation of every building or structure within the area of jurisdiction. Additions, alterations; repairs and changes of use or occupancy in all buildings and structures shall comply with the provisions for new buildings.
Organization	Reg. 1 (2)	There shall be a unit under the Technology Development subdivision of CIDA for formulating regulations, standards, specifications and codes of practice. Implementing monitoring and approval shall be under the control of the respective fire authorities under the local bodies.
Delegation of authority	Reg. 1(3)	The AHJ shall have the authority to delegate powers and assignments to Qualified Persons to carry out and implement relevant provisions of this regulation. Such Qualified Persons shall have the powers to execute the duties as delegated by the AHJ.
System Designer	Reg. 1(4)	The system designer shall be identified on the system design documents. Acceptable minimum evidence of qualifications or certification shall be provided when requested by the authority having jurisdiction. Qualified Persons shall be as defined in this Regulation.
Technical and Compliance Committee on Fire Regulations (TCCFR)	Reg. 1(5)	A Technical and Compliance Committee on Fire Regulations (TCCFR) shall be appointed by CIDA. The committee shall consist of minimum of five members suitably qualified and having adequate training and experience. The committee shall, (a) Advise the AHJ on any technical matters where such advice is required for taking a final decision. (b) Investigate and recommend corrective actions for violations of and non-compliances with the regulations brought to the notice of CIDA. (c) Investigate and recommend the courses of action to be taken with respect to any dispute referred to CIDA.

Responsibility of the Technical and Compliance Committee on Fire Regulations (TCCFR)	Reg. 1 (6)	Technical and Compliance Committee on Fire Regulations (TCCFR) shall perform the functions of the appeal board for any appeals made by the relevant parties. Whenever the AHJ rejects or refuses to approve proposed construction, the owner or his/her duly authorized agent can make an appeal to the AHJ. The AHJ shall refer such appeals to the above committee for advice.
System Installer	Reg. 1 (7)	The installation of the system shall be carried out by a Contractor/ Agency registered with CIDA, under the Fire Detection, Protection & Suppression (FDPS) System speciality of Electro Mechanical (EM) field.
Reports and records	Reg. 1(8)	The AHJ shall keep permanent records, including but not limited to, recommended fire requirements, design drawings, approved drawings, test certificates and approvals giving complete details of premises.
Maintenance of buildings	Reg. 1 (9)	(a) All Active and Passive fire systems in buildings and structures and all parts thereof shall be maintained in a safe condition, and all devices and safeguards required by this regulation shall be maintained in good and proper operating condition (b) Owners/occupiers of the designated buildings falling under the categories of super high rise and high rise buildings which require Fire Risk Assessment as per this regulation shall be required to produce a report issued by a Qualified Person of CIDA or a nationally recognized body acceptable to CIDA and the AHJ, once in every three years, to demonstrate the efficiency of the Fire Protection systems, means of escape and conditions of refuge floors etc.
Stop Work orders	Reg. 1 (10)	(a) Whenever any building work is being done contrary to provisions of this regulation or is being done in an unsafe or dangerous manner, the AHJ shall order such work stopped forthwith. (b) Such violations shall be noticed in writing served on the person(s) engaged in doing or causing such work to be done and such persons shall immediately stop work until authorized by the AHJ to proceed. (c) On receipt of such a notice of stoppage of work the concerned parties shall take action to correct/ rectify all defects listed and inform the AHJ in writing. (d) AHJ on receipt of such notification shall carry out further inspections and withdraw the notice of stoppage of work if the necessary rectification works has been carried out satisfactorily.

Certificate of compliance	Reg. 1 (11)	Certificate of compliance with the regulation shall be issued only by AHJ or by a Qualified Person of CIDA.
Equivalency	Reg. 1 (12)	The use of equivalent or superior Standards systems, equipment and devices complying with this regulation as a basic minimum, is permitted under these regulations. The AHJ has the authority in taking the final decision regarding the use of such alternatives.

CHAPTER 2**MEANS OF ESCAPE**

The provisions of this Chapter of this Regulation shall control the design, construction, protection, location, arrangement and maintenance of required exit facilities to provide safe means of escape from all new buildings and buildings altered or changed in occupancy

Expression		Definition
General	Reg. 2(1)	Areas which are designated as means of escape such as exit staircase, firefighting lobby, smoke stop lobby, exit passageway, escape corridors, shall not be used for any other purpose.
Staircase Identification	Reg. 2(2)	Every staircase forming a part of means of escape shall be clearly identified and the floor levels shall be clearly indicated.
Definitions and General Requirements	Reg. 2(3)	For the purpose of these Regulations the following definitions and associated requirement shall be applicable; (a) " Protected lobby" Is a fire-resisting enclosure providing access to a protected stairway via two sets of fire-resisting and self-closing doors, into which no rooms open, other than toilets or lifts (b) "Dead End " An area from which escape is possible in one direction only. (c) " Place of ultimate safety" ; Place beyond the building in which a person is no longer in danger from fire (d) " Final Exit" Means the termination of an escape route from a building giving direct access to a 'Place of Safety' such as a street, passageway, walkway or open space, and sited to ensure that persons can disperse safely from the vicinity of the building and the effects of fire. (e) "Means of Escape"; Structural means forming an integral part of the building whereby persons can escape from fire by their own unaided efforts to a place of safety. (f) " One way travel " (i) A one-way travel or "common path" exists if a floor space is arranged or provided with partitioning works such that occupants within that space are able to travel in only one direction to reach any of the exits or to reach the splitting point where they have the choice of two or more routes of travel to remote exits. (ii) The travel distance from the most remote point to the splitting point shall not exceed the permissible one-way travel distance allowed in Diagram 1 A. At the splitting point, the angle of divergence between any two alternative routes shall not be less than 90 degrees in order that the routes originating from the splitting point can be considered as two-way travel.

- (iii) The aggregate travel distances of the one-way travel from the most remote point to the splitting point and the continuous two-way travel from the splitting point to the nearest exit shall not exceed the permissible two-way travel distance allowed in Diagram 1 A.
- (g) "Exit" - means of leaving from the interior of the building to an exterior space which is provided with, either singly or in combination, by the following: exterior door openings, protected staircases, exit passageways or exterior stairs.
- (h) The "occupant load" of a building or part thereof means the maximum number of persons that may occupy such building or part thereof at any one time.

The "occupant load" shall be established either;

- (i) By the number of occupants for whom each occupied space of the building is designed as shown on the plans, or
- (ii) By applying to the floor areas available for occupation, the appropriate areas per person as laid down in Table 3, whichever is the greater.
- (i) "Travel distance" means the distance required to be traversed from the most remote point in any room or space to the centre of a door opening directly to a protected staircase, or an open exterior space, or an exit passageway.
- (l) "External exit staircase" means an exit staircase which reaches the ground floor and is located outside the building envelope.
- (k) "Protected Stairway"; A stairway which is adequately protected from fire in adjoining accommodation by fire resisting construction and either discharges through a final exit or a protected route to a final exit.
- (1) "Exit passageway" Is a means of horizontal escape, leading to a place of safety, and is suitably fire rated.
- (m) "Corridor" A passage providing a means of access from rooms or spaces to an exit.
- (n) "Vertical exit" means a staircase or ramp serving as an exit from one or more storeys above or below ground level.
- (o) "Direct distance" means the shortest distance from a point in the floor area, measured within the external enclosure walls of the floor area, to the relevant exits, ignoring internal walls, partitions and fittings other than the enclosure walls of protected staircases

See Diagrams 1-B and 1-C, Where the "direct distance" concept is adopted, the maximum permissible direct distance shall be taken as being $\frac{2}{3}$ of the maximum permissible travel distance given in Table 5.

		(p) "Two-way escape" means a situation where there are 2 independent escape routes and the subtended angle between the 2 lines of travel from the relevant point in the floor area to the exits shall not be less than 45¹⁾. See Diagram 1-D. If the subtended angle is less than 45°, then it should be considered as a one-way escape arrangement, notwithstanding the fact there are 2 exits.
Determination of Exit Requirements -General -	Reg. 2(4)	(a) The determination of exit requirements for a building shall be based upon the type of use or occupancy of the building, the occupant load, the floor area, the travel distance to an exit, and the capacity of the exits as provided in Table 5 and herein. (b) Every storey of a building shall be provided with exit facilities for its occupant load. (c) Vertical exits (staircase or ramps) provided from any storey above ground level may serve simultaneously all storeys above ground level and vertical exits provided from any storey below ground level may serve all storeys below ground level, subject to the provisions of Regulation 6(42) which prohibits basement staircases being continuous with staircases serving upper storey.
Mixed Occupancy	Reg. 2(5)	Where different parts of a building or storey of a building are designed for different types of occupancies or used for different purposes at the same time, the means of escape requirements of the entire building or storey of the building shall be determined on the basis of the type of occupancy or usage having the most stringent means of escape requirements, or the means of escape requirements for each building section shall be determined separately.
Multiple Occupancy or Use	Reg. 2(6)	Where a building, or storey of a building or a part of a building, is used for multiple purposes involving different activities at different times, that purpose or usage involving the greatest number of occupants shall form the basis of determining the means of escape requirements.
Non-simultaneous Occupancy	Reg. 2(7)	The floor areas of toilets, locker rooms, in-house meeting rooms, storage rooms, staff canteens and similar rooms and spaces that serve other rooms and spaces on the same storey but are not occupied at the same time as such other rooms or spaces, may be omitted from the occupant load calculations of that storey of the building on which they are located. In the case of factory and office buildings only, the floor area of a corridor may also be omitted from the occupant load calculations of that storey of the building on which the corridor is located.

Exit Requirements --Capacity of Exits and Exit Facilities	Reg. 2(8)	(a) The capacity of exits, staircases and other exit facilities shall be measured in units of width of one half of a meter and the number of persons per unit of width shall be determined by the type of occupancy and type of exit as listed in Table 5 (I) to (VIII) (b) Fractions of a unit of width less than 250 mm shall not be credited. Where 250 mm or more are added to one or more full units, half of a unit of width shall be credited (c) Where a room or space is required to be provided with two exits, each exit shall be of sufficient width to accommodate not less than half the total occupant load. (d) Exit staircases shall be of such width that in the event of any one staircase not being available for exit purposes the remaining stair cases shall accommodate the highest occupancy load of anyone floor discharging into it calculated in accordance with the stipulations in Table 5 (I) to (VIII)
Minimum Widths	Reg. 2(9)	(a) The width of exits, exit staircase or other exit facilities shall not be less than the minimum as specified in Table 5 (I) to (VIII) (a) The minimum clear width of an exit door opening shall be not less than 1100 mm. (b) The width of exit doors serving a room with an occupant load of 2 persons shall not be less than 750 mm clear. (c) The width of a single leaf swing door along the means of escape shall not exceed 1250mm. (d) Minimum width required for wheel chair access shall not be less than 950mm.
Maximum Widths	Reg. 2(10)	The maximum width of staircases shall not be more than four units of width unless divided by handrails into sections of not less than two units of width or more than four units of width. (a) For the purpose of determining the exit capacity of a staircase that is wider than 4 units of width and forms part of the required means of escape from any storey of the building, that part of its width in excess of 4 units of width shall not be taken into account. (b) The maximum width of single exit staircases shall be not more than 2000mm. (c) Where staircases exceed 2000mm in width, handrails shall be used to divide the staircase into sections of not less than 1000mm of width.

Measurement of Width	Reg. 2(11)	The width of a staircase shall be the clear width between; (a) If the staircase is enclosed on both sides by walls , the finished surfaces of the walls (b) The finished surface of the wall and the inner side of the balustrade or handrail, if the staircase has a wall on one side and a balustrade or handrail on the other side, or (c) The inner sides of the balustrade or handrails if the staircase has balustrades or handrails on both sides.
Number of Exits from Rooms and Spaces	Reg. 2(12)	There shall be at least two door openings remote from each other and leading to exits, from every room or enclosed space in which the total occupant load exceeds the maximum permissible occupancy load for one door as listed in the Table 4. (a) Application of Table 4 shall be subject to the travel distance being in accordance with Regulation 2(3)(i). (b) In a block of residential flats, there shall be not less than two exit doors from each flat, that shall have direct access to the staircase or staircases that are required to be provided for such block of flats under subsequent Reg. 2(13) of this Regulation except; (i) That only one exit door may be provided when there is a smoke dispersal condition within the flat, or (ii) The exit path is sprinkler protected, to the satisfaction of the AHJ and the travel distance from the furthestmost bedroom door to the exit door does not exceed 13 m. When there is an alternative escape route from the furthestmost bedroom door to the single exit door, an extension of the aforesaid travel distance up to 15 m may be permitted. (c) A classroom in any educational institution shall have at least 2 exits, if the occupancy exceeds 50. (d) In a block of small residential units consisting of more than one floor, there shall be at least one exit door from each storey of each unit that shall have direct access to the staircase or staircases that are required to be provided for such block of small residential units under subsequent Reg. 2 (13) of this Regulation. Exceptions; (i) The aggregate floor area of the upper storey of the small residential unit should not exceed 60 m²; (ii) The distance from the furthestmost bedroom door to the main entrance door should not exceed 13 m; (iii) The escape route should not pass through or near an unprotected opening of the kitchen; and (iv) The small residential unit should not comprise more than 2 storeys.

Number of Staircases or Exits per Storey	Reg. 2(13)	There shall be at least two independent staircases or other exits from every storey of a building, unless otherwise permitted under other subsequent provisions of the regulation
	Reg. 2(14)	Staircase serving all buildings (except purpose Group I) shall be provided with a signage not smaller than 300x300mm and within the stairwell at each storey landing with following information: (a) The storey number at least 100 mm in height (b) An identification of the staircase in alphabetical and/or numeric; at least 100 mm in height (c) The letters and numbers on the sign can be of any color that shall contrast with the back ground color
	Reg. 2(15)	Where Scissor exit staircase is provided, each exit staircase shall be separated from the other by non-combustible construction having fire resistance not less than that required for the enclosure.
Exit Reduction	Reg. 2(16)	When a floor area has access to areas of refuge conforming to the provisions of Regulation 2(53) the number of persons for whom vertical exits are to be provided may be reduced to 50 percent of the occupant load of the floor area when one area of refuge is provided and may be reduced to 33 1/3 of the floor area when two areas of refuge are provided.
Location of Exits	Reg. 2(17)	All exits and access facilities shall be located as follows: All exits and access facilities shall be required to comply with the following: (a) Exits and access facilities shall be clearly visible or their locations be clearly indicated with directional signs. All exits must be kept readily accessible and unobstructed at all times. (b) Every occupant within a storey of a building shall have direct access to the required exit without the need to pass through spaces occupied by other occupants (c) When more than one exit is required each exit shall be placed as remote as possible from the other, complying with the travel distance requirements.

Smoke Free Approach Staircase	Reg. 2(18)	Entry at every storey level (including first storey) to an exit staircase of any building of more than six storey's above ground level shall be through: (a) An external passageway or external corridor. The openings for natural lighting and ventilation to the corridor shall be so located that they face and open to : (i) The external space: or (ii) A street, service road or other public space which is open to sky. (b) An internal passageway complying with Regulation 2(40)
Smoke free Lobby	Reg. 2(19)	A lobby that is separated from the adjoining areas of the building by a wall having a fire resistance as per the building fire rating.(Table 8) The exit access door shall have fire resistance of half that of the enclosure fitted with automatic self-closing device. The design of a smoke stop lobby shall not impede the movement of occupants through the escape route. The floor area of a smoke stop lobby shall not be less than 3 m². If a smoke stop lobby also serves as a fire fighting lobby, the floor area shall not be Less than 6 nrand with no dimension less than 2 m. The floor shall be graded from the lift door towards the lobby door with a slope not exceeding 1 in 200.
Firefighting lobby	Reg. 2(20)	A smoke free or firefighting lobby which acts as buffer space for entry in to the protected staircase and used by fire fighters during emergency, shall be maintained as common property. (a) Permanent ventilation openings in an external wall to which the lobby abuts; such openings being not less than 15 percent of the area of the lobby and located not more than 9m away from any part of the lobby; (b) Mechanical ventilation complying with Reg. 2(23). (c) Permanent ventilation openings of similar size as in Clause (a) above of this Regulation, opening to an open air well having a superficial plan area of not less than 10 m² or 0.1 m² for each 300 mm of height of the building whichever is the greater. The enclosure walls of such air well shall be of a similar standard of fire resistance as that required for the enclosures of the protected staircase and shall be imperforate except for the ventilation openings for the lobby or staircase.
Cross Ventilated Corridor	Reg. 2(21)	A cross ventilated corridor having openings in at least two external walls; such openings being not less than 50 percent of the superficial 1 area of the said walls and within 13 m of any part of the corridor.

Smoke free Approach in basement	Reg. 2(22)	Smoke free approach to exit staircase in basement occupancy: (a) In a building comprising more than 4 basements, entry to exit staircase serving the basement storeys at every basement storey level shall be through smoke stop lobbies, one of which shall be designated as fire fighting lobby. The exit staircase connecting to the fire fighting lobby shall be pressurized. (b) In a building comprising 2,3 or 4 basements, entry at every basement storey level to at least one of the exit staircases serving the basement storeys shall be through a smoke stop lobby and where only one smoke stop is provided, it shall be required to serve as a fire fighting lobby.
Pressurized Systems for Stairways	Reg. 2(23)	All fire escape stairways serving high rise and super high-rise buildings shall be provided with a pressurization system.
Pressurization of Exit stairways	Reg. 2(24)	(a) In any building of which the habitable height exceeds 30m, any internal staircases without adequate provision for natural ventilation shall be pressurized (b) where the upper part of the staircase is naturally ventilated, its lower part can be provided with mechanical ventilation or pressurization whichever is appropriate (c) In a building comprising more than 4 basements, exit staircase connected to firefighting lobby in basement stories shall be pressurized.
Pressurization of Internal corridors in hotels	Reg. 2(25)	Where internal corridors are required to be pressurized in compliance with Reg. 2(26) the pressure within such corridors shall be higher than in the guest rooms and the pressure within the internal exit staircases higher than that of corridor. The pressure gradient shall not be less than 50Pa for each area of the pressurized zone
Pressurization level	Reg. 2(26)	All fire escape stairways serving high-rise and super high-rise buildings shall be provided with a pressurization system as follows: (a) The minimum pressurization level of 50 Pa shall be achieved with all doors closed and all pressure relief systems working. (b) When in operation, the pressurization system shall maintain a pressure differential of not less than 50Pa between the pressurized staircase and the adjacent area when all doors are closed. (c) Where a smoke-free lobby is also pressurized, the pressure at the exit staircase shall always be higher than that of the smoke free lobby. (d) The force required to open any door against the combined resistance of the pressurized air and the automatic door closing mechanism shall not exceed 110 N at the door handle.

		(e) When in operation the fan capacity shall be enough to maintain minimum average egress velocity through doorways of 0.75 m/s with any three single leaf entry doors (one on each of three levels) and the largest exit door open, plus leakage allowance for all other doors. (f) When in operation, the system shall maintain an air flow of sufficient pressure when 03 nos of doors open simultaneously to stairway to prevent smoke from entering in to the pressurized area. (g) The minimum fire resistance rating for the enclosure of the pressurization plant shall be the same as that of the staircase served (h) The number and distribution of injection points for the supplying of the pressurization air to the exit staircase shall ensure an even pressure profile. Multiple injection system with air supplies at no greater than 12 m apart to be used. (i) Emergency power supply shall be provided for all equipment serving staircase pressurization, by passing the main distribution panel. The power supply for the pressurization equipment must be fed directly, by-passing the Main Distribution Board and emergency power supply must be directly from the generator output. This requires an Auto Transfer Switch panel at the pressurization equipment room. (j) All wiring, cables, electrical equipment, starters, relays, etc., including building primary and secondary sources of supply shall be suitable for continuous operation at 250 °C ambient for a time period as per the rating given for the building. (k) All cables feeding the pressurization fan located outside fire protected ducts shall be fire resistant and shall have adequate mechanical protection. (l) The pressurization system shall be automatically activated by the building fire alarm system. In addition a remote manual start/stop switch shall be made available for firemen at the fire command centre. Visual indication of the operation status of the pressurization system shall be provided. (m) Supply air for the pressurization system shall be drawn directly from the outside and its intake shall be not less than 5 m from any exhaust discharge openings. (n) The pressurization system shall not be controlled by any the Building Management System. (o) Measures shall be provided to ensure that no excessive pressure buildup occurs to negate the pressure gradient specified.
Exit Doors	Reg. 2(27)	Exit doors giving access to escape corridors shall be so positioned that their swing shall at no point encroach on the required minimum width of the escape corridor. The door must be provided with a door closer.

Smoke Control System	Reg. 2(28)	A "smoke control system" which is a "mechanical ventilation" system, shall be provided for high rise buildings and other buildings where required by the Authority having Jurisdiction" and shall comply with the following stipulations:
	Reg. 2(29)	(A) <u>General Requirements</u> The "smoke control system" shall consist of the following: (a) A "Smoke Purging System" which is independent of all other systems serving other parts of the building, and which will provide an extraction rate of not less than 8 air changes per hour. (i) "Automatic Activation" The smoke purging system shall be automatically activated by the building fire alarm system. (ii) "Manual Control" A manual control system shall be provided at the Fire Control Centre, or at the main Fire Control Panel, which will override the automatic control system. Visual indication of the operation status of the smoke control system shall be provided with the remote control facility. (b) "Supply Air" Supply air for the smoke control system shall be drawn directly from the exterior, and the intakes shall be not less than 5 m from any exhaust discharge openings. (c) The exhaust fan shall be capable of operating efficiently at 250 centigrade for 2 hours and supplied from a secondary source of power.
	Reg. 2(30)	(B) <u>Additional Requirement</u> Basement Smoke Extraction Systems - Basement smoke extraction systems where required by the "Authority having Jurisdiction shall be provided in addition to the stipulations of the "general requirements" above and shall comply with the following stipulations: (i) "Smoke Purging Rate" shall be not less than] 0 air changes per hour for fire conditions. The extract points shall be arranged so that 50 percent are at high level and 50 percent at low level. (ii) "Ducts" Where ducts are used for the smoke extraction system they shall comply with the requirements for mechanical ventilation systems in buildings. The ducting shall be manufactured with metal sheets of suitable thickness of suitable standards in compliance with ANSI standards. (iii)"Melting Point "All components of the smoke extraction system must have a minimum melting point of 800 °C. (iv) Smoke control system shall be activated by the building <u>fire alarm system.</u>

	Reg. 2(31) (C) <u>Ductless Jet Fan system in basement Car park</u> (i) The space shall be divided into smoke control zones not larger than 2000 m² (excluding plant rooms & circulation spaces) (ii) Each smoke control zone shall have its own jet fan system (Fresh air fans, exhaust air fans and jet fans) (iii) The jet fan system shall be activated by the fire detection system/ sprinkler system serving the basement car park level or a manual call point. A fireman over-ride switch shall be provided at the Fire Command centre. (iv) The air velocity within the escape routes shall not exceed 5 m/s. The mechanized air supply fans, smoke exhaust fans, jet fans, duct works and wiring shall be capable of operating effectively at 250 degrees Celsius for 2 hours.
Fire Dampers	Reg. 2(32) Listed fire dampers shall be provided where air ducts penetrate or terminate at openings in walls or partitions required to have the same fire rating as the compartment wall. (a) Fire dampers used for the protection of openings in walls, partitions or floors with fire resistance rating of not less than 3h shall have a minimum of 1.5h fire protection rating. Where the fire resistance rating is 3h or more, the damper shall have a minimum of 3h fire protection rating. (b) All fire dampers shall close automatically using either a fusible link or a closing device activated by a heat detector. (c) Fusible links shall have a temperature rating of 15° C above the normal operating temperature. (d) Dampers shall close against the maximum calculated static air pressure of the portion of the air duct system in which they are installed.
Air duct Access and Inspection	Reg. 2(33) A service opening shall be provided in air ducts adjacent to each fire damper (a) Service opening shall be identified with lettering having a minimum height of 12.5mm to indicate the location of the fire protection devices within (b) Service opening shall be located at 6 m intervals along the length of the duct and at the base of each vertical riser.
Engineered Smoke Control System	Reg. 2(34) Engineered Smoke Control Systems where required by the Authority having Jurisdiction shall be provided in addition to the stipulations of the "general requirements" above and shall comply with the following stipulations; (a) The capacity of an engineered smoke control system shall be capable of handling the largest demand for smoke exhaust.

		(b) The design smoke layer base shall be above the heads of people escaping beneath it. The minimum height shall be 2.5 m (c) Smoke reservoirs to prevent the lateral spread of smoke, and to collect smoke for removal shall be of non-combustible construction capable of withstanding smoke temperatures (d) The smoke reservoir size for a smoke ventilation system shall not exceed: (i) 2000 m² for natural smoke ventilation (ii) 2600 m² for mechanical smoke ventilation (e) The minimum length of the smoke reservoir shall not exceed 60 m. (f) The smoke ventilation system shall be provided with secondary source of power.
Activation	Reg. 2(35)	The engineered smoke ventilation system shall be activated by smoke detectors located in the "smoke control zone."
Shut down of Air-conditioning systems	Reg. 2(36)	Air conditioning systems within the area served by the engineered smoke ventilation system shall be shut down automatically upon the activation of the smoke ventilation system.
Fire Dampers	Reg. 2(37)	Shall not be installed in the engineered smoke ventilation system.
Means of Escape Requirements	Reg. 2(38)	(a) The determination of means of escape requirements for a building shall be based on the type of occupancy, occupant load, floor area, travel distance to an exit and the capacity of exits provided in Table-5 (I) to (VIII). (b) Any building other than Purpose group 1 having total floor area exceeding 800 m² or the one way travel distance exceeding as per table 5 and the height of the occupiable floor exceeding 9 m shall be provided with an alternative means of escape.
Exit Passageways	Reg. 2(39)	All exit passages shall be accessible and kept unobstructed at all times. Exit passageways that serve as means of escape of a building shall have the requisite fire resistance as per Table 8.
Internal exit Passageway	Reg. 2(40)	An internal exit passageway serving as a part of required exit shall be enclosed with fire resistant construction complying with the provision of Table 8. (a) Exit doors opening in to the exit passageway shall have fire rating as required for exit doors opening in to exit staircases fitted with automatic self-closing devices (b) The minimum width and capacity of exit passageways shall comply with the requirements given Table 5 (I) to (VIII). (c) Changes in level along an exit passageway less than two risers shall be a ramp

		(d) Any internal passage way connected to a pressurized exit passageway shall not be naturally ventilated and shall also be pressurized.
External exit Passageway	Reg. 2(41)	The external wall between the exit passageway and the rest of the floor space may provide ventilation openings of non-combustible construction, fixed above a level of 1.8m from the finished floor level of the passageway. Such ventilation openings shall be located not less than 3m from any opening of an exit staircase; (a) May be roofed over provided the depth of the roofed over portion shall not exceed 3m to avoid smoke logging. (b) May be enclosed on the open side only, by a parapet wall of not less than 1100mm or more than 1200 mm in height. (c) Exit doors opening in to an external exit passageway shall have fire resistance for at least 30 minutes and fitted with automatic self-closing device.
Internal Staircases	Reg. 2(42)	Internal staircases serving as fire exits exceeding three storeys in height shall satisfy requirements as for protected staircases given in Regulation 2(3) (k) The minimum width and capacity of staircases shall be as listed in Table 5 and such staircases shall comply with the following:- (i) Headroom: The clear headroom shall not be less than 2200 mm. (ii) Landings: Except for circular or geometric stairs, all staircases shall be in straight flights with landings provided at intervals of not more than 16 risers and at every floor level. Changes in level along an exit stairway less than two risers shall be a ramp. (iii) The minimum width of a landing shall be not less than the width of the existing staircase. (iv) Winders: Winders shall not be permitted in any building other than residential buildings not exceeding 9 m and in such cases there shall be not more than 1 winder per 90 ° turn.
Handrails Balustrades etc.	Reg. 2(43)	Every exit staircase shall have walls, grilles or handrails on both sides, except where the width is less than 1250mm. or less in width, can have a handrail one side only Where the width of the exit staircase exceeds 2000mm, intermediate handrails shall be provided.
Ventilation	Reg. 2(44)	All internal exit passageways shall be naturally ventilated by fixed ventilation openings in an external wall, such ventilation openings being not less than 10 per cent of the floor area of the exit passageway. Internal exit passageways that cannot be naturally ventilated shall be mechanically ventilated in compliance with requirements.

Facility for Re-entry	Reg. 2(45)	Internal staircases where entry is restricted with Fire doors, facility to re-enter the building shall be provided at every 5th floor. Monitoring facilities shall be provided for such doors to prevent unauthorized entry.
External Staircases	Reg. 2(46)	External staircases may be used as exits in lieu of internal staircase provide they comply with all of the requirements of Regulation 2(42) for internal staircases, except enclosure, and also comply with the provisions of Regulation 2(3)(j).
Scissor Exit Staircases	Reg. 2(47)	Where two separate internal exit staircases are contained within the same enclosure, each exit staircase shall be separated from each other by noncombustible construction having fire resistance for a minimum period equal to that required for the enclosure; (a) Such scissor exit staircases shall comply with all applicable provisions for exit staircase. (b) Door opening into scissor exit staircases shall be not less than 7m measured as travel distance between the two closer edges of the staircase doors (c) Door opening into scissor exit staircases shall be at least 7m measured as travel distance between the two closer edges of the staircase doors. (d) Where there is only one pair of scissor exit staircases, the door opening into scissor exit staircases shall be spaced at least 1/3 the diagonal dimension of the area to be served in a sprinkler protected building and 1/2 the diagonal dimension in a non-sprinkler protected building. <u>Exclusions</u> Scissor type staircases shall not be accepted as alternative means of escape for super high rise buildings.
Hardwood Staircases	Reg. 2(48)	Hardwood staircases shall only be permitted as internal staircases in Purpose Group 1 buildings.
Spiral Staircases	Reg. 2(49)	Spiral staircases may serve as required exits from mezzanine and balconies on any storey having an occupant load not exceeding 25 persons, when built externally as an enclosed staircase and also of non-combustible materials and having a tread length of at least 0.75 m. Such spiral staircases shall not be more than 10 m high and shall not be used as an exit from a public area. Spiral staircases shall not be allowed as a fire exit for Purpose Group 2(a).
Exit Ramps	Reg. 2(50)	Internal and external ramps may be used as exits in lieu of internal and external staircases, and also as a "means of escape" for disabled persons, provided their exit capacities are as listed in Table 5 and except where otherwise provided, they comply with the applicable requirements of Regulations 2(42) and 2(46) and with the following: (a) Maximum slope: The slope of such ramps shall not be steeper than 1 in 8 except that in factory buildings and go downs the slope shall not exceed 1 in 12 and except as otherwise provided <u>in any other Regulations.</u>

		(b) Changes in direction: Ramps shall be straight with changes in direction being made at level platforms or landings, except that ramps having a slope not greater than 1 in 12 at any place may be curved (c) Length: The sloping portion of ramps shall be at least 1 m but not more than 12.5 m long between platforms or landings. (d) Platforms: Level platforms or landings at least as wide as the ramp shall be provided at the bottom, at intermediate levels where required and at the top of all ramps. Level platform shall be provided at each door opening into or from a ramp having a minimum length in the direction of exit travel of 1 m. (e) Guards and handrails: Guards and railings of ramps shall comply with Regulation 4(2) (d) except that only ramps having a slope steeper than 1 in 10 need comply with the requirements of handrails and intermediate handrails shall not be required. (f) Surface: All ramps shall be provided with non-slip surfaces. Provided that ramps serving as means of escape to only one basement storey need not be protected by enclosure walls. (g) Exit ramps if enclosed shall be ventilated to comply with the requirements for ventilation of exit staircases (h) Exit ramps serving as means of escape to only one basement need not be protected by enclosure walls.
Exit Doors	Reg. 2(51)	Exit doors and doors providing access to exits shall open in the direction of escape and comply with the relevant parts of the following; (a) Exit doors shall be openable simply by pushing open or by means of a panic bar.(not applicable to buildings under purpose group 1) (b) Exit doors, which are required to have fire resistance rating, shall comply with the relevant provisions for fire resisting doors. (c) Exit doors opening in to exit staircases and exit passageways shall not impede the egress of occupants when such doors are swung open (d) Door opening into the exit path shall close when pushed in the direction of the movement. (e) Any door located in a path of travel shall be side hung and shall be the full width of the exit path. (f) Width and height of doors: The capacity of exit and corridor door openings shall be as listed in Table 5 (I) to (VIII). Door jambs or stops and the door thickness when the door is open shall not reduce the required width by more than 80 mm. The

		minimum open width of the exit door openings shall be 1 m, where the width of the corridor is 1.2 m. The minimum clear height of the door opening shall be 2.1 m. (g) Fire door to protected staircase and smoke stop/Fire lift lobby shall be provided with a vision panel. The vision panel shall have a clear view size of 100mm width by 600mm height and shall have the same fire resisting rating. (h) The provision of vision panel shall not apply to exit doors of residential apartment or maisonette units (i) All compartment doors on long exit paths shall swing both ways and shall be provided with a vision panel, as specified in above (c). (j) Door hardware: Any exit door which has to be kept shut and fastened whilst the building is occupied should be fitted with "panic bar", appropriately shown "Push bar to Open" in letters not less than 100 mm high. (k) An exception to this rule shall be made for exit doors which are kept locked back in the fully open position during the whole time that the building is occupied. All other exit doors shall be fitted with self-closing devices other than rising butt hinges, and all exit doors shall be hung on strong metal hinges and not on any combustible material. The panic bolt shall be not more than 1 m from the floor and shall be capable of being operated by a horizontal thrust. (l) Power operated doors: Power operated doors may be used as exit or corridors provided they remain closed in case of power failure, but shall be manually operable. No such door shall be credited as required exit unless it swings in direction of travel. (m) Revolving doors: Revolving doors shall not be used as required exits or as access doors to exits. (n) Sliding doors: Horizontal and vertical sliding or folding doors shall not be used as required exits or as access doors to exits in places of public resort, shopping centres and departmental stores. (o) Clear opening size of doors i. In the case of an exit door having a single leaf door, the opening shall be measured between the edge of the door jamb and the surface of the door when opened at an angle of 90 degrees, and ii. In the case of an exit door having 2-leaf and fitted with an approved automatic door closer, the clear openings shall be measured between the surface of one leaf to the other door leaf when opened at an angle of 90 degrees, and iii. If one of the door leaves is bolted to the door frame <u>and/or floor by a manually operated bolt</u>, this door leaf
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		shall not be considered for the purpose of determining the exit capacity of the door. the opening of the door leaf shall have a clear width of not less than 900 mm measured between the edge of the bolted door leaf and the surface of the other door leaf; when opened at an angle of 90 degrees.
Area of Refuge	Reg. 2(52)	(a) An area adequately separated from the rest of the building by fire resisting construction and evacuees from the rest of the building enter the area of refuge using external corridor that links this area to the rest of the building. (b) An area of refuge may also be an area in an adjoining building which is separated from the building under consideration by fire resisting construction and evacuees similarly enter this area of refuge using an external corridor (c) An area of refuge shall always be accessible. (d) An area of refuge shall be adequate in size to hold the occupant load it receives from the floor area it serves in addition to its own occupant load on the basis of 0.5 m² per person and shall be provided with at least one protected staircase and exit facilities of adequate width discharging at ground level directly to an exterior open space. (e) Doors providing access to an area of refuge shall be kept unlocked at all times when the floor area served by the area of refuge is occupied. Such doors shall be swinging and self-closing doors having a fire resistance rating of 1V2 hours except that doors in fire divisions of 3 or 4 hours fire resistance rating shall have a fire resistance rating of 3 hours. They shall swing in the direction of exit travel.
Emergency Lighting	Reg. 2(53)	Occupied areas and escape routes in all buildings, except purpose group 1(c) shall be provided with artificial lighting facilities to the satisfaction of the requirements under this regulation. (a) The minimum illuminance to be provided for all exits shall be not less than 5 lux measured at the floor. (b) The delay between the failure of the electrical supply to normal lighting and the energization of the emergency lighting shall not exceed 1 second. (c) Alternate power supply should be adequate to provide the power to maintain the lighting for a period not less than 3 hrs.
Exit and Directional Signs	Reg. 2(54)	In every building or part thereof; except buildings of Purpose Group 1 in Table 2, the locations of every exit door and exit facility on every storey shall be clearly indicated by exit signs using internationally accepted symbols.
	Reg. 2(55)	Such signs shall be placed to be clearly visible at all times. In long corridors, in open floor areas, and in all situations where the location of the exits may not be readily visible, directional signs using internationally accepted symbols shall be provided to serve as guides from all areas of the corridors or floors to the exits.

	Reg. 2(56)	(a) The exit and directional signs shall be in white on a green background with letters and/or symbols and shall be illuminated at all times. (b) Additional low level or floor mounted exit and exit directional signs shall be provided where overnight sleeping accommodation is provided. (c) The legends, dimensions, design and installation of the exit signs and directional signs shall comply with the British Standard. (d) Photo-luminescent exit signs with letters/symbols in white on green background are allowed for use in low rise buildings not exceeding 800 m² of total area.
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CHAPTER 3**STRUCTURAL FIRE PRECAUTIONS**

The purpose of this Chapter of the Regulation is to minimize the risk of the spread of fire between adjoining buildings by a stable and durable form of construction, to prevent the untimely collapse of buildings in the event of fire and to prevent the spread of fire between specified parts of buildings by the division of such buildings into compartments.

In this Chapter of the Regulation and the Tables thereto-

Definitions	Reg. 3(1)	(a) "Basement storeys" has the meaning ascribed to it in Regulation of Chapter 1. (b) "Compartment" means any part of a building which is separated from all parts by one or more compartment walls or compartment floors or both such walls and floors; and for the purpose of this Chapter, if any part of the top storey of a building is within a compartment, the compartment shall also include any roof space above such part of the top storey (c) "Compartment wall" and "compartment floor" means a wall and a floor respectively which is provided for the purpose of dividing a building into compartments to prevent spread of fire and smoke. (d) "Door" includes any shutter, cover or other form of protection to an internal opening in any wall or floor of a building or in the structure surrounding a protected shaft, whether the door is constructed of one or more leaves. (e) "Element of structure" means- (i) Any member forming part of the structural frame of a building or beam or column (not being a member forming part of a roof structure only). (ii) A floor, including a compartment floor, other than the lowest floor (in contact with the ground) of a building. (iii) An external wall. (iv) Party wall. (v) A compartment wall. (vi) Structure enclosing a protected shaft. (vii) A load bearing wall or load bearing part of a wall, (viii) A gallery or mezzanine. (f) "External wall" has the meaning ascribed to it in Chapter 1 and complies with Regulation 3(33) to 3(41). (g) "Fire resistance" has the meaning ascribed to that expression in Chapter 1 and Regulation 3(31) and 3(32) of this Chapter.
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Designation of Purpose Groups	Reg. 3(2)	For the purpose of this Chapter every building or compartment shall be regarded according to its use or intended use as falling within one of the purpose groups set out in Chapter 1 and Table 2 and, where a building is divided into compartments used or intended to be used for different purposes, the purpose group of each compartment shall be determined separately. Provided that where the whole or part of a building or compartment (as the case may be) is used or intended to be used for more than one purpose, only the main purpose of use of that building or compartment shall be taken into account in determining into which purpose group it falls.
Rules for Measurement	Reg. 3(3)	In this Chapter- (a) The height of a building, or (where relevant) of part of a building as described in the code means the height of such building or part, measured from the mean level of the ground adjoining the outside of the external walls of the building to the level of half the vertical height of the roof of the building or part, or the top of the walls or of the parapet (if any), whichever is the higher: (b) The area of ; (i) Any storey of a building or compartment shall be taken to be the total area of that storey bounded by the inner finished surfaces of the enclosing walls or, on any side where there is no enclosing wall, by the outermost edge of the floor on that side; (ii) Any room or garage shall be taken to be the total area of its floor bounded by the inner finished surfaces of the walls forming the room or garage; (iii) Any part of a roof shall be taken to be actual visible area of such part measured on a plane parallel to the pitch of the roof. (c) The cubical extent of a building or compartment shall be ascertained by measuring the volume of space contained (i) The inner finished surfaces or on any side where there is no enclosing wall, a plane extending vertically above the outermost edge of the floor on that side; (ii) The upper surface of its lowest floor (iii) In the case of a building or of a compartment which extends to a roof, the under surface of the roof or, in the case of any other compartment, the under surface of the ceiling of the highest storey within the compartment, including the space occupied by any other walls, or any unprotected shafts, ducts or structure within the space to be so measured, but excluding protected lift walls, staircases etc., and also any accommodation such as lavatory and locker rooms which are enclosed with walls having not less than a one hour standard of fire resistance and Class A fire doors.

Cases where excess height above 4m may be ignored in computing cubical extent	Reg. 3(4)	In computing the cubical extent of compartments in shops, similar premises and in single storey factory building, a height of 4 m may be used where the actual height exceeds that figure provided that this rule shall not be applied when a compartment comprises more than one storey or contains mezzanine, galleries or lofts.
		PROVISION OF COMPARTMENT WALLS AND COMPARTMENT FLOORS
Excess floor area and cubical extent	Reg. 3(5)	Any building other than a building of Purpose Group 1 which has - (a) Any storey the floor area of which exceeds that specified as relevant to a building of that area in column (2) of Table 6, or (b) A cubic capacity which exceeds that specified as so relevant in column (3) of Table 6, shall be so divided into compartments by means of compartment walls or compartment floors or both that- (i) No such compartment has any storey the floor area of which exceeds the area specified as relevant to the building in column (2) of the Table: and (ii) No such compartment has a cubic capacity which exceeds that specified as so relevant in column (3) of the Table.
Compartmentation by height	Reg. 3(6)	(a) In any compartment up to a habitable floor of 30m, no compartment shall comprise more than three storeys, except for atrium spaces. (b) Buildings under purpose group I may consist of more than three storeys if they are occupied by a single household dwelling.
Building Authority may Consent to Greater Sizes	Reg. 3(7)	Where however, the AHJ is satisfied that additional floor area and cubical extent can be allowed for any such building or compartment or part thereof, as aforesaid, the AHJ may consent to such additional floor area and cubical extent provided : (a) Such building or compartment thereof is fitted throughout with an automatic sprinkler system which complies with the relevant requirements of an accepted Code of Practice on Automatic Sprinkler Systems; and (b) There is adequate accessibility of site to Fire Brigade appliances as required in Regulations in Chapter 7; and (c) All other measures have been or will be taken and maintained for lessening so far as is reasonably practicable any danger from fire. Such consent shall continue in force only while such building or compartment thereof as aforesaid is actually used for the purposes in respect of which such consent was given.

Excess Height	Reg.3(8)	In any building which exceeds 18 m in height, any floor which separates one storey from another storey, other than a floor which is- (a) Within a self-contained living accommodation (maisonette); or (b) Within a compartment which is permitted by the provisions of Table 6 or Regulation 3(7) of this Chapter to comprise two or more storeys; shall be construed as a compartment floor.
Other Cases Requiring Compartment Walls and Compartment Floors	Reg. 3(9)	The following walls and floors shall be constructed as compartment walls or compartment floors- (a) Any floor in a building of Purpose Group 2(a): Institutional: Hospitals, Nursing houses, School or other similar building; (b) Any wall or floor separating a flat or maisonette from any other part of the same building; (c) Any wall or floor separating part of a building from any other part of the same building which is used or intended to be used mainly for a purpose falling within a different purpose group in Table 2: and (d) Any floor immediately over a basement storey if such storey- (i) Forms part of a building of purpose group 1 which has three or more storeys or a building or compartment of purpose group 2(b),2(c), 2(d)or 4; and (ii) Has an area exceeding 100 m² (e) In any space below pavement level, no compartment shall comprise more than one storey. Exception : Basement used solely for parking No part of the basement shall be used for the bulk storage of highly flammable liquids or substances of an explosive nature (I) The fire command centre shall be separated from other parts of the same building by compartment walls and floors having fire resistance of at least 2hrs. (g) Kitchen Fire Separation (i) In any eating establishment where kitchen and/or 'open flame' cooking appliances are used, the kitchen shall be separated from other parts of the same building by compartment walls and floor having fire resistance of at least 1 hour (ii) Opening in the compartment wall and floor shall comply with the relevant provisions for protection of openings (Fire Stopping) Doors shall have fire resistance of 30 minutes and shall be fitted with automatic self-closing devices.

Size limitations of buildings and Compartmentation	Reg.3(10)	Compartmentation The spread of fire within a building shall be restricted by subdividing it into compartments separated from one another by walls and/or floors of fire resisting construction.
	Reg. 3(11)	(a) Size Limitations Size limitation for Building and Compartments, except for Purpose Group 1(b) and (c), shall comply with the criterion given in Table 6 For multi-storey Industrial and Storage Buildings (Purpose Group 6 and 7), compartment size shall comply with the criterion given in Table 7 (b) Cubical extent for compartment exceeding 4m in height 1. In computing the cubical extent of compartments in single storey buildings with a pitched roof, the height of 4m shall be used where the actual height exceeds that figure. 2. Where two buildings are connected by external open sided or covered link-bridge, the buildings are considered as separate buildings under following conditions: (i) Within the covered or link —bridge there is no commercial activities or other usage that would pose a fire risk. (ii) The length of the covered way shall not exceed 5m measured from eave to eave. (c) High bay warehouse and Mechanized car parking (i) These buildings shall be treated differently from normal industrial buildings. (ii) The requirement of restricting the compartment to maximum 3 storeys, shall not be applicable. (iii) Automatic sprinkler system and smoke extraction or venting system shall be required for spaces exceeding 15 m in building height. The requirement of restricting the compartment to one storey for buildings above a height 30 m, shall not apply.
	Reg.3(12)	High Hazard Occupancy (a) The compartment shall not exceed one half of the sizes given in Tables 6 and 7, and each compartment shall be limited to one storey only. (b) If habitable floor exceeds 18 m, an Automatic Fire Sprinkler System shall be provided.

	Reg.3(1 3)	Exemption from Size Limitation (a) Buildings used solely for the storage/packaging of non combustible materials. (b) Open sided Car Parking decks having not less than 50% of the sides permanently open to provide cross ventilation.
	Reg. 3(14)	Areas of Special Hazard (a) Boiler Rooms, Transformer Rooms, Generator Rooms, Storage Areas of highly flammable / combustible materials to be separated from compartment walls and floors having fire resistance of not less than 2 hours. If building is provided with an automatic fire suppression system, fire resistance should be a minimum of 1 hour. (b) Rooms having transformer containing flammable liquid and generator rooms shall have easy access for Fire Brigade/ Fire Service Department. (c) Spray Painting Booth Spray painting booth shall be separated from other parts of the same building by compartment walls and floors having fire resistance of not less than 2 hours. Where built-in vapour extraction system is installed, a reduction of the fire resistance requirement may be considered. Where spray paint booth is protected by an automatic sprinkler system, the fire compartment shall have 1 hour fire resistance. (d) Kitchen Separation (i) In an eating establishment where a kitchen is required for the preparation of food and 'open flame cooking' appliances are used, the location shall be separated from other parts of the same building by a fire separation! Compartmentation having fire resistance of not less than 1 hour. (ii) Opening in the compartment wall and floor comply with the relevant provisions for fire protection of openings (iii) Doors shall have fire resistance of 30 minutes with automatic self-closing devices (iv) Where flue or exhaust duct passes through the compartment wall or floor, penetration shall be sealed with fire resistant material having the same fire rating. (v) If automatic fire suppression system is provided to the cooking range, no fire Compartmentation is required. (e) A theatre, cinema or concert hall shall be separated from other parts of the same building, which is of a different purpose group, by compartment walls and floors having a fire resistance of minimum 1 hour. If the building is protected by an automatic sprinkler system, the fire resistance can be half an hour.

		Where openings are provided for access between the theatre, cinema or concert hall and any other part of the same building of a different purpose group, such openings shall be protected by fire doors or provided with a smoke free lobby approach.
	Reg. 3(15)	(a) Other Residential Buildings [Purpose Group 2 (b), 2(c), 2(d)] All floors in buildings of the other residential purpose group should be constructed as compartment floors. (b) Workers Dormitories: Each dormitory bedroom shall be compartmented from adjoining area and other parts of the same building by construction having fire resistance rating of not less than 1 hour, unless otherwise permitted by AHJ.
	Reg. 3(16)	Motor Vehicle Workshop (a) A motor vehicle workshop shall be separated from any other part of the same building by compartment walls and floors having fire resistance of not less than 2 hours (b) If located in a basement, shall be separated by compartment walls and floors having fire resistance of not less than 4 hours.
	Reg. 3(17)	Atrium Spaces The AHJ consent to modify the requirements for compartment size, floor area, cubic extent and compartment height for the design of 'Atrium Spaces' in a building provided following conditions are complied with; (a) The minimum plan area of the atrium void is not less than 90 m² and horizontal dimension between opposite edges of the floor opening is not less than 6 m wide; and (b) Occupancy within the floor space meets low or ordinary hazard content; and (c) The atrium is open and unobstructed; and (d) The building is fitted throughout with an automatic sprinkler system ; and (e) The building is fitted with an engineered smoke control system
	Reg. 3(18)	Residential Institutional Buildings Including Health Care facilities (a) In Residential institutional purpose group buildings, all floors shall be constructed as compartment floors. (b) Compartments should not exceed 2000 m² in multi - storey hospitals and 3000 m² in single storey hospitals.

Exemption from Reg. 3(5)	Reg. 3(19)	Buildings used solely for the sale storage, processing and packaging of goods and substances of a non-combustible nature shall be exempted from the operation of Reg. 3(5) provided; (a) Any portion used otherwise than as described above is separated from the remainder of the building by compartment walls, floors and protected shafts and comply with the limitations of size laid down in Reg. 3(5) (b) The provision of compartment floors are complied with the relevant parts of Reg. 3(8) and Reg. 3(9) (c) Any special risks such as boiler rooms, electrical substations, packing rooms and the like are separated from the remainder of the building by compartment walls and , floors; and (d) The requirements of Regulations in chapter 6 on accessibility of the site to Fire Brigade appliances are complied with. i. Open sided car parking decks having not less than 50% of the sides permanently open and unobstructed, and such openings being evenly distributed along each of the perimeter walls and on every individual floor/deck, so as to provide cross ventilation ii. No part of the floor space shall be more than 12m from the openings on the perimeter walls of the building.
Building of High Fire Risk	Reg. 3(20)	Pursuant to this Regulation the compartments of any building of high fire risk or used for the bulk storage of goods or substances of a combustible nature shall not exceed one half of the sizes given in Table 6 and each compartment shall be limited to one storey only. No storey of a building, the floor level of which is more than 18m above average ground level, shall be used for the storage of combustible goods unless the building is provided with a sprinkler system in accordance with the requirements of Reg. 5(33) and complies with any other requirements which the AHJ may deem fit to impose.
Refuge Floors	Reg. 3(21)	A refuge floor consisting of a refuge area shall be provided for every 10 floors for buildings having the height of occupiable floors exceeding 60 m. (a) The refuge area referred to above shall be of masonry construction having fire resistance rating not less than 2 hours and shall have external walls on at least on two sides to provide adequate openings for ventilation where the total area of opening shall not be less than 25% of the floor area of the holding area. (b) The holding area shall be calculated taking into consideration the total occupant load of 10 floors above and

the area shall be sufficient to accommodate 50% of the above load allowing at least 0.5 m² per person.

- (c) The refuge floor shall be counted commencing from the top most occupied floor. A refuge floor shall be provided for every 10 floors all the way to the ground level.
- (d) The holding area shall be separated from other areas of the refuge floor by compartment wall having fire resistance rating not less than 2 hours. Link of the holding area with other occupied rooms/areas shall be via an external corridor, or a smoke-stop lobby.
- (e) The holding area shall be naturally ventilated with permanent openings on at least 2 sides of external walls. Height of opening shall not be less than 1200mm high and the total area of ventilation openings shall not be less than 25% of the floor area of the holding area.
- (f) All parts of the holding area shall be within 9m of any ventilation opening.
- (g) Ventilation opening shall be located at least 1.5m horizontally and 3m vertically above adjoining unprotected openings. Sprinkler system shall be provided for the refuge floor if there is any non-residential room located on the same floor.
- (h) Escape routes leading to the holding area shall be through smoke stop/fire fighting lobby or external corridor.
- (i) A sign depicting "REFUGE FLOOR" shall be displayed inside the staircase and on wall immediately outside the staircase at the refuge floor. The sign of lettering size not less 50mm shall be displayed at a height of 1500mm above the landing/finished floor level.
- (j) Emergency lighting shall be provided to cover all areas of the holding area. Such lighting shall be connected to secondary power supply, i.e. generator, battery, etc, and shall be able to provide a horizontal luminance at floor level of not less than 5 lux. The delay between the failure of the electrical supply to normal lighting and the energization of the emergency lighting for occupied areas shall not exceed 15 seconds.
- (k) Each area of refuge shall be provided with a two way communication system with the fire command centre.
- (1) The design and layout of exit stair cases shall facilitate discharge of evacuees into the refuge floor before permitted to proceed downwards.

		ELEVATED HELIPORTS
General		This regulation shall apply for Heliports which may be provided in Super High-rise or High-rise buildings.
Heliports Categories	Reg. 3(22)	Fire area requirements for different categories of Heliports classified as per International classifications. H 1-Overall length of Helicopters up to 15m - Practical critical fire area requirement 35 m² H2 -Overall length of Helicopters from 15-24m - Practical critical fire area requirement 78 m² H3 -Overall length of Helicopters from 24-36m - Practical critical fire area requirement 134 m²
Fire requirements for Elevated Hel iports	Reg. 3(23)	(a) Foam hose line using inductor, Foam making branch pipe, and foam concentrate, all stored in a cabinet or; (b) Fixed proportioning system permanently piped to monitors or fixed spray nozzles strategically located around the periphery of the landing pad. (c) In addition to Foam system portable/mobile fire extinguishers shall also be provided. (d) A minimum of two means of access to the landing pad shall be provided for firefighters.
		FIRE RESISTANCE OF ELEMENTS OF STRUCTURE
Minimum Periods of Fire Resistance	Reg. 3(24)	Subject to any express provision to the contrary any element of structure shall be so constructed as to have fire resistance for not less than the relevant period specified in Table 8 having regard to the purpose group of the building of which it forms part and the dimensions specified in that Table, Provided that; (a) Any party wall shall have a fire resistance of not less than one hour. (b) Any compartment wall or compartment floor which separates a part of a building falling within purpose group 1(a) ,l(b) or 1(c) from any other part of the building falling within a purpose group other than purpose group 1(a) ,l(b) or 1(c) shall have a fire resistance of not less than one hour.
Exemption for External Non-load Bearing Wall	Reg.3(25)	Requirement on fire resistance in Reg. 3(24) shall not apply to any part of an external wall which is non-load bearing and in accordance with Regulation 3(33) to 3(41) be an unprotected area.
Exemption for non load bearing walls of standalone car parks	Reg. 3(26)	Standalone Steel structures used exclusively for car parks under following conditions; (i) Where each story is provided with automatic sprinkler protection, and.

		(ii) Building is not more than 30 m habitable height without any basement; and (iii) Steel structure shall meet the specification of BS 5950 Pt.8.
Exemption for Single Storey Building	Reg. 3(27)	In the case of a single-storey building or a building consisting of a ground storey and one or more basement storeys, Requirement on fire resistance in Reg.3(24) shall not apply to any element of structure which forms part of the ground storey and consists of; (i) A structural frame or a beam or column: Provided that any beam or column (whether or not it forms part of a structural frame) which is within or forms part of a wall, and any column which gives support to a wall or gallery, shall have fire resistance of not less than the minimum period specified. (ii) An internal load bearing wall or a load bearing part of a wall, unless that wall or part is, or forms part of, a compartment wall or a separating wall, or forms part of, the structure enclosing a protected shaft or supports a gallery. or; (iii) Part of an external wall which does not support a gallery and which may, in accordance with Regulation 3(33) to 3(41) be an unprotected area.
Interpretation and Application of this Regulation	Reg. 3(28)	(a) In this Regulation and in Table 8 thereto (subject to the provisions of sub-paragraph (b) of this paragraph and any other express provision to the contrary) any reference to a building of which an element of structure forms a part means the building or (if a building is divided into compartments) any compartment of the building of which the element forms a part. (b) In this regulation and in Table 8 thereto, any reference to height means the height of a building, not of any compartment in the building. If any part of the building is completely separated throughout its height, both above and below ground from all other parts by a compartment wall or compartment walls in the same continuous vertical plane, any reference to height in relation to that part means the height solely of that part.
Fire Resistance for Elements of Structure that Carries another Elements	Reg. 3(29)	Any element of structure shall have fire resistance of not less than the minimum period required by this regulation for any element which it carries.
Fire Resistance of Compartment Walls of Flats	Reg.3 (30)	Any compartment wall separating a flat or maisonette from any other part of the same building shall not be required to have fire resistance exceeding one hour unless- (a) The wall is a load bearing wall or a wall forming part of a protected shaft; or

		(b) The part of the building from which the wall separates the flat of maisonette is of a different purpose group and the minimum period of fire resistance required by the provisions of this regulation for any element of structure in that part is one and a half hours or more.
		TESTS OF FIRE RESISTANCE
Meaning of "Fire Resistance" Requirements	Reg. 3(31)	Any requirement in this Chapter that an element of structure door or other part of a building shall have fire resistance of a specified period shall be construed as conforming to the test procedures as specified in BS 476:Pt.20 to 23.
"Deemed to Satisfy" Provisions	Reg. 3(32)	Provided that an element of structure, door or other part of a building shall be deemed to have the requisite fire resistance if it is constructed to the same specification as that of a specimen exposed to test by fire in accordance with the method and procedure under BS 476: Pt.20-30 and satisfied the requirements of that tests for the three performance criteria for stability, integrity and installation for not less than the specific period.
		EXTERNAL WALLS
Unprotected Areas in any Side of a Building	Reg. 3(33)	(a) Except where otherwise provided, any side of a building shall comply with any relevant requirements relating to the permitted limits of unprotected areas specified in Table 10 (b) The extent of unprotected openings in an external wall of a building in relation to its distance from the lot boundary as per table 10 may be doubled when the building is protected by an automatic sprinkler system in accordance with the provision of an accepted Code of Practice. (c) Notwithstanding the requirements of sub-paragraph (a) of this regulation, the unprotected areas in a side of a building or part of a building used for car parking conforming with the provisions of Regulation 6(49) shall be exempted from compliance with the relevant requirements relating to the permitted limits of unprotected areas.
Requirements of External Walls	Reg.3(34)	Any external wall, which constitutes or is situated within a distance of 1m from any point on the relevant boundary, or is a wall of a building which exceeds 18m in height shall- (i) Be constructed wholly of non-combustible materials apart from any external cladding which complies with paragraph (3) of this regulation or any in internal lining which complies with Regulation 3(86); and (ii) Be so constructed as to attain any fire resistance required by this Chapter without assistance from any combustible material permitted by this sub-paragraph:

Exemptions	Reg. 3(35)	Provided that the requirements of this sub-paragraph shall not be applicable to- (i) An external wall of a building which is not divided into compartments and is within the limits of size indicated in Table 8 if, in either case, that building does not exceed 18m in height; (ii) An external wall of a building or part of a building, of purpose group 1(a) which consists of flats or maisonettes if that building has not more than three storeys or that part is separated as described in Regulation 3(28)(b) and has not more than three storeys; (iii) An external wall of a part of a building if that wall is situated 1 m or more from the relevant boundary and that part is separated as described in Regulation 3(28)(b) and does not exceed 18m in height.
	Reg. 3(36)	Any beam or column forming part of, and any structure carrying, an external wall which is required to be constructed of non-combustible materials shall comply with the provisions of Regulation 3(35) above, as to non-combustibility.
Cladding on External Walls	Reg.3(37)	Any cladding on any external wall, shall have a surface complying with the requirements for Class 0 specified in Regulation 3(83) to 3(89); and
	Reg.3(38)	If the building is provided with an external cladding or a cladding is introduced as an external feature with an air gap between the structures, a cavity barrier shall be introduced at every 3rd floor or at the compartment floor, to prevent the vertical spread of fire and smoke. (exemption : facades with no openings or buildings with a height not exceeding 18 m)
Reference to Roofs	Reg. 3(39)	For the purpose of this regulation- Any part of a roof shall be deemed to be part of an external wall or side of a building if it is pitched at an angle of 70° or more to the horizontal and adjoins a space within the building to which persons have access not limited to the purposes of maintenance or repair.
Roof Terrace	Reg. 3(40)	Roof terrace shall not be roofed over. If it is partially or fully roofed over, it shall be considered as a habitable floor
Buildings on land in common occupation	Reg. 3(41)	If two or more detached buildings are erected on land in common occupation, any external wall of any building so erected which faces an external wall of such other building, the relevant boundary shall be a notional boundary passing between those buildings and such boundary must be capable of being situated in such a position as to enable the external walls of those buildings to comply with the requirements of Reg.3(33)

		PARTY WALLS (SEPARATING WALL)
Requirements of Party Walls-	Reg. 3(42)	Any party wall shall conform to the relevant provisions of the Building Regulations with regards to its thickness and non-combustibility of materials and shall have fire resistance for not less than the relevant period specified in Regulation 3(24) to 3(30) and subject to the exceptions specified in paragraph (2) shall be imperforated and shall form a complete vertical separation between any buildings separated (including any roof space therein).
Party Wall - Roof Junction	Reg. 3(43)	A party wall shall in accordance with the relevant provisions of the Building Regulations be either carried up to form a close joint with the underside of a pitched roof of non-combustible covering or carried up to 450mm above the level of such roof covering, and the junctions between such party wall and such roof shall be properly fire stopped, so as not to render ineffective the resistance of such party wall to the effects of fire and the spread of fire.
Party Wall - External Wall Junction	Reg. 3(44)	If any external wall is carried across the end of a party wall, such external wall and party wall shall be bonded together or the junction of such walls shall be fire-stopped.
Prohibition of Combustible Materials in Party Walls	Reg. 3(45)	In addition to the relevant provisions of the Building Regulations, no combustible material shall be carried through into or across the ends of or over the top of any party wall of such a type or in such a way as to render ineffective the resistance of such a party wall to the effects of fire and the spread of fire: Provided that- If a building has a non-combustible roof covering, and is of Purpose Group 1 to 6 the continuation over the top of the party wall shall be allowed with any roof tiling, slating or sheeting battens, if the battens are solidly bedded on mortar or other not less suitable material where they rest on the party wall and the space between them is filled with mortar or other not less suitable material up to the underside of the roof covering.
Non-Combustibility of Party Walls	Reg. 3(46)	In addition any party wall shall be constructed wholly of non-combustible materials apart from any surface finish which complies with Regulation 3(86) and the required fire resistance shall be attained independently of any such combustible surface finish: Provided that the requirements of this paragraph shall not apply to: A wall separating buildings, which are defined as temporary buildings and which are not divided into compartments and are single storey building within the limits of size indicated in Tables 6 and 7

		COMPARTMENT WALLS AND COMPARTMENT FLOORS
Compartment Walls and Floors to be Imperforated	Reg. 3(47)	Any compartment wall or compartment floor shall be imperforated with the exception of any one or more of the following: (a) An opening fitted with a door which has fire resistance for the following minimum period and which complies with the provisions of Regulation 3(68) to 3(76). (i) In the case of a wall separating a flat or maisonette from any space in common use giving access to that flat or maisonette, half an hour; or (ii) In any other case, the period required by the provisions of Regulation 3(24) to 3(30) for the wall or floor; (b) An opening for a protected shaft fitted with a door as in clause (a); (c) An opening for a ventilation duct (other than a duct in, or consisting of, a protected shaft) if any space surrounding the duct is fire-stopped and the duct is fitted with an automatic fire shutter where it passes through the wall or floor; (d) An opening for a pipe, which does not exceed- (i) 25 mm in diameter, if the pipe is of combustible material; or (ii) 150 mm in diameter, if the pipe is of non-combustible material; (e) An opening for a refuse chute, fitted with a self closing smoke stop door. (f) Ventilation openings to an external exit passageway as permitted under Regulation 2(41) and (g) Ventilation openings to an external corridor as permitted under Regulation 2(18)
Compartment Walls and Floors Forming Junction with Other Structures	Reg. 3(48)	Where a compartment wall or compartment floor forms a junction with any structure comprising any other compartment wall, or any external wall, party wall or structure enclosing a protected shaft, such structures shall be bonded together at the junction, or the junction shall be fire-stopped.
Compartment Walls-Roof Junctions	Reg. 3(49)	Where any compartment wall forms a junction with a roof, such wall shall be carried up to form a close joint with the underside of the roof or carried up to 450mm above the level of the roof covering (as the case may be) and the junctions between such party walls and such roof shall be properly fire stopped in accordance with the relevant provisions of Regulations 3(43).

Prohibition of Combustible Material in Compartment Walls	Reg. 3(50)	No combustible material shall be built into or carried through, into or across the ends of any compartment wall or compartment floor or over the top of any compartment wall in such a manner as to render ineffective the resistance of such wall or floor to the effects of fire and the spread of fire:
Use of Fire Shutters and Curtains	Reg. 3(51)	Automatic fire shutters and curtains with the identical fire rating as the compartment wall shall be permitted except for fire Compartmentation of fire command centre and means of escape routes.
Ducts Passing through Compartment Floors and Walls	Reg. 3(52)	Where any duct or pipe passes through a compartment floor or compartment wall such ducts or pipes shall be encased in an enclosure having half the fire rating of the compartment floor/wall.
Non-Combustibility of Compartment Walls and Floors	Reg. 3(53)	Any compartment wall or compartment floor which is required by Regulation 3(24) to 3(30) to have fire resistance of one hour or more (except where that requirement arises solely by virtue of Regulation 3(24)(b)) shall be constructed wholly of non-combustible materials apart from- (a) Any floor finish; or (b) Any surface finish to a wall or ceiling which complies with the requirements of Regulation 3(83) to 3(89) and the required fire resistance of the wall or floor shall be obtained without assistance from any combustible material permitted by this sub-paragraph.
Exemption for Regulation 3(54)	Reg. 3(54)	The requirements of Regulation 3(53) shall not apply to; (a) The following walls and floors in a building or part of purpose group 2(b), 2(c) and 2(d) which consists of flats or maisonettes: (i) If that building has three storeys or that part is separated as described in Regulation 3(28)(b) and has three storeys, any wall or floor other than a wall within a basement storey or a floor immediately over a basement storey; (ii) If that building has four storeys or that part is separated as described in Regulation 3(28)(b) and has four storeys, any floor other than a floor immediately over a basement storey; and (b) Any existing floor in a building or part of purpose groups 3, 4, 5, 6 which is altered or extended if, after alteration or extension, that building does not exceed 18m in height or that part is separated as described in Regulation 3(28)(b) and does not exceed 18 m in height.
Beams and Columns in Compartment Walls and Floors	Reg. 3(55)	Any beam or column forming part of, and any structure Carrying any compartment wall or compartment floor which is required to be constructed of non-combustible materials, shall itself comply with the provisions of Regulation 3(53) as to non-combustibility.

		PROTECTED SHAFTS
"Protecting Structure"	Reg. 3(56)	In this regulation, "protecting structure" means any wall or floor or other structure which encloses a protected shaft other than- (a) A wall which also forms part of an external wall, party wall or compartment wall; or (b) A floor which is also a compartment floor or a floor laid directly on the ground: and (c) A roof.
Use of Protected Shaft	Reg. 3(57)	No protected shaft shall be constructed for use for any purposes additional to those specified in Chapter 7 other than the accommodation of any pipe or duct. Any pipe or duct located within a protected shaft shall have an enclosure having the same fire rating of the shaft.
Enclosure	Reg. 3(58)	Subject to the provisions of this regulation, any protected shaft shall be completely enclosed by walls or floors having the necessary fire resistance rating under Reg. 3(24) to 3(30)
Non-Combustibility of Protecting Structures	Reg. 3(59)	(a) Every protecting structure shall be constructed wholly of non-combustible materials except that floor, wall and ceiling finishes which do not contribute to the fire resistance of such protecting structure. Such surface finishes shall comply with the requirements of Regulation 3(83) to 3(89). (b) Any beam or column forming part of, and any structure carrying a protecting structure which is required to be constructed of non-combustible materials shall itself comply with the provisions of sub-paragraph (a) as to non-combustibility.
Openings in Wall or Floor of	Reg. 3(60)	(a) Any wall or other structure enclosing a protected shaft but not being a protecting structure may contain such openings as shall be in accordance with other provisions of this code. (b) There shall be no opening in any protecting structure other than any one or more of the following- (i) An opening for a pipe; (ii) An opening fitted with a door which has fire resistance complying with the provisions of Regulation 3(62) and complies with the provisions of Regulation 3(68) to 3(76) (iii) (If the protected shaft contains a lift) an opening which complies with the provisions of Regulation 3(63) and (iv) (If the protected shaft serves as, or contains a ventilating duct) an inlet to or outlet from that duct or opening for that duct.

Fire Stopping	Reg. 3(61)	Any opening for a pipe shall be effectively fire-stopped.
Doors in Protected Structure	Reg. 3(62)	Any door fitted to an opening in protected structure shall have fire resistance for not less than half the period required by other provisions of the regulation for the protecting structure surrounding the opening
Protected Shaft for Lift	Reg. 3(63)	Any protected shaft containing a lift or lifts- (a) Shall not contain any pipe conveying gas or oil or any ventilation duct (b) May have an opening in its protecting structure for the passage of the cables operating the lift into the room containing the lift motor: (c) If it serves any basement, it shall be protected by a smoke stop lobby with walls having 1 h fire resistance. The protected lobby shall be mechanically ventilated.
Protected Shaft for Ventilation Duct	Reg. 3(64)	(a) If a protected shaft serves as or contains a ventilating duct. (i) The duct shall be fitted internally with automatic fire shutters so constructed, at such intervals and in such positions as may be necessary to reduce, so far as practicable, the risk of fire spreading from a compartment to any other compartment, or such other provision shall be made as will reduce such risk so far as practicable; and (ii) The duct shall not be constructed, or lined with, any material having a fire resistance less than 1 h. (b) In addition, in the case of a protected shaft containing a ventilating duct, the shaft shall be so constructed with additional barriers to fire between the duct and the shaft as may be necessary to reduce so far as practicable the risk of fire spreading from a compartment to any other compartment.
Ventilation of Protected shaft	Reg. 3(65)	(a) A protected shaft used for the passage of people, shall be ventilated to comply with the relevant provisions of this regulation. Where the staircase is pressurized the differential pressure between the staircase and any adjoining area to which openings are provided from the staircase shall be a minimum of 50 Pa. (b) protected shaft containing a pipe conveying combustible gas shall be adequately ventilated directly to the outside, both at the top and bottom

Protected Shaft of Staircase (Protected shaft containing exit staircase)	Reg. 3(66)	(a) A protected shaft which contains an exit staircase shall not contain any services except for: (i) Cut off sprinkler and pipe for that staircase and (ii) Rain water down pipes serving the roof directly above the exit staircase (iii) Rising mains (b) The protecting structure shall be constructed of masonry or dry wall. If dry wall construction is used, it shall be of non-combustible material. The wall shall have the fire resistance for not less than the relevant period specified in Table-8, having regard to the purpose group of the building.
Protected shaft containing other services	Reg. 3(67)	(a) Kitchen exhaust ducts and mechanical ventilation ducts passing through floor slabs, masonry or dry wall shall be completely compartmented from the rest of the shaft space containing other ducts or any other service installations with a fire resistance rating not less than the rating specified for the building. (b) Protected shaft used for the enclosure of electrical power services shall be interrupted at every floor level with fire barriers of at least 30 minutes (c) Protected shaft used for the enclosure of telecommunication cables shall be interrupted with at least 30 minute fire barriers at vertical intervals of not exceeding 15m.
		FIRE RESISTING DOORS
Application of Regulations	Reg. 3(68)	This regulation shall apply to any door (including the door frame) which is required by the provisions of this Section to have fire resistance.
	Reg. 3(69)	In this regulation- Automatic Self-Closing Device does not include rising butt hinges except in relation to a door to which Regulation 3(74) applies; and Fire doors fitted with automatic devices to hold the door in open position during normal operations shall have devices to close the doors automatically upon the occurrence of each or any one of the following; (a) Detection of smoke by an automatic smoke detector . (b) Manual operation of a switch fitted in a suitable position providing overriding facility. (c) Failure of electricity supply to the device, apparatus or switch; (d) If an evacuation alarm is activated for the building through the fire control panel.

Provisions of Closing Devices and Non-Combustible Hinge	Reg. 3(70)	(a) All fire doors shall be fitted with a positively self-closing devices which are capable of closing the door from any angle against any latch fitted to the door. (b) No means of holding any such door open shall be provided other than a holding device that can be released by the activation of a smoke detector in the vicinity. (c) No part of a hinge on which any such door is hung shall be made either of combustible material or of non-combustible material having a melting point less than 800 °C.
Door to be Manually Openable	Reg. 3(71)	Any door fitted in an opening which is provided as a means of escape in the event of fire or might be so used shall be so constructed and installed that it can readily be opened manually.
Direction of Opening of Door	Reg. 3(72)	Any door to which reference is made in Reg. 3(47) (a)(i) or 3(78)(b) shall be either a single leaf door swinging in one direction only or a double leaf door, each leaf of which swings in the opposite direction from the other leaf, where both open in the direction of egress.
	Reg. 3(73)	Any door which is fitted in protecting structure {as defined in Reg. 3(56) and is not required by the provisions of Reg. 3(62) to have fire resistance of more than half an hour} may consist of any single or double leaf door (the leaf or each leaf of which swings in one or both directions), other than a double leaf door both leaves of which swing in one and the same direction and have rebated meeting stiles, if- (a) The door opens into a hall, lobby or corridor enclosed by walls or partitions having fire resistance of not less than half an hour; and (b) The clearance between the leaf or leaves of any such door and its frame and (if the door has two leaves) between the leaves is as small as is reasonably practicable.
Two Separate Doors to One Opening	Reg. 3(74)	For the purpose of this code, if two separate doors (whether single or double leaf doors) are installed in an opening, it shall be sufficient if the required fire resistance is achieved by the two doors together or by either of them separately.

"Deemed to Satisfy" Provisions	Reg. 3(75)	Where any door is required under the provisions of this Publication to be fire resisting, it shall be deemed to have- (a) The requisite fire resistance if a similar door made to the same specifications as that door is proved to have the requisite fire resistance under the conditions of test prescribed in the foregoing paragraphs of this Publication; or (b) The requisite fire resistance for a minimum period of- (i) 1/2 hour if it is constructed in accordance with one of the relevant specifications for Class A in Table 11 and is of not less standard than that of such class; or (ii) 1 hour if it is constructed in accordance with one or the relevant specifications for Class B in Table 11
Doors in Lift Shafts	Reg. 3(76)	Notwithstanding Regulation 3(70)(a) a door which is not fitted with a self-closing device may be installed in an opening in the structure which encloses a protected shaft containing exclusively a lift or lifts if either- (a) The door has fire resistance for a period of not less than half an hour and there is also installed so as to close the same opening another door which is fitted with an automatic self-closing device, is held open by an automatic closing device activated by smoke detector or fire alarm and has fire resistance for a period not less than that prescribed by the relevant provisions of Regulation 3(24) to 3(30) for the structure surrounding the opening; or (b) Unless the opening is in a compartment wall and is one of the two openings provided at the same level to allow access to a lift from different sides, then, the door shall have a fire resistance for a period not less than that prescribed by relevant provisions of Reg. 3 (24) to 3 (30) for the structure surrounding the opening.
STAIRCASES		
Non-Combustibility of Structure	Reg. 3(77)	Every staircase (including any landing thereof) which forms part of a building shall, whether the staircase is internal or external, be constructed of non-combustible materials except- (a) An internal staircase which is situated- (i) Within a maisonette; or (ii) Within a building of purpose group 1 or 2(b) which has not more than 3 storeys; or (iii) Within the ground storey or an upper storey of part of a building, which part consists of flats or maisonettes if that part is separated as described in Reg. 3 (28) (b) and has not more than three storeys; or

		(b) An external staircase of a building of purpose group 1 and such staircase is situated between the ground and a floor or flat roof the level of which, at the head of the staircase, is not more than 6 m above the finished surface of the ground adjoining the foot of the staircase.
Special Requirements for Buildings for Purpose Group I Exceeding 3 Storeys	Reg. 3(78)	In any building of purpose group 1 which exceeds 9 meters in height any internal staircase (including any hall or landing associated therewith and any part of a floor which affords passage between flight of the staircase) shall be separated from all other parts of the building by structure complying with the following requirements- (a) The structure shall have fire resistance for not less than the minimum period required by Reg. 3(24) to 3(30) for elements of structure forming part of the storey in which it is situated; and (b) Any opening in the structure which gives access to a habitable room, garage, or kitchen shall be fitted with a door which has fire resistance of not less than half an hour and complies with the requirements of Reg. 3(68) to 3(7).
		FIRE STOPPING
Fire Stop to Prevent Passage of Flame	Reg. 3(79)	Any fire stop required by the provisions of this Chapter shall be so formed and positioned as to prevent or retard the passage of flame.
Fire Stop for Pipe Duct or Cavity	Reg. 3(80)	Fire stops shall be provided for; (a) All service ducts provided to locate pipes ducts or cables shall be enclosed with fire resisting materials having the relevant fire resistance. Such ducts must be provided with fire stops having a fire rating equal to the enclosure , at every floor. (b) If provided around a pipe or duct or in a cavity, be made of non-combustible material or if it is in a floor or wall constructed of combustible material of hardwood timber not less than 38 mm thick; and (c) If provided around a pipe or duct, be so constructed as not to restrict essential thermal movement.
Fire Stop at Junction of Elements of Structure	Reg. 3(81)	Fire stop formed as a seal at the junction of two or more elements of structure shall be made of non-combustible material if all such elements are required by this Chapter to be non-combustible.
Fire Stopping of Cavity in an Element of Structure	Reg. 3(82)	A cavity which has a surface of combustible material exposed within the cavity which is of a class lower than Class 0 in Regulation 3(83) to 3(89)., shall be fire stopped; (a) At any junction with another element of structure or with a ceiling under a roof and (b) In such a position that there is no continuous cavity

		RESTRICTION OF SPREAD OF FLAME OVER SURFACES OF WALLS AND CEILINGS
Definitions and Interpretation	Reg. 3(83)	For the purposes of this code and Table 12.
		(a) "Ceiling" includes any soffit and any skylight, or other part of a building which encloses and is exposed overhead within a room, circulation space or protected shaft; "Circulation space" means any space which is solely or predominantly used as a means of access between a room and protected shaft or between either a room or a protected shaft and an exit from the building or compartment; "Small room" means a room which is totally enclosed and has a floor area not exceeding that specified in column (2) of the Table 12, according to the purpose group of the building or compartment; and "Trim" means any architrave, cover mould, picture rail, skirting or similar narrow member; (b) Any reference to the surface of a wall shall be construed as a reference to that surface including the surface of any window, window frame, glazing and frame in which the glazing is fitted but excluding any unglazed portion of a door, door frame, mantel shelf, fitted furniture or trim; (c) Any reference to the surface of a ceiling shall be construed as a reference to that surface excluding the surface of the frame of any roof light or skylight; (d) Any part of a ceiling which slopes at an angle of 70° or more to the horizontal and is not part of a roof light shall be deemed to be a wall; (e) Any reference to a surface being of Class 0 shall be construed as a requirement that: (i) The material of which the wall or ceiling is constructed shall be non-combustible throughout; or (ii) The surface material (or, if it is bonded throughout to a substrate, the surface material in conjunction with the substrate) shall have a surface of Class I and, if tested in accordance with BS 476; Part 6: latest revision, shall have an index of performance not exceeding 12 and a sub-index not exceeding 6.
Requirements for a Class Other than Class 0.	Reg. 3(84)	Any reference to a surface being of a class other than Class 0 shall be construed as a requirement that the material of which the wall or ceiling is constructed shall comply with the relevant test criteria as to surface spread of flame specified in relation to that class in clause 2.8 of BS 476 Part 7: latest revision: and In relation to a requirement of a surface shall be of a class not lower than a specified class, Class 0 shall be regarded as the highest class followed in descending order by Class 1, Class 2, Class 3 and Class 4, as set out hereunder:

Classification	Reg. 3(85)	Class 0 - Surface of no flame spread; Those surfaces that conform to the requirements (i) and (ii) of Reg. 3(84) sub-paragraph (e). Class 1 - Surfaces of very low flame spread; Those surfaces on which not more than 150mm mean spread of flame occurs under the relevant test conditions. Class 2 - Surfaces of low flame spread; Those surfaces on which during the first 1 1/2 minutes of test, the mean spread of flame is not more than 375 mm and the final spread does not exceed 450 mm under the relevant test conditions. Class 3 - Surfaces of medium flame spread; Those surfaces on which during the first 1 1/2 minutes of test the mean spread of flame is not more than 375 mm and during the first 10 minutes of test is not more than 825 mm under the relevant test conditions. Class 4 - Surfaces of rapid flame spread; Those surfaces on which during the first 1 1/2 minutes of test the mean spread of flame is more than 375 mm and during the first 10 minutes of test is more than 825 mm under the relevant test conditions.
Class of Flame Spread to be not Lower than Specified in Table	Reg. 3(86)	The surface of a wall or ceiling in a room, circulation space or protected shaft shall be of a class not lower than specified in the Table 12: Provided that- (a) A wall may have a surface of any class not lower than Class 3 to the extent permitted by Regulation 3(87) and 3(88): (b) A ceiling may have a surface of any Class not lower than Class 3 to the extent permitted by Regulation 3(88) (c) and (d) (c) Where an automatic sprinkler system is provided and installed in accordance with an accepted Code of Practice, in such rooms, circulation spaces or protected shafts, the surfaces of the walls and ceilings therein may be of a class that is two classes lower than specified as relevant in Table 12; and (d) If timber is used as the surface material for the walls along the side gangways of the auditorium or of a cinema, the requirements of this regulation pertaining to the requisite class of flame spread may be relaxed only in respect of those parts of such wall surfaces, which are above a height of 2.1 m (measured from floor level) provided the aggregate area of such wall surfaces does not exceed 50 percent of the whole surface area of the side walls of the auditorium.

Where Class of Flame Spread may be of any Class not lower than Class 3	Reg. 3(87)	Any part of the surface of a wall in a room may be of any class not lower than Class 3 if the area of that part (or, if there are two or more such parts, the total area of those parts) does not exceed the lesser of the following- (a) Half the floor area of the room; or (b) In the case of a building or compartment of Purpose Group 1 and 2, 20 m² or (in any other case) 60 m².
	Reg. 3(88)	Any part of the surface of a ceiling may be of any class not lower than Class 3 if that part of the surface is the face of a layer of material the other face of which is exposed to the external air and- (a) (i) The ceiling is that of a room in a building or compartment of Purpose Group 1, 2, 3, 4, or 5 or that of a circulation space in a building or compartment of any purpose group; and (ii) The area of that part does not exceed 2.5 m² and (iii) The distance between that part and any other such part is not less than 3.5 m; or (b) (i) The ceiling is that of a room in a building or compartment of Purpose Group 6 or 7(a); and (ii) The area of that part does not exceed 5 m² and (iii) The distance between that part and any other such part is not less than 1.8 m; and (iv) That part and all other such parts are evenly distributed over the whole area of the ceiling and together have an area which does not exceed 15% of the floor area of the room; or (c) The ceiling is that of a balcony, veranda, open carport, covered way or loading bay which (irrespective of its floor area) has at least one of its longer sides wholly and permanently open; or (d) The ceiling is that of a garage, or outbuilding which (irrespective of whether it forms part of a building or is a building which is attached to another building or wholly detached) has a floor area not exceeding 40 m². (e) The internal non-load bearing walls, or partitions of a building of Purpose Group 3, 4 or 5 occupancy, which are exempted from being constructed of non-combustible materials under Reg. 3(94) and 3(95) may be of any class not lower than Class 3.
"Deemed to Satisfy" Requirements	Reg. 3(89)	Any type of material shall be deemed to be of a specific class relating to flame spread if the same type of material (manufactured to the same specifications) is proved to satisfy the relevant tests criteria as to flame spread specified in relation to such class in Clause 2.8 of BS 476, latest revision.

		ROOFS
Roof Coverings and Construction to be Non-combustible	Reg. 3(90)	Surface of materials for roof covering and roof construction shall have a surface spread of flame rating not lower than class I except buildings protected throughout with automatic sprinkler system.
	Reg. 3(91)	When a pitched roof is deemed to be part of an external wall under Reg. 3(39) or when otherwise required by the Urban Development Authority, such roof shall have a fire resistance of not less than that of an external wall in the building or compartment of which such roof forms a part, as laid down in Reg. 3(24) to 3(30).
Special Requirement for Other Types of Roofs	Reg. 3(92)	When approving the use of any other type of roof construction or roof covering materials not less than class I, the AHJ may require a ceiling of a class higher than that specified in Table 12 relating to flame spread, to be provided under a roof of such construction or roof covering materials, in addition to imposing other fire preventive requirements that he may deem fit to impose.
Roof terrace	Reg. 3(93)	Roof terrace shall not be roofed over. If it is either partially or fully roofed over, it shall be considered as a habitable floor.
Exemption from the Requirement for Non-combustibility of Partition in Office Buildings	Reg. 3(94)	The provisions requiring all walls of a building to be constructed of non - combustible materials may be waived by the AHJ in respect of the internal non — load bearing walls or partitions of a building or part of a building of Purpose Group 3 occupancy (offices), provided they do not form any part of the means of escape of the building and- (a) The said building or part of the building is fully sprinklered or (b) The building does not exceed 18 meters in height and such walls or partitions are within a compartment which has a floor area of not more than 250 m² and is enclosed by fire walls and doors having a fire resistance of at least one hour.
Exemption from the Requirements for Non-combustibility of Partitions in Shops and Places of Public Resort	Reg. 3(95)	The provisions requiring all walls of a building to be constructed of non- combustible materials may be waived by the AHJ in respect of the internal non - load bearing walls or partitions of a building or part of a building of purpose group 4 (Shops) or (Place of public resort) provided: (a) Such building or part of the building is fitted throughout with a sprinkler system; (b) Such walls and partitions do not form any part of the means of escape of the building; and (c) All other measures have been or will be taken and maintained for lessening so far as is reasonably practicable any danger from fire as directed by the AHJ

		FIRE LIFTS AND FIRE FIGHTING SHAFTS
General	Reg. 3(96)	In any building or part thereof, in which the floor level of any storey exceeds 30m in height, there shall be provided at least one fireman's lift, which shall be contained within a separate protected shaft or a common protected shaft containing other lifts subject to such other lifts being served at each storey by the protected ventilated lobby, which is required by the provisions of Reg.3(97) of this regulation. In the event of a fire all lifts shall be automatically brought down to ground level and remain there with doors open.
Siting	Reg.3(97)	A fireman's lift shall be adjacent to a staircase enclosure and be approached by a protected ventilated lobby at each storey, which if not naturally ventilated, shall be mechanically ventilated in accordance with the requirements of the AHJ.
Firemen's lift requirements	Reg. 3(98)	(a) The Fireman's lift shall be provided with the primary power supply connected to a sub main circuit exclusive to the lift and a secondary power supply from a different source. The distribution network for both supplies shall be adequately protected from fire and water damage. The supplies shall be separated from each other such that a failure in cables or equipments in one system will not affect the other. The cables supplying power to the fireman's lift shall be either a fire resisting cable or in an enclosure, both of which should have a fire rating equal to or better than the fire rating of the fire enclosure. (b) Fireman's lift car shall be not less than 1100 mm wide by 2100mm deep with a rated load of 1000 Kg. minimum entrance with 800 mm. (c) Fireman's switch shall be located within 2m horizontally from lift, and shall be located between 1.8 and 2.0m above the finished floor level. (d) Lift shall be capable of reaching the uppermost habitable floor from fire service access level within 60 seconds. (e) Electrical equipment and fittings within the lift well and lift car shall be IP 65 rated.
Fire fighting Shafts	Reg.3(99)	Fire fighting shafts shall be provided in High Rise, Super High Rise buildings and buildings with basements below 09 meters. Fire fighting shafts shall be located such that they allow access to every part of every storey served, and shall, wherever possible, be sited against an exterior wall. Access to the fire fighting shaft shall be provided at all levels served

		The "PURPOSE" of a Fire Fighting Shaft is to provide a quick, safe, and protected access route for Fire Fighters when fighting fires in tall buildings. It may also serve as a means of escape for the occupants of the building in the event of a fire. "Fire Fighting Shaft" shall be installed in all high-rise buildings and shall consist of the following: (a) Components of a Fire Fighting Shaft - The Fire Fighting Shaft shall consist of: (i) A protected enclosure containing a fire fighting stairway, and (ii) A fire fighting lobby, and (iii) A fire fighting lift. (b) Protected Enclosure - The protected enclosure shall comply with the requirements stipulated in Fire Reg. 2(3)(k) and Chapter 5 of these regulations. (c) Fire Fighting Lobby - The fire fighting lobby shall comply with the requirements stipulated in Fire Regulation 2(19) of these regulations. The fire fighting lobby shall be provided with a Fire Main as stipulated in Fire Reg. 5(8) to 5(14) of these regulations. (d) Fire Fighting Lift - The fire fighting lift shall comply with the stipulations of Fire Reg. 3(96) and 3(97) of these regulations.
		EVACUATION LIFTS
General	Reg. 3(100)	An evacuation lift shall be provided for buildings categorized as Super High Rise buildings. The evacuation lift shall be different from the Fire fighter's lift and shall comply with the following condition. (a) Protected lobby and lift well shall be designed to restrict the ingress of smoke. Approach shall be through a smoke free passage. (b) The floor of the fire fighting lobby shall be so designed to prevent flow of water to the emergency exit stairway and the lift shaft (c) If openings are available in the lift car on more than one side approaches from every sides shall be through a protected smoke free approach.
Power supply to fireman's lift and evacuation lift	Reg. 3(101)	(a) Power supply shall be available directly from the main breaker of the generator via a dedicated Automatic Transfer Switch (ATS) panel with manual overriding facility located within the fire enclosure. (b) Main and auxiliary power supply cables for these systems shall be located in a fire protected shaft. Any part of the cable located outside a fire protected shaft shall be with enhanced fire resistance and protection against mechanical damage.

Evacuation Lift Requirements.	Reg. 3(102)	(a) Serve every floor of the building. (b) Evacuation lift car shall be large enough to accommodate a stretcher or a bed. The load shall be not less than 1000 kg and the dimensions shall be minimum 1100 mm wide by 2100 mm deep. (c) Lift shall be capable of reaching the uppermost habitable floor from fire service access level within 60 seconds. (d) Electrical equipment and fittings within the lift well and lift car shall be IP 65 rated. (e) The elevator lobby shall be adequate to accommodate not less than 50% of the occupant load of the area served by the lobby and also include one wheel chair space of 760mm x 1220mm for each 50 persons or part thereof. (f) Two-way communication systems shall be provided between elevator lobbies and the fire command centre and between elevator cars and the fire command centre (g) Wiring used for the two way communication systems shall be fire rated for not less than one hour and shall be provided with mechanical protection.
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CHAPTER 4**FIRE DETECTION AND ALARM SYSTEMS**

Purpose of this Chapter	The purpose of this Chapter is to provide minimum requirements of Fire Detection Alarm and Voice Evacuation systems that shall be installed in building to enhance the life safety of the occupants and to mitigate damage to property. The requirements describe below are stated as the minimum requirement to be complied with and any improvements based on any international standard is accepted provided that the designer/Installer could justify its performance is superior to what has been stated in this regulations, to AHJ
Definitions/Abbreviations	
Fire Alarm Control Panel (FACP)	A component of the fire alarm system, provided with primary and secondary power source, which receive signals from initiating devices or other fire alarm control units, and processes these signals to determine part or all of the required fire alarm system output functions.
Addressable System	A system, in which input and output devices have a unique address that can be read, recognized and controlled by the control panel.
Alarm Signal	An audible and visual signal activated by the alarm system to warn of emergency conditions that require immediate action by all occupants of the affected area.
Alarm Zone	Geographical sub-division of the protected premises, in which the fire alarm warning or signal can be given separately, and independently, of a fire alarm warning or signal in any other alarm zone
Combination /Multi-sensor Detector	A combination / multi-sensor that either responds to more than one of the fire phenomena or employs more than one operating principle to sense one of these phenomena. Typical examples are the combination of a heat detector with a smoke detector or a combination rate of rise and fixed temperature heat detector.
Decibel	Decibel (dB) is a measurement unit of sound pressure level
Voice Evacuation system	A system that is interlinked with the fire alarm system, intended to evacuate the building occupants to a safe refuge by broadcasting alert and evacuation messages.
Flame Detector	A device used for detecting infrared and ultraviolet rays emitting from flames.
Fixed Temperature Detector	A device that responds only when its sensitive element heated up reaches a predetermined temperature.
Heat Detector	Fire detector that detects either abnormally high temperature or rate of temperature rise.

Manual Call Point		Manual operation device used to activate the fire alarm.
Optical Beam-Type Smoke Detector		A smoke detector comprising a light source and a receiver to detect the obscuration of light as a result of smoke. The transmitter and receiver may be at opposite ends or they may be incorporated into a single housing with a reflector at the opposite end.
Phased evacuation		A system of evacuation in which different parts of the premises are evacuated in a controlled sequence of phases, those parts of the premises expected to be at greatest risk being evacuated first.
Rate-of-Rise Detector		A device that responds when the temperature rises at a rate exceeding a predetermined level.
Smoke Detector		A device used for detecting visible and invisible particles of smoke resulting from combustion.
Spot / Point Detector		A device in which the detecting element is concentrated at a particular location. detector which responds to a phenomenon sensed in the vicinity of a fixed point
Sounder base		Base used for detectors with an inbuilt sounder
Standby Supply		A secondary power supply connected to the fire alarm system and automatically changes over in case of main supply failure.
Voice Alarm system		Dedicated manual or automatic system for originating and distributing of voice instructions, alert and evacuation signals for the safe evacuation of occupants. Sound distribution system that provides means for automatically broadcasting speech messages and warning signals.
Zone		A part of the protected building which contains one or more fire detectors, the zone is defined by a unique alphanumeric identification which is indicated at the control panel.
		FIRE DETECTION AND ALARM SYSTEMS
General	Reg. 4(1)	Every building or part of a building shall be provided with an automatic fire detection system which shall comply with the regulations stipulated in this chapter.
System design	Reg. 4(2)	Buildings shall be divided into a number of detection zones for easy recognition and short search time. (a) In conventional systems each detection zone shall be supplied by a separate circuit. (b) In addressable systems, several detection zones may be supplied by a single loop.

Detection Zone	Reg. 4(3) The area and the number of detection zones in a given building shall comply with the following: (a) The area of any single detection zone to which the building has been divided shall not exceed 2000 m²; except for a single, open plan area, which should not exceed 10 000 m². (b) If the floor area of a given building is less than 300 m² and height less than 15 m, a zone may cover more than a single story, therefore the entire building may be considered a single zone even if it is a multiple storey building. In case the total building floor area exceeds 300 m², each detection zone shall be restricted to a single storey. (c) Automatic fire detectors within any enclosed stairwell lift shaft or other similar enclosed structure shall be considered as a separate detection zone. (d) Spaces above or below the floor area of a room may be included within the same detection zone of the room, provided that the spaces and the room constitute a single fire compartment and the floor area is less than 1000m². (e) Remote indicator shall be clearly labeled to indicate the detectors located in the voids. They should be sited and/or labeled in such a way as to assist in determining the location of the detector. (f) The search distance to be travelled by anyone responding, after entry to the zone, should not exceed 60 m.
Alarm Zones	Reg. 4(4) Alarm zones shall be clearly defined in complex buildings where phased evacuation is required or in buildings where a stage alarm is to be provided (a) Alarm zone may include several detection zones, but not vice-versa. (b) Alarm zone boundaries shall match those of fire compartments and/or detection zones. (c) A common signal should be used throughout all alarm zones to convey the need for evacuation. A different common signal should be used throughout all alarm zones for any alert signal that can be given by the system
Categorization of Systems	Reg. 4(5) Systems are divided into different categories as follows: (a) Category L - Automatic fire detection systems intended for the protection of "Life". They are further sub-divided as follows; - Category L1- Systems installed throughout the building - Category L2 - Systems installed only in defined parts of the building - Category L3 - Systems installed covering the escape routes.

		(b) Category P - Automatic fire detection systems intended for the protection of property. They are further subdivided as follows; i. Category P1 - Systems installed throughout the building ii. Category P2 - Systems installed only in defined parts of the building iii. Category P3 - Systems installed covering the escape routes. (c) Category M - Manual systems which does not incorporate automatic fire detectors.
Requirement for Fire Detection system	Reg. 4(6)	The fire detection systems shall be installed as per Table 14 in this regulation.
Types of fire detectors and selection	Reg. 4(7)	(a) Fire detectors shall be selected to detect one or more of the following Characteristics of fire: i. Heat ii. Smoke iii. flame iv. Combustion gases (such as carbon monoxide) v. radiation (b) The "detector" shall be selected taking the following into consideration: i. The speed of response required ii. The need to minimize false alarms iii. The nature of the fire hazard iv. The possibility of presence of smoke/heat under normal operating conditions. (c) The detector shall not contain any hazardous material
Location and Siting of Spot/ Point Detectors	Reg. 4(8)	(a) The selection and placement of smoke detectors shall take into account both the performance characteristics of the detector and the areas into which the detectors are to be installed to prevent nuisance alarms or improper operation after installation. (b) Smoke detectors shall not be installed if any of the following ambient conditions exist at the detector location: i. Temperature below 0°C (32°F) ii. Temperature above 38°C (100°F) iii. Relative humidity above 93 percent iv. Air velocity across the detector is greater than 1.5 m/sec (300 ft/min) v. The height of the detector from the floor level is more than 10.5m. (c) The location of spot-type smoke detectors shall be based on an evaluation of potential ambient sources of smoke, moisture, dust, or fumes, and electrical or mechanical influences to minimize nuisance alarms.

Location and Spacing.	Reg. 4(9)	The location and spacing of smoke detectors shall be based upon the anticipated smoke flows due to the plume and ceiling jet produced by the anticipated fire as well as any pre-existing ambient air flows that could exist in the protected compartment. The design shall account for the contribution of the following factors in predicting detector response to the anticipated fires to which the system is intended to respond: i. Ceiling shape and surface. ii. Ceiling height. iii. Configuration of contents in the protected area. iv. Combustion characteristics and probable equivalence ratio of the anticipated fires involving the fuel loads within the protected area. v. Compartment ventilation. vi. Ambient temperature, pressure, altitude, humidity, and atmosphere.
Air-sampling type smoke detectors	Reg. 4(10)	Air-sampling type smoke detectors are particularly applicable in the following situations: (a) Where very early detection is required (b) Where the protected area is particularly challenging in terms of environment (hot, cold & humidity) and/or contamination (dust, & dirt) (c) Where smoke is difficult to detect due to high airflows or large volume/high ceiling spaces
Optical beam-type smoke detectors	Reg. 4(11)	Optical beam-type smoke detectors shall be provided where point detectors cannot be effectively used due to their limited performance, such as; (a) Where a large open area requires detection (b) Where ceiling heights are very high and there is a high risk of stratification
Manual Call Points	Reg. 4(12)	(a) The manual call points shall be used only for fire alarm initiation. In addition, all manual call points within occupancy shall be of similar design. Deviation from single design shall be justified based on special needs and shall be approved by AM. (b) Time from the operation of call point to the actuation of alarm signal shall not exceed three (3) seconds. (c) Where call points are installed in combustible, explosive environments these devices shall be listed for such application. (d) The manual call points shall be installed on all escape routes and in particular all stairwell entrances and all exits to open air. (e) The manual call points shall be installed so that they are conspicuous, unobstructed and easily accessible.

		(f) The maximum travel distance to the nearest manual call points shall not be more than 45m. (g) In limited mobility areas and areas where processes may result in rapid fire development the maximum travel distance to the nearest call point may be reduced to 25m and 16m respectively. (h) Manual Call Point shall be installed within 1.5m from exit door way opening and shall be mounted on both sides of the grouped opening if the width is more than 12 m. (i) The manual call points shall be installed at the height of (1.1 — 1.4) m, above floor level. Where wheel chair access is used, the height shall be 0.9m -1.2m
Above ceiling coverage	Reg. 4(13)	Fire detection shall be provided in any area that contains a horizontal void of 800 mm or more in height.
		FIRE ALARM CONTROL PANEL
Location	Reg. 4(14)	The fire alarm control panel shall be located in the Fire Command Centre (FCC) or in the absence of such FCC, in a location manned and monitored 24 hours of the day, on the ground floor close to the entrance to the building likely to be used by the Fire service.
Power Requirement	Reg. 4(15)	(a) At least two independent and reliable power supplies shall be provided. Each power supply shall be of adequate capacity for the application. Integrity of power supplies shall be monitored by the FACP. One supply shall be a dedicated circuit of the main supply and the other shall be from a dedicated standby battery. (b) The secondary power supply shall automatically provide power to the protected premises system instantaneously. (c) The secondary power supply shall have sufficient capacity to operate the system under quiescent load (system operating in a non-alarm condition) for a minimum of 24 hours and, at the end of that period, shall be capable of operating all alarm notification appliances used for evacuation or to direct aid to the location of an emergency for an additional 5 minutes. (d) Under normal operating condition the standby power source shall be capable of operating all alarm notification appliances used for 30 minutes.

		AUDIBLE AND VISUAL ALARMS																
Sound Level	Reg. 4(16)	(a) A fire alarming device shall have a sound that is continuous and readily distinguishable from any other alarm system on the same floor level, and shall be adequate to provide 65 dB(A) or 10 dB(A) above any ambient back ground noise likely to persist for more than 30seconds , whichever is higher. Refer following table for Ambient sound levels. Ambient Sound Levels for Different Occupancies <table><tr><th>Ambient Condition</th><th>Sound Level (dBA)</th></tr><tr><td>Business Occupancies</td><td>55</td></tr><tr><td>Educational Occupancies</td><td>45</td></tr><tr><td>Industrial Occupancies</td><td>80</td></tr><tr><td>Institutional Occupancies</td><td>50</td></tr><tr><td>Places of Assembly</td><td>55</td></tr><tr><td>Residential Occupancies</td><td>35</td></tr><tr><td>Storage Occupancies</td><td>30</td></tr></table> (b) The maximum sound level shall not exceed 120dB(A). (c) The sound pressure level shall not be less than 75 dB (A) at the bed head with all doors closed where overnight sleeping accommodation is provided. (d) Arrangements shall be made to ensure that at least one Sounder in each alarm zone remained functional during any fault condition.	Ambient Condition	Sound Level (dBA)	Business Occupancies	55	Educational Occupancies	45	Industrial Occupancies	80	Institutional Occupancies	50	Places of Assembly	55	Residential Occupancies	35	Storage Occupancies	30
Ambient Condition	Sound Level (dBA)																	
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Visual Alarms	Reg. 4(17)	(a) Strategically located visual alarms shall be provided in addition to audible alarms where necessary i. Purpose group 2(a) shall have visual alarm at the nurses' station or supervisory station. Audible and visual alarm shall be available in wards and private rooms to be activated by nursing staff / supervisory staff as required. ii. Where accommodation is provided for disabled persons, both visual and audible alarms shall be provided. iii. Where the ambient sound level is high, which compels the occupants to wear ear protection, strategically located visual alarms shall be provided. (b) The visual alarm signal should flash at a rate within the range of 30-130 flashes per minute (c) Visual alarms should be mounted at a minimum height of 2.1 m, but no closer to the ceiling than 150 mm																
Warning in The Event of a Fire	Reg. 4(18)	The "warning" given by a fire alarm shall comply with the following stipulations: i. The warning "sound" to indicate a fire situation should be distinct from all other alarm sounds.																

		ii. The automatic fire alarm shall be activated as follows. I. For LOW fire hazard buildings - Fire Floor only. 2. For MEDIUM fire hazard buildings - Fire floor, and the floor immediately above and below. 3. For HIGH fire hazard buildings - All floors
Alarm audible requirement	Reg. 4(19)	(a) Public Mode Audible Requirements To ensure that audible public mode signals are clearly heard, they shall have a sound level at least 15 dB above the average ambient sound level or 5 dB above the maximum sound level having a duration of at least 60 seconds, whichever is greater, measured 1.5 m above the floor in the area required to be served. Refer to Reg. 4(16) above for the sound level based on location. (b) Private Mode Audible Requirements To ensure that audible private mode signals are clearly heard, they shall have a sound level at least 10 dB above the average ambient sound level or 5 dB above the maximum sound level having a duration of at least 60 seconds, whichever is greater, measured 1.5 m above the floor in the area required to be served. (c) Sleeping area requirement Audible appliances shall be installed to provide signals for sleeping areas. They shall have a sound level of at least 15 dB above the average ambient sound level or 5 dB above the maximum sound level having duration of at least 60 seconds or a sound level of at least 75 dBA, whichever is greater, measured at the pillow level in the area required to be served.
		VOICE EVACUATION SYSTEM
Requirement for One way emergency voice evacuation and communication system	Reg_ 4(20)	(a) One way emergency voice evacuation and communication system and an emergency command centre shall be provided as follows: i. For all large buildings with gross floor area greater than 2800 m² or having a total occupant load exceeding 1000 persons and large industrial and warehouse buildings with gross floor area greater than 5000 m². ii. For all buildings which are categorized as high rise and super high rise buildings. iii. For hotel or health care occupancies exceeding 18 m height where phased evacuation is necessary (b) The emergency voice evacuation and communication system shall override any public address systems (c) Speakers for emergency voice evacuation shall be provided in every lift lobby, staircase enclosure, corridors and other strategic positions within audible distance of all parts of all storeys of the building

		(d) Speakers with flashers (strobe lights) shall be provided for basement car parks, mechanical or machine rooms, large machining areas of factories (more than 5000m ²) and at locations where the ambient noise level is 75db or more
Activation of Voice Evacuation System	Reg. 4(21)	(a) The voice/alarm communications system shall be used to provide an automatic response to the receipt of a signal indicative of a fire alarm or other emergency. (b) The system shall permit, where applicable, the application of an automatic evacuation signal to one or more evacuation signaling zones and, at the same time, shall permit manual voice paging to the other evacuation signaling zones selectively or in any combination.
Voice Evacuation Message	Reg. 4(22)	(a) In response to an initiating signal indicative of a fire emergency, the system shall transmit a voice recorded message. (b) Evacuation messages shall be preceded and followed by a minimum of two cycles of the emergency evacuation signal.
Control of Voice Evacuation System	Reg. 4(23)	(a) Controls for the emergency voice/alarm communication system shall be at the Fire Command Centre or a central location accessible by building staff and emergency responders. (b) Controls shall be located or secured to allow access by only trained and authorized personnel. (c) Operating controls shall be clearly identified. (d) If there are multiple emergency voice/alarm communications control locations, only one shall be in control at any given time. (e) The location having control of the system shall be identified by a visible indication at that location. (f) If live voice instructions are provided, they shall override previously initiated signals to the selected notification zone(s) and shall have priority over any subsequent automatically initiated signals to the selected zone(s).
Cabling for Voice Evacuation System	Reg. 4(24)	All circuits necessary for the operation of the notification appliances shall be protected until they enter the evacuation signaling zone that they serve. Any of the following methods shall be considered acceptable as meeting the requirements of this subsection: i. Cables of 3-hour fire rated circuit integrity at 950 °C, Cat CWZ ii. Cables installed inside 2-hour fire rated enclosure

Two-way telephone communication system with Fire Command Center	Reg. 4(25)	Two-way telephone communication system with Fire Command Centre (FCC) shall be provided for all buildings which are categorized as high rise and super high rise buildings and any other building as requested by AHJ. Two way communication systems shall be provided between FCC and the following areas or any area specified by the AHJ. i. Every firefighting lobby ii. In basements, where the number of basements exceeds 2. iii. Firefighting related mechanical rooms iv. Fire service lifts v. Rooms having smoke control equipment vi. Each area of refuge (a) Monitoring of the integrity of two-way telephone communications circuits shall be provided. (b) Two-way telephone communications service shall be capable of permitting the simultaneous operation of any five telephone stations in a common talk mode. (c) A notification signal at the control equipment, distinctive from any other alarm, supervisory, or trouble signal, shall indicate the off-hook condition of a calling telephone circuit. (d) In high rise and super high rise buildings equipped with a fire pump(s), a telephone station or jack shall be provided in each fire pump room. (e) If telephone jacks are provided, at least five handsets shall be stored at fire command center. (f) All circuits necessary for the operation of two-way telephone communication systems shall be installed using one of the following methods: iii. Cables of 3-hour fire rated circuit integrity at 950 °C, Cat CWZ iv. Cables installed inside 2-hour fire rated enclosure
Fire Command Centre.	Reg. 4(26)	(a) Any High Rise and Super High Rise Building shall be provided with a Fire Command Centre (FCC). (b) The FCC shall be located in the ground floor or any other raised floor level where fire service access is provided as approved by AM, and in close proximity to fire fighting shaft. (c) The FCC shall be manned 24 h.
Temporary Buildings in Construction Sites	Reg.4(27)	Building under construction shall be provided with manually operated/ hand crank fire siren audible to construction workers
	Reg.4(28)	Clear access path with proper illumination shall be provided <u>for evacuation in an emergency.</u>

CHAPTER 5**FIRE EXTINGUISHING APPLIANCES,
FIXED FIRE PROTECTION AND SUPPRESSION SYSTEM**

Purpose of this Chapter	The purpose of this Chapter is to provide minimum requirements of portable and fixed Fire Protection systems that shall be installed in building to enhance the life safety of the occupants and to mitigate damage to property. The requirements described below are stated as the minimum requirements to be complied with and any improvements may be accepted provided that the designer/Installer could justify its performance is superior to what is specified in this regulation.	
Definitions		For the purpose of this Code the following definitions shall be applicable:
Classification of Fires		Fires shall be classified as follows: (a) Class A: Fires involving solid materials, usually of organic nature, in which combustion normally takes place with the formation of glowing embers. (b) Class B: Fires involving liquids or liquefiable solids. (c) Class C: Fires involving gases. (d) Class D: Fires involving metal. (e) Class F: Fire involving cooking oil. (or Class K)
FAFA (First Aid Fire Appliances)		Fire fighting appliances assigned for the use of occupants of the building (ex: Fire extinguishers, fire blanket, fire hose reels)
Dry Rising Main		A water main, normally empty, installed in the building or within the site of the building, with an inlet or inlets at street level through which water can be pumped to provide a water supply for branch mains to hydrants on various storeys.
Wet Rising Main		A water main permanently charged with water of adequate pressure and flow, installed in the building or within the site of the building to provide a water supply for branch mains, hydrants and hose reels.
Hydraulic Hose Reel		A flexible reinforced tubing having an internal diameter of 19 to 25 mm, permanently connected to the water supply and wound on a metal drum.
Landing Valve		Valves that are installed on hydrants, where a branch pipes with or without a control device and hose can be connected to a coupling on it.
Fire Hose		Rubber lined canvas flexible hose of diameter 65 mm, with instantaneous coupling.
Jet nozzle (Jet/spray/shut off Nozzle)		Nozzle used to be coupled to the end of fire hose forming a water jet/Spray, with shut off facility.

Protected Fire Lobby		An enclosed fire/smoke proof space into which a fire lift opens and having a direct access to a protected staircase for the purpose of fire fighting. The elements of structure of such a lobby shall conform to the provisions of Chapter 3 of the regulation.
Protected Ventilated Fire Lobby		A lobby as defined above but with a provision of natural ventilation or with permanent ventilation to the open air
		PORTABLE FIRE EXTINGUISHERS
General	Reg. 5(1)	Portable Fire Extinguishers shall be approved by the AHJ, after complying with the relevant performance test procedures
	Reg.5(2)	(a) Portable fire extinguishers, effective to the class of fire which may occur and the nature of the processes or contents within a building, shall be provided and maintained in fully charged and operable condition as may be required by the AHJ. (b) All-purpose groups except Group I (c) shall be provided with portable fire extinguishers as per regulation 5(3). (c) In any building where rooms or spaces designated as electrical switch rooms, transformer rooms, generator rooms, lift machine rooms, fuel storage etc and any other room designated for specific uses shall be provided with fire extinguishers of suitable quantity, type and capacity. (d) Portable fire extinguishers shall be charged, tested and maintained in fully operational conditions, at all times. (e) Portable fire extinguishers intended for different classes of fire, shall be marked conspicuously according to their intended use..
Selection of Fire Extinguishers	Reg. 5(3)	Selection of fire extinguishers shall be as follows: (a) Fire extinguishers for protecting Class A fire hazards shall be selected from the following: Water, foam and Dry powder (b) Fire extinguishers for protection of Class B fire hazards shall be selected from the following: Carbon dioxide, dry chemical, multi-purpose dry chemical, foam. (c) Fire extinguishers for protection of Class C fire hazards shall be selected from the following: Carbon dioxide, dry chemical, multi-purpose dry chemical powder (d) Fire extinguishers and extinguishing agents for the protection of Class D fire hazards other than foam shall be of types approved by the AHJ. <i>Note: Water based extinguishers excluding water mist type, shall not be used for Tres involving live electrical e • wimenl.</i>

Siting	Reg. 5(4)	Extinguishers shall be located where they will be readily accessible and immediately available in the event of fire. Except where otherwise stated in this regulation, they shall be located along normal paths of travel, preferably on escape routes in such a way that no person need travel more than 30 m to reach an extinguisher or as otherwise stated in Regulation 5(5) and Table 15 & 16.
Distribution and Minimum Sizes and Quantity of Extinguishers	Reg. 5(5)	Minimum sizes of fire extinguishers for the listed grades of Class A fire hazards shall be provided and located so that the maximum travel distances shall not exceed those specified in Table 15 except that for hazards connected with energized electrical equipment, any of the types of fire extinguishers specified in Regulation 5(3) and of size and at locations as determined by the AHJ, shall be provided.
	Reg. 5(6)	Minimal sizes of fire extinguishers for the listed grades of Class B fire hazards other than for fires in flammable liquids of appreciable depth shall be provided and located so that the maximum travel distances shall not exceed those specified in Table 16.
	Reg. 5(7)	For flammable liquid fire hazards of appreciable depth such as in dip or quench tanks, extinguishers listed in Table 16 shall be provided on the basis of 80 litres foam extinguishing potential per square metre of flammable liquid surface of the largest tank hazard within the area. If automatic fire protection systems are installed for flammable liquid hazards, extinguishers may be waived.
		RISER
General Design Condition	Reg. 5(8)	Any building in which the floor level of the highest storey is higher than 18 metres above pavement or ground level, shall be equipped with Wet Rising Mains.
Maximum Design Pressure of Wet Riser System	Reg. 5(9)	System shall be designed such that maximum system pressure does not exceed 20 bar.
Number of Mains Required	Reg. 5(10)	Where rising mains have to be provided under 5(8) of this regulation or are otherwise provided, the number of mains required shall be such that: (a) There shall be at least one rising mains for every 900m² or part thereof of floor area on each storey, which is higher than 18 meters above pavement or ground level. If any part of a building is physically separated and with no or restricted access, such areas shall be provided with an additional riser. (b) They may be so positioned that all parts of any floor space shall be within 60 metres from a landing valve on any storey which is higher than 18 metres above pavement or ground level
	Reg. 5(11)	Where the building has access from more than one ground level or road level, the height measurements for the purpose of paragraphs (1) and (2) of this regulation shall be taken from the lowest level at which access for fire brigade engines is required to be provided.

Size	Reg. 5(12)	(a) Diameter of the riser shall not be less than 100mm where the rising main does not exceed 30 m in height and only one landing valve is provided at each floor (b) Diameter of the riser shall not be less than 150mm where the rising main either i. Exceeds 30 m in height or ii. has more than one landing valve connected in a floor.
Pipe Work	Reg. 5(13)	(a) The rising main and all associated pipe work shall be steel complying with the required system pressure rating. The pipes should withstand the testing pressure of the system which shall be 1.5 times of the system design pressure (b) All rising main on high rise and Super high-rise building shall be of seamless pipes
Automatic Air Vent Valve	Reg. 5(14)	Each rising main should be provided with an automatic air vent valve at the highest point
		HOSE REEL
Hydraulic Hose Reel - General	Reg. 5(15)	(a) At least one hydraulic hose reel conforming to the provisions of this regulation shall be provided in every storey of every building except the following: (b) Purpose group 1 (c) buildings (c) A shop on the ground storey of a shop-cum-residential building provided the shop does not exceed 280m² in floor area and is properly separated from the residential floors or parts of the building by compartment walls and floors. (d) A petrol service station comprising a small office of the like. (e) Any building which is less than 800 m² of total habitable area does not require a hose reel.
Size	Reg. 5(16)	The hose reel shall be of 19mm diameter not exceeding 45 meters length with a 4.8 mm nozzle, or 25mm diameter not exceeding 30m in length with 6.4 mm nozzle. It shall be made of non-kinking reinforced flexible material and terminating in "shut-off" nozzle.
Pressure/flow rate	Reg. 5(17)	(a) The pressure at the end of the nozzle shall be not less than 1.25 bar for 4.8 mm nozzle and 3.0 bar for 6.1 mm nozzle. (b) shall be capable of generating a jet of water having a throw of not less than 6m (c) Shall provide a flow rate of not less than 30 litres per minute.
Siting	Reg. 5(18)	(a) Hose reels shall be sited in prominent and accessible positions at each floor level preferably adjacent to exits in corridors on exit routes,

	(b) Hose Reel shall be located in a position such that the nozzle of the hose can reach every room and within 6 m of each part of a room, having regard to any obstructions and normal access path.
Reg. 5(19)	The Hose reel shall be mounted such that the inlet to the hose reel is at 1200 mm above finished floor level.
Reg. 5(20)	(a) Colour and Signage of Hose Reel cabinet shall be painted red or shall be provided with a sign with Red letters 100 mm high on a white background. The international Hose reel symbol in red on a white background shall also be accepted. (b) The red colour used for fire hose reel shall be RAL 3000.
Reg. 5(21)	The hose reel cabinet shall not be locked. If the requirement arises that it shall be locked and the key shall be kept in a place visible to the user, in an enclosure with a break glass on the hose reel cabinet door.
	LANDING VALVES
Siting of Landing Valve Reg. 5(22)	The landing valve shall be located, (a) Within the protected ventilated or smoke free lobby/ fire fighting lobby. (b) External corridor immediately outside the door of the exit staircase. (c) Position of rising mains and associated landing valves shall be located in the following order of priority i. Within smoke stop lobby ii. In the common area and within a protected shaft, immediately outside the exit staircase, if there is no smoke-stop lobby iii. Inside staircase where smoke-stop lobby and common area are not provided. The location of the rising main shall not cause obstruction to the escape path inside the staircase. (d) In the case of existing buildings where there are no fire fighting lobby, smoke stop lobby or external corridor, it shall be located in the common area and within a protected shaft, immediately outside the door of the exit staircase. (e) Horizontal distance between two landing valves shall not exceed 60m.

Landing valves pressure and flow rate	Reg. 5(23)	Landing Valves shall be provided off the rising mains on each level at a height of 1m above the floor level. The pressure and flow rate at each outlet shall be as given below. (a) Maximum pressure at the landing valve should not exceed 5 bar. (b) Minimum pressure at the landing valve should not be less than 4 bar. (c) The flow rate shall be not less than 500 litres per minute.
Location of Wet Riser	Reg. 5(24)	The locating of Wet Risers should be as follows: (a) Rising Mains should be housed in a protected shaft or in a protected duct where additional risers are required. (b) Rising Mains shall not be more than 60 meters apart. (c) No part of any floor shall be more than 60 meters from a Landing Valve.
Installation	Reg. 5(25)	The installation shall be in accordance with the provisions of the accepted code of practice on rising mains. Landing valves should be installed as follows: (a) They should be protected from mechanical damage. (b) The lowest point approximately between 750 mm to 1200 mm above floor level, not less than 150 mm clearance on both sides, not less than 230 mm below the centre line of the outlet of the landing valve, not less than 200 mm clearance above the hand wheel, and the front edge of the female coupling of the landing valve should be not more than 75 mm behind the face of the wall.
EXTERNAL PILLARHYDRANTS		
External Hydrants	Reg. 5(26)	External hydrants are required for buildings, (a) For Purpose groups 4,5,6,7 and 8 where the foot print area exceeds 1000 m². (b) For buildings under purpose group 1, where more than 100 units are located in a designated dedicated complex. (e) For buildings of purpose groups other than 1, where access for fire vehicle appliances is restricted.
Siting of Pillar Hydrants	Reg. 5(27)	(a) Each external Pillar Hydrant shall consist of two Nos of 65 mm outlets. (b) Each Pillar hydrant shall be provided with a fire hose cabinet consisting of 3x15 m 65 mm dia fire hoses, with instantaneous coupling and one jet/ spray shut off branch pipe. (c) The Hydrant shall be located as part of a ring main system and

		be positioned such that a hydrant is located not more than 18M from the main entrance. (d) The hydrant shall not be more than 90m apart and shall not be less than 6m from the building or other risk
Material for piping	Reg. 5(28)	Piping material shall be steel and rated for minimum 10 bar. All underground piping shall be adequately protected against corrosion. Non-metallic piping rated for minimum 10 bar may be used exclusively for underground piping only. Any section of piping including tap-off or branches to connect pillar hydrants or risers shall be metallic if located above ground.
Water Supply	Reg. 5(29)	(a) Water supply shall be such that the Pillar hydrant furthestmost from the fire pump shall have a minimum running pressure of 4 bar at a flow rate of 500 litre/ min. (b) The ring main system shall be common to both the external pillar hydrant and internal landing valves. (c) Isolating valves with tamper proof devices shall be provided for the ring main to facilitate maintenance without compromising the operational functions of the system. (d) The minimum internal diameter of the ring main shall be 100mm.
		BREACHING INLET
Breaching Inlets and Access	Reg. 5(30)	(a) Buildings having rising mains shall be fitted with a Breaching Inlet located within 18 meters from the main entrance with clear access for Fire Brigade Vehicles. (b) The Breaching Inlet shall be suitably located on an external wall of the building at a height of 1m above the surrounding ground road or pavement level and a notice with the words "WET RISER INLET" shall be prominently displayed adjacent to every inlet. (c) A marking stating "No Parking" to be made on the floor area of the breaching inlet preventing vehicle to park and block Fire Vehicle access. (d) For high rise and super high rise buildings a breaching inlet shall be provided to fill the fire suction tank.
		FIRE PROTECTION FOR BUILDING UNDER CONSTRUCTION
Building Under Construction	Reg. 5(31)	All building under construction shall be equipped with following Fire Fighting facility (a) A Dry Type rising main shall be installed where the building habitable floor height is less than 60 meters.

		(b) A WET Type rising main shall be installed where the building habitable floor height is more than 60 meters (c) Breeching inlets (2 way or 4 way) shall be provided and connected to the riser as per approved building plan (d) Power Supply for the fire pumps from power grid or generator is acceptable (e) Access to Fire Vehicles shall be provided where clearance of 4meter width and 4meter overhead clearance, to the breeching inlet shall be provided. (f) Portable fire extinguishers shall be provided at each floor level (g) Rising main and landing valve shall be provided to every floor, except the top most 3 floors as the building gains height and made operational (h) Temporary fire water storage tank of 12 m³ of water shall be provided for fire fighting until the arrival of fire engine to replenish water. The storage tank must be constructed before the building reaches 60 m height (I) Pump shall have a capacity of 1500 ltr/min with a minimum of 4 bar pressure at the highest hydrant location. (i) The main contractor for the project shall be responsible and accountable for the provision and maintenance of the rising main in the building under construction.
Water Supply	Reg. 5(32)	(a) The water supply for the landing valves and the hose reels shall be independent, and shall not be combined with any other services supplies. (b) Landing valve and Hose Reel shall be connected in a single riser or in different riser.
		AUTOMATIC SPRINKLER SYSTEMS
General	Reg. 5(33)	The following shall be provided with an automatic sprinkler system: (a) All indoor car parks (b) All high rise & Super High Rise Buildings as defined by these regulations. It should be noted that the sprinklers systems should cover all storeys including the basements and not limited to storeys above 30m. Exemptions are toilets, electrical switch gear rooms, server rooms or any other area where water damage will be significant and that will not escalate the fire spread to adjoining areas. (c) Any part of a basement storey which is used as a place of public resort, shops, office, factory or warehouse irrespective of whether or not the compartmentation requirements are <u>complied with.</u>

		(d) Where a building, or a compartment of a building except that of purpose groups 1 and 2 and school buildings, is allowed to comprise more than one storey and there are unprotected openings in the floor between such storeys, each storey shall be sprinklered irrespective of compliance with compartmentation requirements, provided that these requirements shall not be applicable if the means of escape of the building or compartment are properly protected in accordance with Chapter 2 of the Regulation. (e) In buildings or parts of buildings used for special purposes or hazards, where required under provisions of Chapter 6 of the Regulation.
Commencement of the Installation Work	Reg. 5(34)	No person shall commence work on the installation of the sprinkler system unless plans for such system required to be submitted in accordance with the provisions of regulations 4(35) and 4(36) of this chapter, have been approved by the AHJ.
Submission of plans	Reg. 5(35)	Plans for the installation or extension of a sprinkler system shall be signed by a Qualified Person and by the owner of the building in which the sprinkler system is installed and bear the certification of the Qualified Person that: (a) Such plans have been prepared in accordance with the accepted code of practice on automatic sprinkler systems, and (b) The Qualified Person shall be personally responsible for the proper installation of the sprinkler system in accordance with such plans and shall carry out such tests and submit such test reports as and when required by the AHJ
Information Required to be given on design drawings	Reg. 5(36)	Design drawings of sprinkler system shall consist of sprinkler locations, detailed piping layout and valve positions with a legend to the similar scale as the building floor plans.
Inspection Maintenance and Testing	Reg. 5(37)	The inspection, maintenance and testing of a sprinkler system shall be in accordance with Chapter 8 of this regulation document.

Occupancy Hazard Classification	Reg. 5(38)	Buildings or areas to be protected which contain one or more of the following occupancies and fire hazards shall be classified as belonging to the appropriate hazard class as follows; (a) Light Hazard (LH) - Occupancies with low fire loads and low combustibility and with no single compartment greater than 125 m² with a fire resistance of at least 30 min. (b) Ordinary Hazard (OH) - Commercial and Industrial occupancies handling, processing and storage of ordinary combustible materials unlikely to develop intensely burning fires initially. May be classified as : Ordinary Hazard Group I or Ordinary Hazard Group II or Ordinary Hazard Group III or Ordinary Hazard Group IV (c) High Hazard (HH) - High Hazard Process (HHP) or Storage (HHS) covers occupancies where the materials concerned have a high fire load and high combustibility and are capable of developing a quickly spreading or intense fire.																														
Design Density	Reg. 5(39)	The following Minimum Design density shall be taken into consideration when installation of sprinkler system <table border="1"> <thead> <tr> <th>Hazard</th><th>Minimum Design Density mm/min</th><th>Assumed Maximum Area of Operation in m²</th></tr> </thead> <tbody> <tr> <td>Light</td><td>2.25</td><td>84</td></tr> <tr> <td>Ordinary Group I</td><td>5</td><td>72</td></tr> <tr> <td>Group II</td><td>5</td><td>144</td></tr> <tr> <td>Group III</td><td>5</td><td>216</td></tr> <tr> <td>Group IV</td><td>5</td><td>360</td></tr> <tr> <td>High Hazard Group I</td><td>7.5</td><td>260</td></tr> <tr> <td>Group II</td><td>10.0</td><td>260</td></tr> <tr> <td>Group III</td><td>12.5</td><td>260</td></tr> <tr> <td>Group IV</td><td>Deluge System</td><td>Complete deluge protection for each building</td></tr> </tbody> </table>	Hazard	Minimum Design Density mm/min	Assumed Maximum Area of Operation in m ²	Light	2.25	84	Ordinary Group I	5	72	Group II	5	144	Group III	5	216	Group IV	5	360	High Hazard Group I	7.5	260	Group II	10.0	260	Group III	12.5	260	Group IV	Deluge System	Complete deluge protection for each building
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Minimum Water requirements
For Low Hazard and Ordinary Hazard occupancies

Reg. 5(40)

The following Minimum Water Volume shall be considered for sprinkler systems for a pre calculated Light Hazard (LH) and Ordinary Hazard (OH) occupancies.

The water volume indicated shall be reserved solely for the use of the sprinkler system.

Group	Height h of the highest Sprinkler above the lowest Sprinkler ^a (m)	Minimum water volume (m ³)
LH — (Wet or Pre-action)	$h < 15$	9
	$15 < h < 30$	10
	$30 < h < 45$	11
OHI — Wet or Pre-action	$h < 15$	55
	$15 < h < 30$	70
	$30 < h < 45$	80
OH I — Dry or alternate OH2 — Wet or Pre-action	$h < 15$	105
	$15 < h < 30$	125
	$30 < h < 45$	140
OH2 — Dry or alternate OH3 — Wet or Pre-action	$h < 15$	135
	$15 < h < 30$	160
	$30 < h < 45$	185
OH3 — Dry or alternate OH4 — Wet or Pre-action	$h < 15$	160
	$15 < h < 30$	185
	$30 < h < 45$	200
0114 — Dry or alternate	Use HIT protection ^b	
^a Excluding sprinklers in the sprinkler valve room		
^b Hydraulically calculated only		

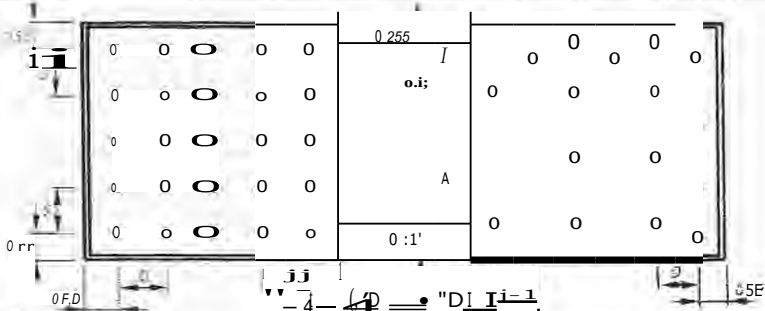
Minimum Water requirements
High Hazard occupancies

Reg. 5(41)

The following Minimum Water Volume shall be considered for sprinkler systems for pre calculated High Hazard (HHP and HHS) occupancies.

The water volume indicated shall be reserved solely for the use of the sprinkler system.

Design density not exceeding mm/min	Minimum water volumem ³	
	Wet systems	Dry systems
7.5	225	280
10.0	275	345
12.5	350	440
15.0	425	530
17.5	450	560
20.0	575	720
22.5	650	815
25.0	725	905
27.5	800	1000
30.0	875	1090

Pressure/ Flow Requirements	Reg. 5(42)	The following flow / pressure requirements shall be maintained for Light Hazard (LH) and Ordinary Hazard (OH) occupancies. <table><tr><th>Group</th><th>Minimum Running Pressure kPa</th><th>Flow Rate //min</th></tr><tr><td>LH</td><td>225+ P_s</td><td>225</td></tr><tr><td>01-1 I</td><td>100+ P_s 70 + P_s</td><td>375 540</td></tr><tr><td>OH II</td><td>140+ P_s 100+ P_s</td><td>725 1000</td></tr><tr><td>OH III</td><td>170+ P_s 140+ P_s</td><td>1100 1350</td></tr><tr><td>OH IV</td><td>200+ P_s 150+ P_s</td><td>1800 2100</td></tr></table> Ps : Static System Pressure Loss	Group	Minimum Running Pressure kPa	Flow Rate //min	LH	225+ P _s	225	01-1 I	100+ P _s 70 + P _s	375 540	OH II	140+ P _s 100+ P _s	725 1000	OH III	170+ P _s 140+ P _s	1100 1350	OH IV	200+ P _s 150+ P _s	1800 2100							
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Maximum coverage and spacing for sprinklers other than side wall sprinklers	Reg. 5(43)	A. The following maximum coverage and spacing shall be considered for sprinklers other than side wall sprinklers. <table><tr><th rowspan="3">Hazard class</th><th rowspan="3">Maximum area per sprinkler (m²)</th><th colspan="3">Maximum distance as shown in figure below(n1)</th></tr><tr><th rowspan="2">Standard Layout S and D</th><th colspan="2">Staggered layout</th></tr><tr><th>S</th><th>D</th></tr><tr><td>LH</td><td>21.0</td><td>4.6</td><td>4.6</td><td>4.6</td></tr><tr><td>OH</td><td>12.0</td><td>4.0</td><td>4.6</td><td>4.0</td></tr><tr><td>HHP and HHS</td><td>9.0</td><td>3.7</td><td>3.7</td><td>3.7</td></tr></table>  Key S Distance between sprinklers D Distance between sprinklers Ceiling sprinkler spacing	Hazard class	Maximum area per sprinkler (m ²)	Maximum distance as shown in figure below(n1)			Standard Layout S and D	Staggered layout		S	D	LH	21.0	4.6	4.6	4.6	OH	12.0	4.0	4.6	4.0	HHP and HHS	9.0	3.7	3.7	3.7
Hazard class	Maximum area per sprinkler (m ²)	Maximum distance as shown in figure below(n1)																									
		Standard Layout S and D			Staggered layout																						
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OH	12.0	4.0	4.6	4.0																							
HHP and HHS	9.0	3.7	3.7	3.7																							

Maximum coverage and spacing for side wall sprinklers		B. The following maximum coverage and spacing shall be considered for wall sprinklers							
		LH	17.0	Spacing along walls		w < 3.7	any	1	single line
				4.6	2.3				
		OH	9.0	3.4 (see note 2)	1.8	w < 3.7	any	1	single line
						3.7 < w < 7.4	< 9.2	2	standard
							9.2	2	staggered
						w > 7.4	any	2 (see note)]	standard
						w < 3.7	any	1	single line
						3.7 < w < 7.4	< 6.8	2	standard
							6.8	2	staggered
						w > 7.4	any	2	Standard] (see note)]
NOTE 1 An additional row of rows of roof or ceiling sprinklers is required.									
NOTE 2 This can be increased to 3.7 m provided the ceiling has a fire resistance of no less than 120 min.									
NOTE 3 The sprinkler deflectors should be located between 0.1 and 0.15 m below the ceiling and between 0.05 and 0.15 m horizontally from the wall.									
NOTE 4 There should be no obstruction at the ceiling within a square extending along the wall 1.0 m on each side of the sprinkler and 1.8 m perpendicular to the wall.									
Clearance Space below Sprinkler	Reg. 5(44)	Following clearance space shall be provided below sprinklers (a) High piled combustible stocks - 1 m (b) Others - 0.5 m							
Protection of Concealed spaces	Reg. 5(45)	All concealed spaces between floors (or concrete slab roofs) and ceilings below, exceeding 800 mm in depth shall be fully protected by sprinklers in accordance with the requirements for the particular hazard class.							

Information to be Provided for Approval of Sprinkler Systems	Reg. 5(46)	Following Information shall be Provided for Approval of Sprinkler Systems. (a) Detail of Water Supplies. (b) Details of Automatic Pump Sets. (c) Fully detailed and dimensioned design drawings specifying the material to be used.
SOURCES OF WATER SUPPLY		
Sources of Water Supply	Reg. 5(47)	Water supply for fire fighting installations shall comply with the conditions stipulated below. Where the water supply from a public service main is not adequate in quantity, or pressure, an alternative water source shall be provided. Tanks supplying water for domestic purposes shall not be used as suction tanks for fire systems, unless arrangements are made in such a manner that the requisite reserve of water for the rising main is always maintained. Minimum capacities shall be as specified below. These capacities relate to stored water sources entirely reserved for the sprinkler system (including fire hose reels). The minimum capacity must be sufficient to supply the pump for one hour when running at its normal rating. (a) Fire Hydrant and Hose reel systems shall be fed by a Fire storage tank of minimum capacity of 67,500 litres. This storage capacity may be revised to account for additional flow requirement where multiple risers are installed. (b) Sprinkler systems shall be fed by a Fire storage tank of minimum capacity as stipulated in Regulations 4(40) and 4(41). This storage capacity may be revised to account for additional flow requirement where multiple risers or combination systems are installed (c) In buildings where both sprinklers and wet riser are provided, the minimum total water storage volume shall be the higher of the individual requirement for each system. (d) In the event of the fire water storage tank being emptied for maintenance work or otherwise, arrangements shall be made for required minimum water volume to be available for the fire fighting system.
Combination Systems	Reg. 5(48)	(a) A combination system (combined automatic Sprinkler/Hydrant riser) shall be allowed where the buildings with partially automatic sprinkler protection system and falling under the following purpose groups, where the total building height does not exceed 30m.

		Title	Purpose Group	Purpose for which the Building or Compartment is intended to be used
		1. Residential (Dwelling)	1(a)	Residential Apartments or flats: Having multiple residencies in a single block of building, using common facilities.
			1(b)	Residential Unit: A unit having single or multiple residencies in a single block of building exceeding 300 m ² .
			1(c)	Single Residential House: A unit not exceeding 300 m ² .
		2. Residential (Institutional)	2 (a)	Establishments used for treatment, care or maintenance of persons suffering from disabilities, or educational purposes and Accommodations, including hospitals, clinics, polyclinics orphanages, and children's homes, elder's homes, day-care centres, infant care, kindergartens.
		Residential (Other)	2(b)	Hotels
			2(c)	student hostels, dormitories, army camps, schools, colleges, commercial schools, vocational institutions and universities
			2(d)	Detention & correction centres
		3. Office	3	Offices or premises used for administration, clerical work, handling money, communications, recording or performance.
		4. Commercial	4(b)	Shops or premises used for super markets, department stores, food courts, restaurants, bars & pubs, betting halls, show rooms , total floor area exceeding 800 m ² .
			4(c)	Shops or premises used for retail trade hire or repair, restaurants, bars, pubs , salons, total floor area not exceeding 800 m ² .

		5. Place of public Assembly and Recreation	5(b)	Cinemas, theatres, Exhibition centres, stadium, sports complexes, private clubs(night clubs), casinos, public halls.
			5(d)	Colleges, universities, class rooms, libraries.
			5(e)	Conference centres, gymnasias, skating rinks, churches.
		6. Industrial / Factories	6(c)	Mechanical plant rooms.
		7. Storage	7 (a)	Warehouse located in Basements.
			7 (b)	Place for the storage of goods or materials other than 7 (a) and any building not within groups 1-6.
			7(c)	Container terminals, freight stations, logistic centres, general goods
		8. Car parks	8(b)	Automated mechanized car parks
		8(c)	Parking structures above ground, naturally ventilated car parks.	
		(b) Combined system shall consist of one pump set in a building which is partially sprinklered.		
		(c) Combination system pumps shall be supplied with intake water under positive head.		
		(d) Pump sets shall be capable of satisfying the flow and pressure requirements of any assumed areas of operation in the system under consideration		
		(e) The total flow rate required for a combined system shall be the flow rate required for the hydrant system and the minimum flow rate of the sprinkler system.		
		The residual pressure required for the combined system shall be the minimum residual pressure required for landing valve/ hydrant at the hydraulically most remote location.		
		(g) If a dedicated Fire water sump is used, a facility for monitoring the water level and raising an alarm if the level drops shall be provided. This alarm shall be integrated with the Fire Alarm Control panel to give an alarm at the fire command centre		

		FIRE PUMP INSTALLATIONS FOR SPRINKLER, HYDRANT AND HOSE REEL SYSTEMS
General	Reg. 5(49)	Separate pumps sets shall be provided for Sprinkler and hydrant installations other than for combination system.
Pumps, drivers, and controllers	Reg. 5(50)	Pumps, drivers and controllers shall comply with the following. (a) Drivers for pumps at a single installation shall be either electric motors or diesel engines. (b) Fire pumps shall be listed for fire protection services and certified by an independent testing authority. (c) A pump shall not be equipped with more than one driver. (d) Each fire pump shall have its own dedicated driver (e) Each driver shall have its own independent dedicated controller.
Location of Fire Pumps	Reg. 5(51)	(a) Fire pump units shall be protected from surrounding occupancies by a minimum of 2-hour fire-rated construction or physically separated from the protected building by a minimum of 50 ft (15.3 m). (b) The location of and access to the fire pump room shall be approved by the AHJ. (c) Outdoor installations shall be protected against unauthorized entry. (d) Fire pump rooms not directly accessible from the outside shall be accessible through an enclosed passageway from an enclosed stairway or exterior exit. The enclosed passageway shall have a minimum 2-hour fire-resistance rating.
Testing Pressure	Reg. 5(52)	Sprinkler piping are required to be tested hydrostatically at a minimum pressure of either 15 bar or 1.5 times the system pressure and Hydrant/Hose Reels systems for 10 bar or 1.5 times the operating pressure. Combination systems shall be tested as for Hydrant/sprinkler system whichever is higher.
Contractor's certificate	Reg. 5(53)	The installing contractor shall furnish a certificate for flushing and hydrostatic test prior to the start of the fire pump field acceptance test. Both flushing and hydrostatic tests must be documented and witnessed by the AHJ, the project engineer or the commissioning agent.
Electrical equipment testing	Reg. 5(54)	The fire pump controller and its operation must be checked and verified by a Qualified person prior to start up. The pre-acceptance testing can include starting and operating the pump and the pressure maintenance pump and performing other control-related activities, including operating time, alarms, and so on. All electric wiring to the fire pump motor(s), including control (multiple pumps) inter-wiring, normal power supply, alternate power supply where provided, and jockey pump, shall be completed and checked by a Qualified person prior to the initial start-up and acceptance test.

Performance Results	Reg. 5(55)	The fire pump system shall be commissioned and the following parameters tested. (a) Churn: The pump is operating with no flow. (b) Rated capacity: The pump is discharging water at the specified pressure. (c) Overload: The pump is discharging water at a rate of 150% of rated flow at a pressure of 65% of rated pressure.
		FIRE PUMP CONFIGURATIONS
Hydrant Systems-Fire Pump Set	Reg. 5(56)	The Hydrant Pump "Set" shall consist of a "Duty" Pump, a "Stand-by" Pump, and a "Jockey Pump". (a) Jockey Pump - The purpose of the jockey pump is to compensate for a small loss of water pressure and to maintain the system pressure. Flow rate of the jockey pump shall not be less than 60 lit/min or 10 % of the capacity of the duty pump. (b) Duty Pumps - The purpose of the duty pump is to provide the water for all Fire fighting operations at the required pressure and flow rate. This pump shall be capable of delivering not less than 1500 l/min at the system design pressure. When more than one fire riser is installed in a building, the potential need for additional pumping capacity shall be taken in to account. A minimum of 500 l/min shall be add to each riser having more than 03 landing valves. Maximum flow rate shall not exceed 3000 l/min. Where the building is sprinklered the maximum flow rate shall be 2500 l/min. (c) Standby Pumps - The purpose of the stand-by pump is to take over operations immediately in the event the main pump fails for any reason. The capacity of the stand by Pump shall be equal to the main pump. In the event of Super High Rise Buildings, the stand by Pump shall be an engine driven pump or an electric pump with an auxiliary power supply from a generator which will start on a signal received from the pump control panel, when there is a failure of the main power supply. (d) Fire Pumps shall have positive suctions. If this is not feasible, vertical turbine pumps shall be used. (e) Automatic and Manual Operation - The pumps shall start "automatically" on a fall in pressure or demand in water, and once started automatically shall be stopped only manually. Facility for starting the pump manually, overriding the automatic controls, must be available. (f) Pumps in Operation - An audible and visual alarm shall be installed in the control room to indicate that the pumps have started. (g) Draining of the System - Arrangements shall be made to "drain" the wet riser system at a suitable location at a lowest point.

		(h) Excess Pressures and Flows -In order to deal with any excess pressures and flow in the system, a bypass line operated by a pressure relief valve shall be installed where the discharge shall be directed to the fire sump. (i) Each pump shall be capable of providing a flow of at least 1500 l/min at the required design pressure
Sprinkler System- Fire Pump Sets	Reg. 5(57)	The Sprinkler Pump "Set" shall consist of a "Duty" Pump, a "Stand-by" Pump, and a "Jockey Pump (a) Jockey Pump - The purpose of the jockey pump is to compensate for a small loss of water pressure and to maintain the system pressure. Flow rate of the jockey pump should be not less than 30 lit/min or 10 % of the capacity of the main pump (b) Duty Pump - The purpose of the duty pump is to provide the water for the Sprinkler system at the required pressure and flow rate. The pump shall be capable of delivering not less than the minimum design flow rate as per the sprinkler hazard classification. When more than one distribution pipe is installed in a building, the potential need for additional pumping capacity shall be taken in to account (c) Standby Pumps - The purpose of the stand-by pump is to take over operations immediately in the event the main pump fails for any reason. The capacity of the stand by Pump shall be equal to the main pump. In the event of Super High Rise Buildings, the stand by Pump shall be an engine driven pump or an electric pump with an auxiliary power supply from a generator which will start on a signal received from the pump control panel, when there is a failure of the main power supply. (d) Sprinkler pumps shall have positive suction. (e) Automatic and Manual Operation - The pumps shall start "automatically" on a fail in pressure or demand in water, and once started automatically shall be stopped only manually. Facility for starting the pump manually, overriding the automatic controls, must be available. (f) Pumps in Operation - An audible and visual alarm shall be installed in the control room to indicate that the pumps have started. (g) Draining of the System - Arrangements shall be made to "drain" the sprinkler system at a suitable location at a lowest point. (h) Excess Pressures and Flows -In order to deal with any excess pressures and flow in the system, a bypass line operated by a pressure relief valve shall be installed where the discharge shall be directed to the fire sump

		(i) Each pump shall be capable of providing a flow of at least the design flow rate as per hazard classification
Fire Hose Reel Pump Set	Reg. 5(58)	The Fire Hose Reel Pump "Set" shall consist of a "Duty" Pump and a "Stand-by" Pump. The stipulations in relation to the Fire Pump Set is as follows: (a) Duty Pump - The purpose of the duty pump is to provide the water for all Fire fighting operations at the required pressure and flow rate. Each pump shall be capable of delivering not less than 60 l/m at the system design pressure. (b) Stand by Pump - The purpose of the stand-by pump is to take over operations immediately in the event the main pump fails for any reason. (c) Automatic and Manual Operation - The pumps shall start "automatically" on a fall in pressure or demand in water, and once started automatically shall be stopped only manually. Facility for starting the pump manually, overriding the automatic controls, must be available. (d) Pumps in Operation - An audible and visual alarm shall be installed in the control room to indicate that the pumps have started. (e) Draining of the System - Arrangements should be made to "drain" the Hose Reel system at a suitable location at a <u>lowest point</u>.
Power Supply for Fire Pumps	Reg. 5(59)	(a) An electric motor—driven fire pump shall be provided with normal power as a continually available source. (b) The power supply to the fire pump shall be by-passing the main DB of the installation. The stand-by pump shall be either engine driven or a motor driven pump provided with an auxiliary power source by-passing all normal distribution systems with a dedicated auto transfer switch. (c) In the event of failure of the main power (where both Fire pumps are electrically driven) alternative power shall be provided for the operation of the Fire pump. The alternative power should be adequate to start the pump even using emergency start up procedures. (d) Transfer of power to the fire pump controller between the normal supply and one alternate supply shall take place within the pump room, via the ATS with manual overriding facility. (e) Cables laid between the Fire Pump and the main power distributing station must be carried out in 03 Hour Fire resistance cables with adequate mechanical protection or laid inside a 03 hour fire resistant enclosure for the entire length.

Alternate Power Supply.	Reg. 5(60)	On installations with an alternate source of power and an automatic transfer switch, loss of primary source shall be simulated and transfer shall occur while the pump is operating at peak load.
Anti-water-hammer check valve	Reg. 5(61)	Where conditions can be expected to cause objectionable water hammer, a listed anti water- hammer check valve or anti water hammer device shall be installed in the discharge line of the fire pump.
Fuel capacity for engine driven pumps	Reg. 5(62)	Sprinkler system The fuel tank shall contain sufficient fuel for the engine to operate on full load, without having to replenish the fuel for a period of 3 h/ 4h and 6h for light/ ordinary and high hazards respectively. Hydrant system The fuel tank shall contain sufficient fuel for the engine to operate on full load, without having to replenish the fuel for a period of 4 h. In the events the pumps are powered by a generator, same fuel capacities as above, shall apply.
Acceptance Testing, Performance, and Maintenance	Reg. 5(63)	(a) Hydrostatic Test.
		Suction and discharge piping shall be hydrostatically tested at not less than 10 bar for wet riser systems and 15 bar for sprinkler systems or 1.5 times the design pressure whichever is higher. The test pressure shall be maintained for a period not less than 2 hours. The installing contractor shall furnish a certificate for flushing and hydrostatic test prior to the start of the fire pump.
		(b) Field acceptance test
		The pump manufacturer's representative, representative from AHJ and contractor's representative shall be present for the field acceptance test.
		(c) Certified Pump Curve
		A copy of the manufacturer's certified pump test curve shall be available for comparison with the results of the field acceptance test.
		(d) Controller Acceptance Test for Electric and Diesel Driven Units
		Fire pump controllers shall be tested in accordance with the manufacturer's recommended test procedure.
		(e) As a minimum, no fewer than six automatic and six manual operations shall be performed during the acceptance test.

		(f) An electric-driven fire pump shall be operated for a period of at least 05 minutes at rated flow during each of the above operations. (g) The Electrically Driven fire pump shall be started once from each power source and run for a minimum of 10 minutes. (h) The Engine Driven fire pump shall be started and run for a minimum of 30 minutes (i) Facilities shall be provided to test and demonstrate the performance of the fire pump.
Piping material	Reg. 5(64)	(a) Piping material used for Sprinkler installations shall be Steel Piping rated for minimum 15 bar. (b) Non-metallic piping shall not be used for sprinkler installations coming under the categories of ordinary hazard and high hazard. (c) The use of non-metallic fire rated piping shall be restricted to light hazard classification only and for range pipes and branch pipes of diameters 65mm or less.
Exemptions of sprinkler protection	Reg. 5(65)	The following areas are exempted from sprinkler protection in a sprinklered building (a) Canopies/car porches (b) External corridors not exceeding 4m in width provided there is no commercial activities or storage within these areas (c) External open sided link ways not exceeding 5 m in width provided there are no commercial activities or storage within these areas. (d) Toilets and wash rooms
		TOTAL AUTOMATIC FLOODING SYSTEMS
Total gaseous flooding Systems	Reg. 5(66)	Installation of a total gaseous flooding system for property protection shall be considered in addition to the protection system as specified in this document. Such flooding systems shall not compromise the fire protection systems as specified in this document or be considered as substitutes. (a) For occupied places CO₂ shall not be used as the extinguishing gas. (Ex: Data Centres, Archives)
	Reg. 5(67)	(a) In the event a total flooding system is installed to protect special purpose rooms for the use as communication nerve centres, data process centres and process control rooms composing of high value computers or telecommunication equipment, the enclosure to the hazard or occupancy shall comply with the following :

		i. the room shall be constructed to have a minimum of 1-hour fire resistance rating ii. Any door opening shall be protected with minimum 1-hour fire door iii. The total gaseous fire extinguishing system shall use a clean agent. iv. Components of total gaseous flooding systems shall be listed by a recognized testing laboratory.
Automatic Fire detection system	Reg. 5(68)	The total flooding system shall be activated by an automatic fire detection system.
Alarms before discharge	Reg. 5(69)	An audible and visual alarm must be given prior to the discharge of the suppression agent.
System Design	Reg. 5(70)	(a) The system shall be designed by the manufacturer or by a person certified and authorized by the manufacturer. (b) Suppression agents used in occupied areas shall be human friendly (c) Agents used for protecting equipment shall not cause collateral damage to the protected equipment. (d) Selection of the appropriate fire suppression agent shall include consideration of the following items: (1) Potential environmental effect of a fire in the protected area (2) Potential environmental impacts, including, but not limited to, ozone depletion potential (ODP) and global warming potential (GWP) of the clean agents that could be used.
Testing and commissioning	Reg. 5(71)	Testing shall be carried out by a suitably trained and qualified engineer certified and authorized by the manufacturer.
Automatic water mist spray system	Reg. 5(72)	Water mist spray systems shall be used as a flooding system where collateral damage due to water is not a matter of concern. (a) Components of water mist systems shall be listed by a recognized testing laboratory. (h) The system shall be designed by the manufacturer or designed based on the manufacturer's recommendation and should be certified by a Qualified person. (c) Testing shall be carried out as by a competent engineer and approval shall be obtained by the AHJ.
Fire suppression system for kitchen hood	Reg. 5(73)	Any commercial, display or show kitchen which is not separated (compartmented) shall be provided with an automatic fire suppression system for the kitchen hood.

		(a) Fire suppression system for kitchen hood shall cover the entire kitchen hood. (b) Fire suppression system shall be installed in compliance with the manufacturer's specifications and recommendations. (c) Automatic fire suppression system shall have a manual override facility. (d) A suitably sized fire blanket shall be provided for each cooking station. (e) Automatic shutdown of gas supply by the activation of a gas leak detector or activation of kitchen hood fire suppression system shall be provided for each cooking station. (f) A manual isolating valve shall be provided in an easily accessible location close to the cooking station for the main incoming gas supply. If there are more than one gas supply lines each supply line shall be provided with isolation facility. (g) Piping of fire suppression system shall be of noncombustible materials. (h) <u>No damper shall be fitted in kitchen exhaust ducts.</u>
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CHAPTER 6**SPECIAL USES, OCCUPANCIES AND STRUCTURES**

Application	Reg.6(1)	This chapter of the code shall apply only to those buildings types, uses, occupancies and structures specifically regulated herein, and the requirements of this chapter shall be in addition to the general requirements of the other chapters of the regulation governing the size, location, structural fire precautions, means of escape, construction, ventilation, fire-extinguishing equipment and other service equipment of buildings
Occupancies not Regulated by this Chapter	Reg.6(2)	Chemical plants, refineries and similar special occupancies which are not regulated by this chapter shall conform to the requirements of the other chapters of this regulation where applicable and in addition thereto shall be constructed in accordance with the practices and requirements of the particular trade or industry, subject to the approval of the AHJ.
		PLACES OF PUBLIC ASSEMBLY AND RECREATION
Separation of Theatre, Cinema or Concert Hall from Other Parts of the Building	Reg.6(3)	Notwithstanding the provisions of Regulation 3(48) to 3(56), an opening may be permitted in a compartment wall separating a theatre, cinema or concert hall in a part of a building from any other part of the same building which is of a different purpose group, if such opening is provided with A door having the requisite fire resistance as laid down in Regulation 3(69) and such doors shall be held in the open position by a magnetic door holding device connected to the alarm system so that on operation of the alarm the holding device would be de-energized and thus cause the door to be closed automatically, or, be provided with a lobby enclosed by walls having fire resistance of at least one hour, naturally ventilated or mechanically ventilated. All doors to the lobby shall have fire resistance not less than half an hour and fitted with automatic self-closing devices.
Gangways in Places of Public Assembly and Recreation.	Reg.6(4)	Gangways not less than 1050 mm in clear width, shall be provided intersecting the rows of seating in a place of Public Assembly and Recreation in such a manner that no seat shall be at a greater distance than 7 seats from a gangway measured in the line of seating having a minimum seat way clearance of 305 mm. There shall be no projection in to the gangway which would diminish its clear width, and the ends of all rows and seats should be so aligned as to maintain a uniform width of gangway throughout its length.

	Reg.6(5)	An external wall which faces the access way and is windowless or blank-wall shall be provided with access openings at each storey level.
	Reg.6(6)	Gangway not less than 1050mm wide running parallel to the rows of seating in a place of Public Assembly and Recreation shall be provided where required by the AHJ.
Floors	Reg.6(7)	The floor of a balcony or gallery in a place of Public Assembly and Recreation shall have a fire rating not less than that of the structure.
Steps and Slopes of Gangways	Reg.6(8)	Steps shall not be used to overcome differences in level in a gangway in a place of Public Assembly and Recreation unless the slope of such gangway exceeds one in ten.
Handrails	Reg.6(9)	Where steps of a pitch exceeding 30 degrees or ramps of a slope exceeding one in ten are provided in gangways flanking the seating in a place of Public Assembly and Recreation, suitable handrails shall be provided
Treads of Steps of Gangways	Rcg.6(10)	The treads of steps in gangways in a place of Public Assembly and Recreation shall have a non-slip surface and the edges of such steps shall be illuminated so that each step will be clearly visible.
Guardrails	Reg.6(l I)	Along the fascia of a balcony or gallery in a place of Public Assembly and Recreation there shall be provided a guardrail which may be fixed on a parapet wall or otherwise, having a height of not less than 1100 mm above the floor level and where the incline towards the guard rail exceeds 15 degrees, the guard rail shall be not less than 1100mm in height at the feet of any gangway that descends such incline.
Slope of Tiers	Reg.6(12)	The slope of tiers in places of Public Assembly and Recreation shall not exceed 30 degree.
Seats	Reg.6(13)	The seats shall be either fixed to the ground or battened together in numbers not less than seven. No seat shall be more than 4250 mm from an aisle.
Exit Doors External Doors	Reg.6(14)	All doors used by the public as exit door from any part of a place of Public Assembly and Recreation or leading to the open air, shall open only in the direction of exit.
	Reg.6(15)	In a place of Public Assembly and Recreation all exit doors and doors through which the public pass on the way to the open air, shall be without locks, bolts or other fastenings while the public are in the building; provided that floors used for exit only may be fitted with Panic Bolts or Panic Bar complying with the provisions of Regulation 4(8) (d).
	Reg.6(16)	In a place of Public Assembly and Recreation every external door used by the public shall be kept locked in the fully open position when the premises is occupied. If these doors are to be kept locked for security reasons they shall be provided with panic hardware and signage/ instructions for emergency exit.

Stages in places of public assembly and recreation where Theatrical or Live Shows, Pop concerts and similar events are performed.	Reg.6(17)	(a) Areas capable of seating more than 500 persons and extensive stage sets be installed on the stage side, the stage shall be separated from the seating area by proscenium wall of not less than 1h fire resistance. The organizer shall agree the occupant capacity with the AHJ (b) Stairways width shall not be less than 1000 mm and the aggregate capacity of stairways shall be sufficient for the number of people likely to have to use them at the time of fire (c) Where ramps are used the gradient shall be constant and not broken by steps The maximum gradient shall not exceed 1 in 12 (d) Every venue shall be provided with exits which are sufficient for the number of people present in relation to their width, number and siting (e) Where doors have to be kept fastened while people are present , they shall be fastened only by pressure release devices such as panic bolts, panic latches or pressure pads (f) Exit and directional signs shall be illuminated whenever the public are present. Signs at outdoor events shall be weather proof and shall be clearly visible above the audience (g) All parts of the venue to which the public have access and all escape routes shall be provided with normal and emergency lighting
Openings in Proscenium Wall	Reg.6(18)	The proscenium opening shall be protected by a fire curtain with fire resistance of not less than 1h, automatically operated by a fusible link or a smoke detector. A smoke curtain is acceptable, if engineered smoke control and automatic sprinkler system is provided to the stage area. Not more than three openings may be provided in the proscenium wall, not exceeding 2 m² in area fitted with 30 minute fire doors.
Provision of Sprinklers	Reg.6(19)	A sprinkler system shall be provided for the whole of the fire risk area on the stage side of the proscenium wall.
Cinema Projection	Reg.6(20)	Cinematic apparatus using celluloid films in places of Public Assembly and Recreation shall be operated or set up for operation only within approved enclosures.
Rooms using Celluloid films	Reg.6(21)	Any such enclosure used for cinematograph apparatus shall be outside the auditorium and shall be of non-combustible construction having fire resistance for a period of not less than 2 hours. No openings other than projection and observation apertures or ventilation ducts protected by approved type of shutters or fire dampers shall be permitted between the enclosure for projectors and the auditorium.

	Reg.6(22)	Two exits shall be provided to each enclosure for projectors and such exits shall (a) Be outside the auditorium, and (b) Each opening shall be fitted with a self-closing, close-fitting door having a fire resistance of a standard not lower than that of Class A in Table 11
	Reg.6(23)	For each projector used, there shall be provided in the enclosure for projectors- (a) An observation aperture not larger than 1300cm ² , and (b) A projection aperture not larger than 800cm ²
	Reg.6(24)	Each such opening shall be provided with an automated self-closing device activated by a fire detector.
	Reg.6(25)	In a place of Public Assembly and Recreation, where cinematograph apparatus is installed, an enclosure for the purpose of rewinding films and an enclosure for battery installation shall be provided and each such enclosure shall be separated from the enclosure for projectors and from each other by walls having fire resistance for a period of not less than 1 hour.
	Reg.6(26)	All openings, bushes and joints in an enclosure where any cinematograph apparatus is installed shall be so constructed and maintained as to prevent, as far as possible, the escape of any smoke or noxious fumes into the auditorium.
	Reg.6(27)	All enclosures prescribed in this regulation shall be adequately ventilated in accordance with the relevant provisions stipulated by the AHJ. When ventilation of the enclosures is by means of an air conditioning system which also serves the auditorium, the air from the enclosures shall not be re-circulated.
		HOTELS, SHOPPING, OFFICES AND RESIDENTIAL COMPLEXES
Pressurization of Internal Corridors and Staircases	Reg.6(28)	Internal corridors between bedrooms and the internal staircases of a hotel shall be provided with natural ventilation by means of permanent openings, such as fixed louvers, grills, etc. located at a minimum of 900mm above ground level. Such corridors where natural ventilation cannot be provided shall in addition to being mechanically ventilated or air-conditioned, be pressurized so that the pressure within such corridors will be higher than that in the bedrooms and the pressure within the internal staircases higher than that of the corridors.
Walls between Hotel Bedrooms and Corridors	Reg.6(29)	The bedrooms shall be separated from the corridors by walls having a fire resistance rating of at least one hour, which shall be imperforate except for door openings protected by self-closing doors having a fire resistance rating of at least half an hour to ensure the integrity of the protected route or other openings as permitted under Regulation 3(48) to 3(56).

Areas of Special Risk	Reg.6(30)	Boiler rooms, mechanical plant rooms, electrical substations and other areas of special risk within hotel building or a complex development of shopping areas, offices, residential units and the like, shall conform to the provisions of Regulation 6(66) to 6(71).
Rooms using flammable substances	Reg.6(31)	Mechanical ventilation shall consist of exhaust and supply providing a minimum of 20 air-changes per hour. The system shall be independent from those serving other parts of the building Ducts serving other areas shall not pass through rooms involving the use of flammable and explosive substances.
Separation of Kitchen Area from Dining Area	Reg.6(32)	In a hotel, restaurant, or other similar place of Public Assembly and Recreation in which the seating area exceeds 75, the main kitchen area shall be separated from the dining area by a wall having fire resistance of not less than 1 hour and openings in such wall shall be protected by self-closing and close-fitting doors having a fire resistance of not less than 1/2 hour.
Display / show kitchens	Reg.6(33)	Any display or show kitchen which is not separated (compartmented) from the restaurant shall be provided with an automatic fire suppression system.
Exhaust canopy	Reg.6(34)	An exhaust canopy with grease filter shall be provided over the entire cooking range of a kitchen referred to in Reg. 6 (32) and such exhaust shall be fitted with an extractor fan discharging outside the building and away from neighbouring premises. Where the duct passes through any other part of a building, it shall be of a construction having a fire resistance rating of not less than 2 hours or equivalent to that required for the walls or floors through which it passes as specified in Table 8, whichever is the greater. There shall be no damper installed in such ducts. The hood and ducts for the exhaust shall have a clearance of 500mm from unprotected combustible materials. The exhaust shall be discharged directly to the external and shall not be less than 5 m from any air intake openings to ensure that smoke or fumes exhausted will not be drawn in with intake air.
		BASEMENT OCCUPANCIES
Smoke Vents	Reg.6(35)	Where the total aggregate floor area of all semi basements does not exceed 1900 m ² , a building shall be provided with smoke vents openings in well distributed positions in external walls along the street frontages or where they are easily accessible to the Fire Brigade and conforms with the following requirements;
	Reg.6(36)	The number and their sizes shall be such that the aggregate effective vent openings shall not be less than 2.5% of the semi basement floor area served. Such outlets shall be so arranged that a through draught can be created.

	Reg.6(37)	The vent outlets if covered under normal conditions shall be openable in case of fire.
	Reg.6(38)	The position of all smoke vents and the areas they serve shall be suitably indicated on the external face of the building adjacent to such vents.
	Reg.6(40)	Smoke extraction shafts where extending through storeys above, shall be enclosed with imperforate walls having fire resistance for a period of not less than that of the storey served by such shafts.
	Reg.6(41)	Separate smoke extraction shafts and vents shall be provided for each basement storey. In large basement areas where there may be difficult to comply with the provisions of this Regulation, a smoke extract system by mechanical means may be provided to the satisfaction of the AHJ.
Basement Staircase	Reg.6(42)	In addition to being constructed in accordance with the provisions of Regulation 2(3) (k), any staircase which serves a basement storey of a building shall not be made continuous with any other staircase which serves an upper storey of such building or other building and shall be separated from such other staircase by walls having the requisite fire resistance under Regulation 3(24) to :3(30).
Smoke Stop Lobby for Lift Opening	Reg.6(43)	In any basement storey served by a passenger lift, there shall be provided a lobby enclosed by a wall having fire resistance for a period of not less than 1 hour and a self-closing, close fitting, fire resistance door of a standard not lower than Class A in Table 11, directly in front of and adjoining the lift door opening.
Fire Fighting Protected Lobby and Staircase	Reg.6(44)	The entry to all protected staircases serving the basement storeys of a building comprising more than four basement storeys shall be through a smoke free lobby at every such basement storey floor level provided that- (i) When the building has only three or four basement storeys, there need be only one smoke-free lobby adjoining a protected staircase at each floor level thereof for the use for the firemen in the event of a fire, and (ii) When the building has more than four basement storeys, the fire fighting staircase shall be pressurized to the satisfaction of the AHJ.
Rising Mains in Deep Basements	Reg.6(45)	In any basement storey, the floor level of which is 12m or more below the lowest pavement level adjoining the building, there shall be a wet rising main provided in accordance with the applicable parts of Reg. 5(9) to 5(14).
Fire Brigade Foam Inlets	Reg.6(46)	When flammable liquids are used in basement or locations not easily accessible for fire fighting, the following shall be provided. (i) Foam inlets, pipe works and nozzles for the purpose of delivering foam to the area referred above.

		(ii) The piping at ground level with a pipe run of minimum 65 mm bore terminating at the area specified above. (iii) A secondary containment for the storage tank having a capacity of not less than 110% of the fuel capacity. (iv) Fixed foam spreaders as part of a foam pouring system with a Foam inlet box built in to the wall of the building at a point approved by AHJ where it is easily accessible with a pipe run of minimum 65 mm bore terminating at the area referred to above. The box shall be placed at a height of 700 to 1000 mm above ground level and shall be clear of any openings through which heat, smoke or flame can pass. All pipe work shall have a slope towards the spreader. The distance between the inlet and the furthestmost nozzle shall not exceed 18 m. (v) Signage: 'FOAM INLETS' in red lettering, at least 100mm high, on a white background to be displayed on the foam inlet box. (vi) In the event a dedicated Automatic Foam Fire Suppression System is provided, fire brigade foam inlet shall not be required.
Non- Permissible Uses	Reg.6(47)	No part of a basement storey shall be used for the storage of highly flammable liquids or substances of an explosive nature.
		COVERED CAR PARKS AND PARKING GARAGES
Definitions		(a) "Ramp type" means that the building or the part of the building used as a car park is provided with inclined driveways or ramps for driving the cars to various parking levels. (b) "Hoist type" means that the building or part of the building used as a car park is installed with mechanical lift equipment which provides a vertical operation for mechanical parking. (c) "Automated mechanized parking" means a parking system fully automatic in operation and transporting the vehicles both horizontally and vertically. (d) "Parking level" in this regulation means a floor in a building or part of a building on which cars are parked.
Covered Car Parks Distances from Other Buildings or Lot Boundaries	Reg.6(48)	The distances between the external walls of a building or a part of a building that is used for car parking and any other building or the relevant lot boundaries shall be as follows- (a) If the unprotected openings in the external walls of such building or part of the building are in accordance with the provisions of Reg. 3 (33), the clear distance between such

		external walls, and any other building shall not be less than 9 m. (b) If the openings in the external walls of such building or part of the building are protected by an automatic drencher system installed in accordance with an accepted code of practice, the clear distance between such external walls and any other buildings shall be- (i) Not less than 4.5m or (ii) Such that the maximum distance between any part of such building and its lot boundaries (if applicable) as required by the AHJ, whichever is the greater.
Exemptions	Reg.6(49)	This regulation shall not apply to (a) A garage used in connection with a building of Purpose Group 1 (b) and 1(c) for housing or parking motor vehicles kept for private use only, and (b) An open-sided ground floor car park of a building of Purpose Group 2(b) , 2(c) and 8(c) , provided (i) The relevant requirements pertaining to separation of the car park from areas of special risk or of other purpose groups within the same building are complied with, and (ii) Such car park is only used by the residents of such building
Multi-Storey Car Parking Facilities	Reg.6(50)	The car parking facilities in multi-storey car parks may be of either the ramp type, hoist type or automated mechanized type and shall be provided with fire protection systems as specified in this regulation.
Automatic Sprinkler System Requirement	Reg.6(51)	Any parking facility having more than 40 parking slots shall be provided with automatic sprinklers.
Application of Ventilation, Means of Escape and Fire Safety Requirements	Reg.6(52)	For any building or part of a building used for car parking the normal requirements for lighting , ventilation , means of escape, structural fire precautions and fire protection systems under Chapter 2, 3 and 4 of this Regulation shall be applicable except for a building with automated car parking facilities- (a) A reduced scale for calculating the occupant load under Table 3, for whom means of escape facilities shall be provided, may be adopted subject to the approval of the AHJ. (b) Hoist-ways and elevator shafts used solely for the movement of cars from one parking level to another need not be enclosed by protecting structures in accordance with the provisions of Reg. 3(57) to 3(68).

Carbon Monoxide detection for Car parks	Reg.6(53)	(a) All basement and enclosed car parks shall be provided with a carbon monoxide detection system. (b) The carbon monoxide detection system shall be integrated with the basement car park ventilation system and building fire detection system.
Openings in Compartment Floor of a Multi-storeyed Ramp Type Car Park	Reg.6(54)	In addition to the provisions of Reg. 3(48) to 3(56), the ramp openings in the compartment floors of a multi-storeyed car park shall be enclosed by walls having a fire resistance of not less than 2 hours and automatic fire resisting doors or shutters of equivalent fire resistance.
Ramps	Reg.6(55)	Driveways and ramps used as means of escape leading in or out of any garage or car park shall be graded to a slope of not more than 1:8
		VEHICLE REPAIR GARAGES
Separation of a Repair Garage from Other Parts of Buildings used for any other Purpose	Reg.6(56)	A repair garage shall not be located within or attached to a building used for any purpose other than a garage unless completely separated by compartment floors or walls having fire resistance for a period of not less than - (i) 3 hours or, (ii) As required under Reg.3(24) to 3(30) and Table 8, whichever is the greater. Provided that a garage in a basement storey of a building shall be separated from any other part of the building which is used for any purpose other than a garage, by compartment walls, and floors having fire resistance of not less than 4 hours.
Separation within a Repair Garage	Reg.6(57)	The motor repair areas in a repair garage shall be separated from the parking areas, showroom, office and other such areas by walls or floors having fire resistance for a period of not less than- (i) 2 hours, or (ii) As required under Reg. 3(24) to 3(30) and Table 8, whichever is the greater Provided that the motor repair areas in a repair garage that is located in a basement storey of a building, shall be segregated from the parking areas, showroom and other ancillary areas by walls and floors having a fire resistance of not less than 4 hours.
Separation of Spray Painting Booth	Reg.6(58)	Areas in which spray painting or other allied processes are performed or carried out, if contained in the main building, shall adjoin external walls and shall be completely separated from the other areas of the buildings by walls and floors or cover slabs having a fire resistance of not less than 2 hours. The doors in such wall shall have a fire resistance of not less than 2 hours and shall be self-closing or so installed as to close automatically in case of fire.

		Where spray painting booths that have built-in vapour extraction system the fire resistance requirement is not applicable.
Ventilation of Spray-painting Booths	Reg.6(59)	Spray-painting booths, drying and baking areas shall be well ventilated with mechanical exhaust fans fitted approximately 1m above floor level and suitably arranged air inlets communicating directly with the external air.
Minimum Number of Exits for a Garage	Reg.6(60)	Notwithstanding the provisions of Regulation 2 (12) every building or part of a building used as a garage shall be provided with at least 2 exits conforming to the applicable requirements of Chapter 2 of this regulation.
Drying or Baking Oven	Reg.6(61)	A drying or baking oven shall be provided with any of the following- (a) Explosion relief hatches, which shall be of light construction and readily breakable, (b) Permanent openings having an effective area of not less than 0.1m² for each 0.3m³ of oven capacity, or (c) An oven top which is loose in its entirety being held in its place by its weight only.
	Reg.6(62)	No less than 2 doors shall be provided for each oven
	Reg.6(63)	A drying or baking oven shall be so ventilated as to prevent the accumulation of inflammable gases or vapors.
	Reg.6(64)	A drying or baking oven shall- (a) Not be in contact with any combustible materials, and (b) Be located on floors constructed of fire resisting material
		AREAS OF SPECIAL RISK IN ANY BUILDING
Separation and Permissible Openings	Reg.6(65)	A boiler room, generator or transformer room or any other area of special risk in any building shall be separated from the other parts of the building by a compartment having fire resistance for a period of not less than 4 hours notwithstanding the provision of Reg.3(24) to 3(30) and Table 8. Such compartment wall or compartment floor shall be imperforated except for the following- (a) Openings in the wall fitted with doors having fire resistance for a period of not less than that of the compartment, or (b) Openings in the floor for protected shafts which shall conform to the requirements of Reg. 3(57) to 3(68), or (c) Openings for ventilation ducts which shall comply with the requirements of Reg. 3(53), or (d) Openings for pipes which shall comply with the requirements of Reg. 3 (48) (d).

Areas of Special Risk in Basements	Reg.6(66)	Where the areas of special risk are located in basement storeys, the additional requirements of Reg. 6(46) shall also be complied with.
Fuel Storage	Reg.6(67)	(a) For boiler rooms or generator rooms that use flammable liquid as fuel, the amount of fuel that is allowed to be stored within the same compartments shall not be more than the requirement of 8 hour operation and such storage shall be provided with a secondary containment tank having a volume of not less than 110% of the storage capacity. Any fuel in excess of this quantity shall be stored in separate compartments in conformance with the requirements of Reg. 6(67), and if such fuel qualifies as "Petroleum" under the provisions of the Petroleum Act, the storage compartments shall also conform to the requirements of the Petroleum Ordinance Regulations, or if such fuel does not qualify as "petroleum", then the storage compartment shall conform to the requirements of Reg. 6 (72) (b) Generators and associated fuel supplies located in buildings other than in basement or ground floor level shall be protected by a an automatic fire suppression system.
Storage Areas in Factories	Reg.6(68)	Factories processing combustible materials shall be provided with proper storage areas for raw materials, waste materials and finished products, and such areas shall be separated from the production areas and be of adequate size in accordance with the practices and requirements of the particular trade or industry, subject to the approval of the AHJ. Storage areas inside buildings shall be of non-combustible construction, having a fire rating not less than 1 h. Storage areas shall be separated from any building and other areas by walls and/or floors of fire resistance not less than 1 h. No single storage compartment shall be larger than 2000 m² in floor area, unless fitted with automatic fire extinguishing systems. Flammable liquids with flash points below 32 degree Celsius shall not be stored within the factory building. Such material shall be stored in separate well ventilated single story building made of non-combustible material and situated not less than 15 meters away from any building. Storage of flammable gases shall be separated from all other materials and shall have adequate low and high-level ventilation, as relevant to the material stored. There shall be a space of at least 1 meter between the ceiling and the top of goods stored and clear space on all sides of stacks and racks to form gangways not less than 1 meter wide. Storage areas shall always be kept tidy with no waste or other litter lying around.

Storage Areas in Departmental Stores and Shops	Reg.6(69)	Departmental stores or shops shall be provided with proper storage areas and such areas shall be separated from the sales, display and other stores and public areas and escape routes, with compartments of minimum 1 h fire rating.																																		
When Storage Areas are to be Treated as Areas of Special Risk	Reg.6(70)	When the storage areas referred to in 6(68) and 6(69) are required for the storage of combustible materials or goods, they shall be treated as areas of special risk and the provisions of 6(65) and 6(69) of this Regulation shall be applicable.																																		
		LIQUIFIED PETROLEUM GAS - STORAGE, SAFE USE, AND SALE																																		
Siting of Gas Piping Above ground Tanks and Equipment	Reg.6(71)	Siting of GAS PIPING Tanks and Equipment																																		
		(a) The separation distance of tanks with respect to boundaries, buildings, and fixed sources of ignition shall be as stipulated in Table given below:																																		
		<table><tr><th>Of any single vessel in a group (Litres)</th><th>Tons</th><th>Separati^{on} Distance (Boundaries, Buildings etc)</th><th>With Fire Wall</th><th>Between Vessels</th></tr><tr><td>150 to 500</td><td>0.05 to 0.25</td><td>2.5 m</td><td>0.3 m</td><td>1 m</td></tr><tr><td>>500 to 2500</td><td>0.25 to 1.1</td><td>3 m</td><td>1.5 m</td><td>1 m</td></tr><tr><td>>2500 to 9000</td><td>1.1 to 4</td><td>7.5 m</td><td>4 m</td><td>1 m</td></tr><tr><td>>9000 to 135000</td><td>4 to 60</td><td>15 m</td><td>7.5m</td><td>1.5 m</td></tr><tr><td>>135000 to 337500</td><td>60 to 150</td><td>22.5 m</td><td>11 m</td><td rowspan="2">1/4 of sum of the dia. of 2 adjacent vessels</td></tr><tr><td>>337,500</td><td>150</td><td>30 m</td><td>15 m</td></tr></table>	Of any single vessel in a group (Litres)	Tons	Separati ^{on} Distance (Boundaries, Buildings etc)	With Fire Wall	Between Vessels	150 to 500	0.05 to 0.25	2.5 m	0.3 m	1 m	>500 to 2500	0.25 to 1.1	3 m	1.5 m	1 m	>2500 to 9000	1.1 to 4	7.5 m	4 m	1 m	>9000 to 135000	4 to 60	15 m	7.5m	1.5 m	>135000 to 337500	60 to 150	22.5 m	11 m	1/4 of sum of the dia. of 2 adjacent vessels	>337,500	150	30 m	15 m
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Reg.6(72)	(a) The ground providing standing for the tank shall be compacted or paved with suitable material. The area should be kept free from weeds, grass, and combustible materials.																																			
	(b) Gas piping, storage vessels pumps, etc., shall not be located directly beneath electrical power and telephone cables. Vessels shall not be sited less than 1.5 m from a plane drawn vertically downwards from the cables.																																			
	(c) It shall be sited away from possible enclosed areas where natural dissipation of any escaped gas could not take place.																																			

Fire Walls	Reg.6(73)	The purpose of the Fire Wall is to protect the tank from thermal radiation from a fire nearby, and to ensure an adequate dispersion distance to boundaries, buildings, etc. (a) Fire Walls shall be imperforate, and of solid masonry, concrete, or similar construction. (b) Fire Walls shall be not less than 2 metres high, or as high as the top of the tank whichever is the greater, and shall be sited between 1 metre and 3 metres from the nearest point of the tank. (c) With the provision of a fire wall the separation distance may be reduced to the values given in Reg. 6(71). A fire wall shall be provided on any side/s of the separation distance of a vessel or group of vessels where a risk of thermal radiation exists.
Underground LPG Tanks	Reg.6(74)	The tank shall be, (a) Placed on a concrete base strong enough to carry the weight of the tank and its contents, and the cavity be backfilled with sand. (b) The tank shall be coated against corrosion. (c) All pipe work shall be taken from the top of the tank. (d) Manhole of the tank shall be minimum 15m away from all the openings of adjacent buildings.
Minimum underground LPG tank separation distances	Reg.6(75)	(a) Distance to air in-take for air conditioning and ventilation system: 6 m (b) Distance to building openings, doors, shafts and windows: 3m (c) Distance to any point of the nearest structure or building: tank capacity 0 — 7500 litre - 3 m tank capacity above 7501 litre - 15 m
Fire Protection for external above ground storage LPG Tanks	Reg.6(76)	A manually operated water deluge system shall be installed to protect the tank surface area and the piping.

Fire Fighting Equipment	Reg.6(77)	The following fire-fighting equipment shall be provided: (a) Two numbers 9 kg. Dry Chemical Powder portable fire extinguishers, sited such that they are protected from inclement weather. (b) One numbers 19 mm diameter x 30 metre length Hose Reel, with a jet and spray nozzle capable of providing at least 30 lpm output and a jet throw of at least 6 metres. The fire water sump capacity shall be not less than 2000 litres. (c) An automatic water spray system shall be installed to provide exposure cover for the entire surface area of the tank. The application rate shall be 10.2 lpm per square metre of total tank surface area for at least 60 minutes. The above system shall be provided with an independent pump set comprising duty and standby pumps, of adequate capacity and pressure head to meet the requirements as specified. The standby pump shall comply with Reg. 5(57). (d) An access road of 4 metre wide hard standing and 4 metre clearance height, for a 15 ton fire appliance shall be provided.
		INSTALLATION OF GAS PIPING AND OTHER FUEL GAS PIPING SYSTEMS AND APPLIANCES
Installation of Gas Piping Systems and Appliances	Reg.6(78)	Any building having residential apartments more than 16 in number or, is more than ground plus two storeys, or has commercial gas operated cooking stations, gas operated incinerators, laboratories with multiple gas operated burners shall be required to install a centralized Gas Piping distribution system. The use of individual Gas Cylinders is not permitted. The installation of Gas Systems and Appliances shall comply with the following regulations. The Gas Piping system shall be designed by a qualified and competent Engineer.
Piping Plan	Reg.6(79)	Before proceeding with the installation of a Gas Piping system, a piping plan shall be prepared showing the proposed location of piping, the size of the different branches, etc. for approval by the Authority having Jurisdiction. The installation of the system shall be carried out by a Contractor/ Agency registered with CIDA. The Gas Piping installation shall be carried out in accordance with national or international standards applicable to such installations and acceptable to the AHJ. Liquid petroleum gas used as fuel shall never enter into any building in liquid form. Gas shall not enter residential apartment with a pressure higher than 28 mbar. A mechanically or electrically operated automatic shutoff valve shall be provided to shut off the gas supply automatically when the gas supply rate increases above 1.2 kg/h.

Safety Mechanisms	Reg.6(80)	The following safety mechanisms shall be installed. (a) Pressure Relief devices shall be installed and shall be located so that the discharge from the container pressure relief device is at least 1.5 metres horizontally away from any building opening below the level of such discharge and 1.5 metres in any direction from any source of ignition. (b) "Emergency Shutoff Valves" shall be installed, and shall incorporate all of the following means of closing: (i) Automatic shutoff through a gas leak detector or a fire detector. (ii) Manual shutoff from a safe remote location. (iii) Manual shutoff at the installed location. (c) "Shutoff Valves", readily accessible, shall be installed immediately downstream of the Gas Piping container end, and upstream of the entry point of the pipe into the building, and at any other locations that would facilitate quick manual shutoff of the gas in an emergency. (d) Automatic Shutoff Valves actuated by a gas leak detector shall be installed in the kitchen of every apartment, such shut off devices shall have only manual re-set facility. (e) Gas leak detectors suitably spaced shall be installed in dedicated risers where central gas piping is located to activate automatic isolating valves. Where gas piping is located above false ceilings gas leak detectors shall be installed as necessary (f) Power supply to the automatic shutoff valve in any apartment/ kitchen shall be the same electrical circuit supplying power to the Gas leak detector of the apartment/ kitchen. If this circuit is failed or switched off, the automatic valve shall be shutoff. (g) In apartments with central gas piping where gas leak detectors are installed a remote indicator shall be provided at the main entrance and in the Fire Command Centre (h) All gas leak detectors shall provide a supervisory audible alarm locally and shall integrated with the fire alarm system of the building. (I) For high rise and super high rise buildings, the gas leak detection system shall be of the addressable type having a separate addressable detection and alarm panel. This shall be installed in the fire command centre near the main fire alarm panel of the building. The two panels shall be interfaced. When the main fire alarm panel detect fire, the main gas solenoid valve shall be set to close automatically.
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		I (j) Gas detectors shall have in built Power indicator, Fault indicator, Leak indicator, Test button and Alarm sounder. "Electrical Bonding and Grounding." Each above ground portion of a Gas Piping system upstream from the equipment shutoff valve shall be electrically continuous and bonded to any grounded electrode. Gas Piping shall not be <u>used as a grounding electrode</u>
		LPG STORAGE IN RETAIL OUTLETS
LPG Storage in Retail Outlets	Reg.6(81)	The "storage" of LPG at "Retail Outlets" shall conform to the following requirements. Location-Cylinders shall not be located near exits, stairways, or in areas normally used, or intended to be used for the safe egress of occupants. Storage - The "storage" of LPG Cylinders shall comply with the following stipulations: (a) Cylinder storage shall be in such a manner as to minimize exposure to excessive temperature rises, physical damage, or tampering. (b) Cylinders shall be stored "upright" with the valves uppermost to avoid possible leakage. (c) Cylinders shall not be stored within 3 metres from a store containing oxygen cylinders or highly flammable liquids. (d) Cylinders shall preferably be stored in the open air with a security fence, and not within 1 metre from other buildings or the boundary fence. (e) If the LPG store is to be a part of a building used for other purposes, the store shall be: (i) In the ground storey in a room against an outside wall, with a door or doors leading to the outside. (ii) The store shall be provided with high and low level ventilation facilities. (iii) The store shall be entirely of non-combustible construction. (iv) The store shall be separated from other parts of the building by a fire resisting compartment of not less than 2 hours fire resistance. (f) Protection of Residence — Any residential parts of a building in which LPG is stored should be protected with a well maintained fire alarm system which triggers automatically on detection of an LPG leak by the LPG detector and is audible in every room in the building. (g) Explosion relief - This should be located in an outside wall or the roof with discharge to a safe place outside the building. It should give explosion relief equivalent to half the area of one of the walls and make use of light weight panels or industrial wire mesh that will not disintegrate <u>during an explosion</u>

Signage	Reg.6(82)	The storage area and rooms shall be clearly marked, "Highly Flammable," in red lettering on a white background. Smoking, Naked Lights, and Cellular Telephones shall be prohibited.
Fire Protection	Reg.6(83)	Two numbers 9kg. Dry Chemical Powder portable fire extinguishers with an adequate BC fire rating shall be provided. The extinguishers should be located not more than 15 metres from the risk.
		STORAGE OF PETROLEUM IN UNDERGROUND TANKS AND PUMPS
Storage of Petroleum in Underground Tanks and Pumps	Reg.6(84)	Storage of Petroleum Fuel in Underground Fuel Storage Tanks and Pumps (Filling Stations.) The storage of petroleum fuel in underground tanks and the pumps for the discharge of the fuel at "filling stations" shall comply with the following stipulations.
Conditions Applicable to the Site	Reg.6(85)	The following stipulations must be complied with in relation to the "site." (a) The location of the pumps and the tanker stand shall be as far as possible on level ground. (b) Every petroleum fuel storage tank shall be installed below ground with appropriate environmental protecting methods (c) Petroleum dispensing pumps and any openings to a petroleum fuel storage tank shall be installed not less than 4.3 metres from any building, boundary, or access route used by the public on the premises. (d) Adequate means of escape shall be provided from all parts of the premises in the event of a fire.
Vent Pipes	Reg.6(86)	The following stipulations must be complied with in relation to "vent pipes." (a) Each tank or tank compartment shall have a separate vent pipe of not less than 50 mm in diameter. (b) The upper end of the pipe shall discharge upwards into the open air, should be out of reach, should be not less than 1.5 metres above ground level, and should not be located less than 1.5 metres of windows or other openings. (c) Electrical fittings within 1.5 metres in any direction of the vent discharge shall be to Zone 1 Standard. Elsewhere within 1.5 metres of the vent riser pipe Zone 2 Standard fittings shall be used. Note : Zone Details Zone 0 An area in which an explosion gas-air mixture is continuously present, or present for long periods; Zone 1 An area in which an explosive gas-air mixture is likely to occur in normal operation; and

		Zone 2 An area in which an explosive gas-air mixture is not likely to occur in normal operation, and if it occurs it will exist only for a short time. By implication an area which is not classified as zone 0,1 or 2 is deemed to be non-hazardous or safe with respect to the selection of electrical apparatus.
Electrical Installations	Reg.6(87)	The following stipulations must be complied with in relation to electrical installations. (a) Electrical ducts must not communicate with ducts or manholes containing fuel lines or tank openings. (b) Electrical chambers should be filled with sand and covered with a screed of sand and cement. (c) Electrical cables shall not be laid adjacent to or across metal pipe work unless run in a conduit of non-conducting material
Illuminations	Reg.6(88)	The filling area of the forecourt shall be provided with a general illumination of intensity less than 110 lux when measured at the top of each pump.
Fire Fighting Equipment	Reg.6(89)	The following fire fighting equipment shall be provided: (a) At filling stations with up to 4 pumps: 2 x 9 litre portable foam fire extinguishers, and 2 x 9 kg. dry chemical powder portable fire extinguishers, and 2 x 9 litre water portable fire extinguishers. (b) At filling stations with 5 to 8 pumps: 3 x 9 litre foam portable fire extinguishers, and 2 x 6 kg. dry chemical powder portable fire extinguishers, and 2 x 9 litre water portable fire extinguishers. (c) For each additional 3 filling pumps, 1 additional portable fire extinguisher of each type.
Notices	Reg.6(90)	The following notices shall be prominently displayed in the vicinity of the pumps. "No Smoking", "Switch off Engines", and Cellular Phones." The notice shall be on a white background in block red letters of 50 mm height. The notice shall not display any advertisements, etc
Physical Protection	Reg.6(91)	Heavy guard rails shall be installed for the protection of the filling Pumps. The guard rails shall provide protection to a height of not less than 1.2 metres.

CHAPTER 7**ACCESSIBILITY FOR FIRE FIGHTING**

Purpose of this Chapter is to ensure provision for external access to building for fire fighting and accessibility of fire fighting appliances.

Introduction	Reg. 7(1)	Access way shall be provided for accessibility to site for fire fighting appliances. To permit fire-fighting appliances to be deployed, the access way shall have a minimum width of 4.5m and height of 4.5 m throughout its entire length. Access openings shall be provided along the external walls of buildings fronting the hard standing to provide access into the building for fire-fighting and rescue operations. Access way shall be provided to within minimum visible distance of 18m of breeching inlet for buildings that exceed the habitable height of 18m.
Hard standing requirement	Reg. 7(2)	(a) In any residential building if the height of the habitable floor does not exceed 18m no hard standing is required other than a clear working space 4m x 4 m wide, along the side of the building where access opening is located. (b) In buildings where habitable floor exceeds 18m, hard standing to be provided to accommodate the entry and manoeuvring of Fire Engines, Aerial Appliances etc. i. The minimum width of the hard standing shall be 7m wide and the minimum length shall be 15m long. Other parts of the access way for fire appliances shall be not less than 4.5m wide. ii. Hard standing shall be positioned so that the nearer edge shall be not less than 2m or more than 10m from the centre position of the access opening. iii. The hard standing shall be constructed to sustain the stationary load of 40 tonnes. Hard standing shall be laid on a level platform or if on an incline, the gradient shall not exceed 1:15. Access way may be laid on an incline not exceeding a gradient of 1:8:3. iv. Overhead clearance of hard standing or access way shall be at least 4.5 m and the outer radius for turning of access way shall be not less than 10.5 m. 1 (c) Public roads may serve as hard standing provided the location complies with Reg. 7(2) b) i. (d) In the case of basement the fire engine access road shall be provided within a travel distance of 18m to the entrance of all exit staircases that are provided with landin: valve. The

		measurement of 18m shall be between the fire engine access road and the entrance of exit staircase. (e) The podium of a building where used, shall not restrict or impede the reach of the boom of the fire fighting appliance, when parked on the designated hard standing.																								
Appliance accessibility to buildings	Reg. 7(3)	In the case of any building that exceeds total floor area given in the table shall conform to the vehicle access requirement in such table. <table><tr><th>Total Floor Area (m²)¹</th><th>Height of Floor of top storey above ground (m)²</th><th>Provide vehicle access to</th><th>Type of Appliance</th></tr><tr><td>Up to 2000</td><td>Up to 18 Over 18</td><td>See Reg.7.2 (a) 15% of perimeter</td><td>Pump High reach</td></tr><tr><td>2000 -8000</td><td>Up to 18 Over 18</td><td>15% of perimeter 50% of perimeter</td><td>Pump High reach</td></tr><tr><td>8000 -16000</td><td>Up to 18 Over 18</td><td>50% of perimeter 50% of perimeter</td><td>Pump High Reach</td></tr><tr><td>16000 - 24000</td><td>Up to 18 Over 18</td><td>75% of perimeter 75% of perimeter</td><td>Pump High Reach</td></tr><tr><td>Over 24000</td><td>Up to 18 Over 18</td><td>100% of perimeter 100% of perimeter</td><td>Pump High Reach</td></tr></table> Notes 1. The total floor area is the aggregate of all the floors in the Building. 2. In the case of storage buildings, height should be measured to mean roof level. (a) For buildings under Purpose Group I, access way will not be required, provided that the buildings are not more than 45m from a roadway, and the height of a building shall not exceed 9m. However, in cluster housing developments, (Cluster housing is landed housing with shared communal facilities), fire engine access road with a minimum within a travel distance of 45m from every point on the projected plan area of any building in the housing development. (b) The outer radius for turning of access way and fire engine access road shall comply with the requirements as shown in Diagram 2 A and 3A. (c) Overhead clearance of access way and fire engine access road shall be at least 4.5m for passage of fire fighting appliances.	Total Floor Area (m ²) ¹	Height of Floor of top storey above ground (m) ²	Provide vehicle access to	Type of Appliance	Up to 2000	Up to 18 Over 18	See Reg.7.2 (a) 15% of perimeter	Pump High reach	2000 -8000	Up to 18 Over 18	15% of perimeter 50% of perimeter	Pump High reach	8000 -16000	Up to 18 Over 18	50% of perimeter 50% of perimeter	Pump High Reach	16000 - 24000	Up to 18 Over 18	75% of perimeter 75% of perimeter	Pump High Reach	Over 24000	Up to 18 Over 18	100% of perimeter 100% of perimeter	Pump High Reach
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Over 24000	Up to 18 Over 18	100% of perimeter 100% of perimeter	Pump High Reach																							

		(d) Public roads can serve as access way provided the location of such public roads is in compliance with the requirements of distance from access openings. (e) Access way shall be laid on a level platform or if on an incline, the gradient shall not exceed 1:15. Access road shall be laid on an incline not exceeding a gradient of 1:8.3. (f) The access way shall have a minimum width of 6m throughout. Such access way must be able to accommodate the entry and manoeuvring of fire engine, extended ladders pumping appliances, turntable and/or hydraulic platforms. (g) For buildings under Purpose Group II that exceed the habitable height of 18m, fire engine access road shall be provided within a travel distance of 18m to the entrance of all exit staircases where the landing valves are provided. The fire engine access road shall have a minimum 4.5m width and designed to sustain the load of stationary 40 tonnes fire engine fighting appliance, and shall be provided within 18m of breeching inlets of the building. The breeching inlets shall be located at the exterior, readily visible and accessible from the fire engine access road. (h) Dead end access way and fire engine access road shall not exceed 20 m in length or if exceeding 20 m, be provided with turning facilities. (i) Access way and fire engine access road shall be kept clear of obstructions and other parts of the building, plants, trees or other fixtures shall not obstruct the path between access way and access openings.
Access openings to building for fire fighting	Reg. 7(4)	Access opening shall include unobstructed external wall openings, windows, balcony doors, glazed wall panels or access panels must be readily openable from the inside or outside unless fitted with breakable glazing
Access Opening	Reg. 7(5)	(a) Inside and outside of access openings shall be unobstructed at all times during the occupancy of the building. Access panels shall be placed at habitable spaces and not be located at protected staircases or lobbies (b) An external wall which faces the access way and is windowless or blank-wall shall be provided with access openings at each storey level.
Signage	Reg. 7(6)	Access openings shall be posted with red triangle (150mm on each side) on the external side of the wall with wordings 'Fire Fighting Access- Do Not Obstruct' of at least 25mm height in the internal side

Size of Access openings	Reg. 7(7)	Access openings shall be not less than 850mm wide by 1000mm high with sill height of not more than 1100mm and head height not less than 1800mm above the inside floor level. It shall be placed against an occupied space. Access openings shall be remote from each other and located along the side of the building. Such access openings shall be spaced at not more than 20m apart measured along the internal wall from centre to centre of the access openings.
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CHAPTER 8

INSPECTION MAINTENANCE AND TESTING OF ACTIVE / PASSIVE FIRE PROTECTION FACILITIES

The purpose of this Chapter is to ensure that owner/occupier will "inspect, maintain and test" their Active and Passive Fire Protection Facilities as listed below, but not limited to, in a proper operational condition meeting the intended use and as stipulated in the regulation at all times. In the event that there are multiple occupants occupying the same complex, the owner, the condominium management corporation and/or the organization appointed to maintain and upkeep the building shall be responsible for the above.

Systems to be inspected, maintained and tested	Reg. 8(1)	The following shall be subjected to regular inspection and testing. a) Fire Detection and Voice Evacuation Systems b) Fire Protection and Suppression Systems c) The Means of Escape d) Smoke Control Systems e) Stairway "pressurization" systems f) Portable Fire Extinguishers g) Exit Signage h) Emergency Lighting i) Total Flooding Systems j) Fuel Gas Leak Detection Systems k) Fire Brigade Access Routes l) Emergency Access Openings m) Fire Water Storage
Inspection and Maintenance of Systems	Reg. 8(2)	All systems shall be subjected to Routine Inspection and Maintenance, Periodic Maintenance and Inspection by AHJ. (a) Routine Inspection & Maintenance - This involves a visual check" of the complete facility to ensure that the facility is in good operating condition. (b) Periodic Inspection & Maintenance - This involves the systematic and periodic functional and operational checks and tests to ensure that the facility is in a reliable and efficient operating condition. This shall be carried out by Qualified Persons having access to the necessary special tooling and equipment and trained and certified in the use of same, required for such activities. (c) Inspection by AHJ - All buildings falling under all Purpose Groups except 1(b) and 1(c) shall be subjected to an annual inspection and issuance of a certificate.

Frequency of Inspection and Maintenance	Reg. 8(3)	(a) Routine Inspection and Maintenance —Minimum once a month. (b) Periodic Inspection and Maintenance - Once in six (06) months or as per manufacturer's recommendation whichever is shorter (c) Inspection by AHJ - Once a year
Servicing of Portable Fire Extinguishers	Reg. 8(4)	(a) Routine Inspection & Maintenance - Once a month. (b) Periodic Inspection & Maintenance - Once in six (06) months or as per manufacturer's recommendation whichever is shorter (c) Inspection by AI-IJ - Once a year (d) Maintenance Intervals (Portable Fire Extinguishers) (i) Water & water based - Basic Service —Every 12 months - Extended Service — Every 5 years (ii) Dry chemical powder - Basic service — every 12 months - Extended service — every 5 years (iii) Carbon dioxide - Basic service — every 12 months - Overhaul - every 10 years Extended service shall be carried out 5 years of the date of commissioning or 6 years from the date of manufacture whichever is sooner Servicing intervals may be shortened on the recommendation of the competent person or at the request of the responsible person Replacement of parts should not affect these intervals Intervals for carbon dioxide extinguishers require that the stamped date of manufacture or last overhaul date
Testing of Active Fire Systems	Reg. 8(5)	(a) All active Fire Systems stated in Reg. 8 (1) shall be tested for its functionality and ensure that they perform to their intended purpose. (b) Testing has to be carried out by a trained competent person. (c) Testing methodology shall be in accordance with manufacturer's recommendations and to comply with stipulated fire regulations.

Frequency of Testing of Fire Systems	Reg. 8(6)	(a) Fire Detection & Fire alarm System (i) Weekly testing by the user. (ii) Monthly testing by the user. (iii) Testing by competent person every six months. (iv) Annual Testing by a competent person and certified by AHJ. (b) Fire Protection Systems (i) Monthly testing by the user. (ii) Testing by competent person every six months. (iii) Annual Testing by a competent person and certified by AHJ (c) Sprinkler Systems (i) Pressure switch type water flow alarms/ Mechanical water flow alarm devices including, but not limited to water motor gongs, shall be tested quarterly (ii) Annual Testing by a competent person and certified by AHJ. (d) Automatic Fire suppression system (i) Inspection and functional test by specialist manufacturer trained person every six months. (ii) Annual Testing by a competent person and certified by AHJ. (e) Fire Pumps (i) Diesel engine driven fire pumps shall be operated bi-weekly for a minimum duration of 30 minutes. Electric motor driven pumps shall be operated monthly for duration of 15 minutes. (ii) Tests shall be carried out to ensure that the system controls are operating in good order. (iii) Annual Testing by a competent person and certified by AHJ (f) Fire Door Assembly (i) Fire door operating mechanisms and assemblies shall be operated and tested and recorded monthly by the user. (ii) Fire door operating mechanisms and assemblies shall be operated and tested annually for functionality and a written record of the inspection shall be signed and kept for inspection by the AHJ
Responsible Person for Fire Systems in <u>buildings</u>	Reg. 8 (7)	Any Building coming under all-purpose groups except, 1(b) and 1(c) shall have a person responsible for Active and Passive Fire Protection Systems in the building.

		Such person shall be appointed by the property owner or the operator or the occupier or the Condominium Management Corporation. Such person will be responsible for maintaining of fire systems in good and proper working condition and conducting fire risk assessments. Liabilities arising from malfunctioning of active and passive fire protection systems in buildings shall remain with the property owner or the operator or the occupier or the Condominium Management Corporation, as applicable.
Servicing contract	Reg. 8(8)	Responsible person of the Fire Service Installations (FSI) shall employ a registered contractor to inspect FSI of their buildings at least once every 12 months
Availability of Spares	Reg. 8(9)	Responsible person shall ensure that necessary spares are available for any replacement of any damage or unserviceable equipment.
Documentation	Reg. 8(10)	(a) Routine Inspection and Maintenance activities shall be recorded and such records shall be available for inspection by AHJ on request. (b) A certificate of compliance shall be issued by the certified inspector on successful completion of the periodic inspection. The owner/ occupier shall retain such certificate for inspection by AHJ when required.

DIAGRAMS

1. CONCEPT ILLUSTRATIONS OF ESCAPE ROUTE DISTANCES

DIAGRAM I — (A —

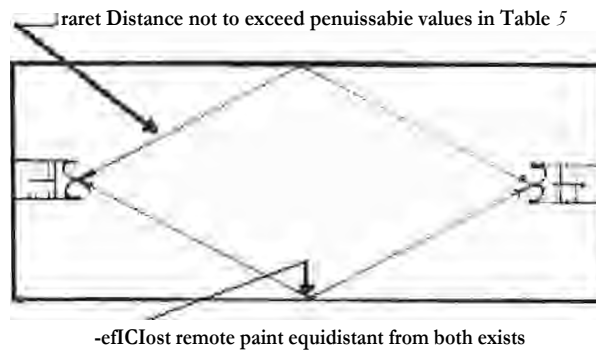


DIAGRAM I A (in reference to Reg. 2 (c) (i))
 Illustrating application of Travel Distance elements to 2 exits)

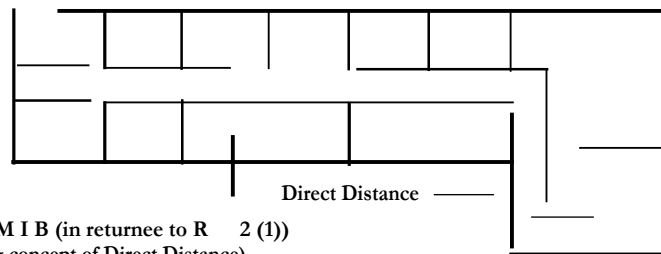


DIAGRAM I B (in reference to Reg. 2 (1))
 Illustrating concept of Direct Distance
 Note In this concept internal walls and partitions are ignored

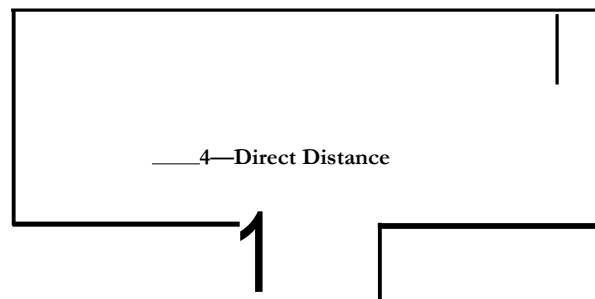


DIAGRAM I C (in reference to Reg. 2 (1))
 Illustrating concept of Direct Distance

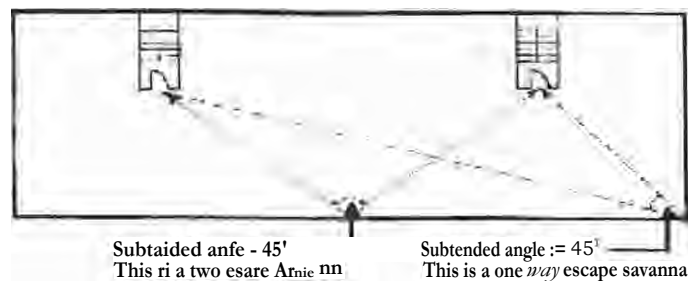
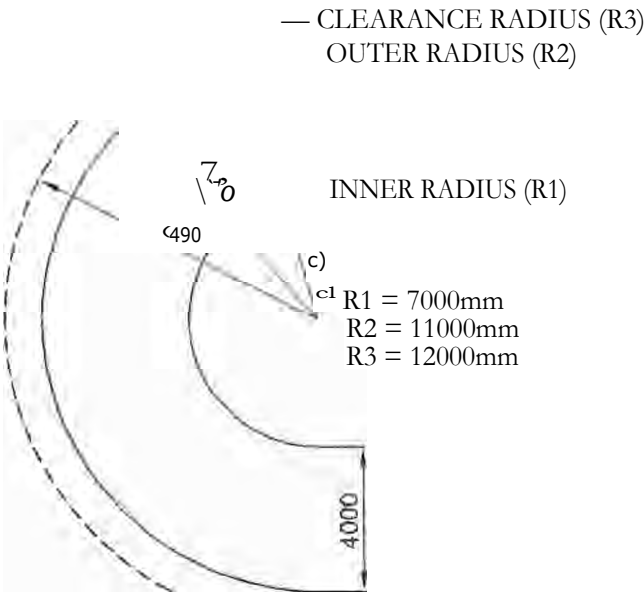


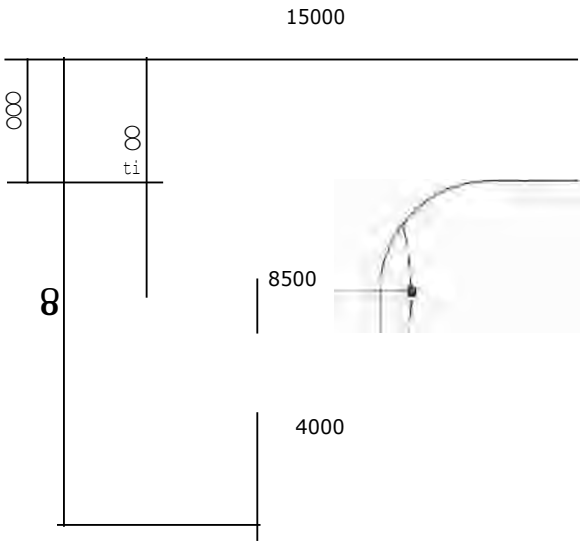
DIAGRAM I D (in reference to Reg. 2 (m))
 Illustrating concepts of one way or two way escapes

2. CLEARANCE RADII

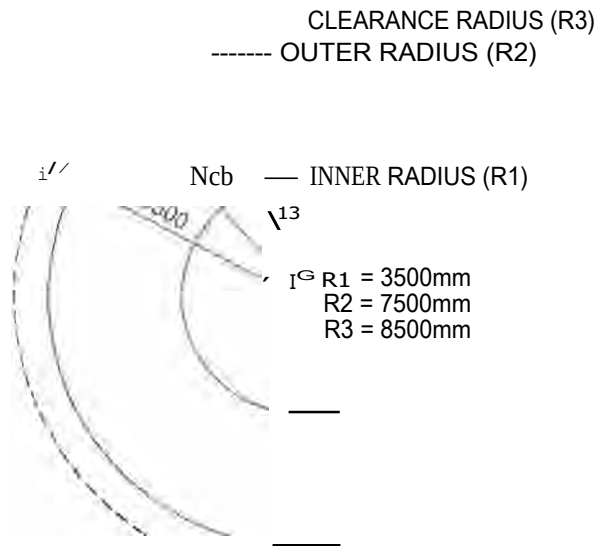
2.1 Diagram 2(A) U-Turn Radii of Pumper Appliances (24 tonnes)



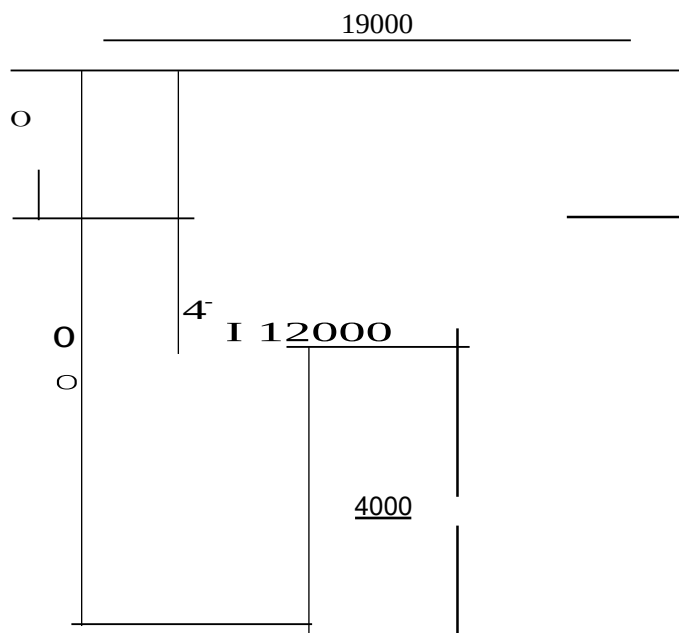
2.2 Diagram 2(B) Turning Facilities for Pumper Appliances (24 tonnes)



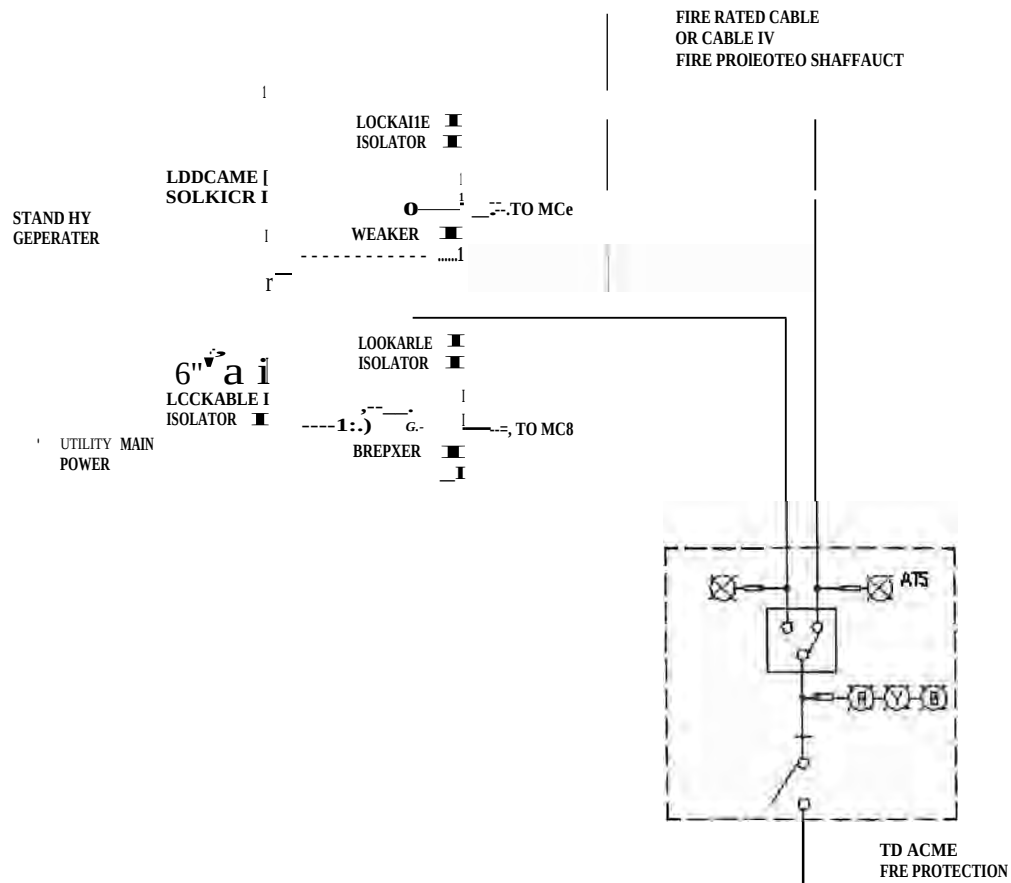
3.1 Diagram 3(A) U-Turn Radii of Aerial Appliances (30 tonnes)
Applicable to buildings exceeding the habitable height of 10m



3.2 Diagram 3(B) Turning Facilities for Aerial Appliances (30tonnes)
Applicable to buildings exceeding the habitable height of 10m



3. POWER SUPPLY TO FIRE PROTECTION SYSTEMS



TYPICAL ARRANGEMENT OF POWER SUPPLY TO ACTIVE FIRE PROTECTION SYSTEMS

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TABLE 1
Abbreviations and Symbols

Abbreviation or Symbol	Definition
BS	British Standard
NFPA	National Fire Protection Association
C	Celsius
CP	Code of Practice
dB	Decibel
o	Degree
kg	Kilogram
kN	Kilo Newton
W	Watt
kW	Kilowatt
m	Meter
m ²	Square Meter
m ³	Cubic Meter
mm	Millimeter
mm ²	Square Millimeter
min	Minute
N	Newton
Reg.	Regulation
UDA	Urban Development Authority

TABLE 2
Classification of Purpose Groups

Title	Group	Purpose for which the Building or Compartment is intended to be used
1. Residential (Dwelling)	1(a)	Residential Apartments or flats: Having multiple residencies in a single block of building, using common facilities.
	1(b)	Residential Unit : A unit having single or multiple residencies in a single block of building exceeding 300 sqm or the habitable floor height from ground exceeding 6m and not exceeding 18m.
	1(c)	Single Residential House: A unit not exceeding 300 sqm and the habitable floor not exceeding 6m.
2. Residential (Institutional)	2 (a)	Establishments used for treatment, care or maintenance of persons suffering from disabilities, or educational purposes and accommodations, including hospitals, clinics, polyclinics orphanages, children's homes, elder's homes , day-care centers, infant care, kindergartens.
Residential (Other)	2(b)	Hotel
	2(c)	student hostels, dormitories, army camps, schools, colleges, commercial schools, vocational institutions and universities
	2(d)	Detention & correction centers
3. Office	3	Offices or premises used for administration, clerical work, handling money, communications, recording or performance.
4. Commercial	4(a)	Underground shopping centers
	4(b)	Shops or premises used for super markets, department stores, food courts, restaurants, bars & pubs, betting halls, show rooms, total floor area exceeding 800 sqm.
	4(c)	Shops or premises used for retail trade hire or repair, restaurants, bars, pubs, salons, total floor area not exceeding 800 sqm.

5. Place of public Assembly and Recreation	5(a) 5(b) 5(c) 5(d) 5(e)	Underground subway terminals. Cinemas, theatres, Exhibition centers, stadium, sports complexes, private clubs (night clubs), casinos, public halls. Passenger terminals, airports, bus terminals, railway stations. Colleges, universities, class rooms, libraries. Conference centers, gymnasias, skating rinks, churches.
6. Industrial / Factories	6(a) 6(b) 6(c) 6(d)	Flammable/hazardous production & processing facilities. Manufacturing & processing, power generation, commercial kitchen/laundries/ laboratories. Mechanical & electrical plant rooms. IT equipment room, server rooms, air traffic control towers.
7. Storage	7 (a) 7 (b) 7(c)	Warehouse located in Basements. Place for the storage of goods or materials other than 7 (a) and any building not within groups 1-6. Container terminals, freight stations, logistic centers, general goods
8. Car parks	8(a) 8(b) 8(c)	Underground car parks and above ground enclosed car parks Automated mechanized car parks Parking structures above ground , naturally ventilated car parks.

TABLE 3

Occupancy Density Factor

Purpose Group	Types of Accommodation	Areas of occupied floor space per person (m ²)
1. Residential (Dwellings) 1 (a)	Flats including service apartments	Refer note
2. Residential (Institutional) 2(a)	Hotel — Guest room Restaurant Lobby Dancing hall / Frankfurt hall Hostel Colleges, universities, class rooms, libraries	Refer Note 03 12 01 08 02
2(b)	Hospital — General ward Room ward Channeling floor Elders' home Nursing home	04 15 03 10 20
2(c)	Detention Centre Correction Centre	03 05
3. Office	Administrative office Clerical work — witting, book-keeping etc. Handling money — banking etc. Communication — Postal	03 04 05 03
4. Commercial 4(a) > 800m ²	Shops, supermarkets and departmental stores Food courts Restaurants, pubs and bars, Betting halls Show rooms	04 03 02 08
4(b) < 800m ²	Shops for retail trades Hire or repair Restaurants, pubs and bars Salons	02 04 1.5 03
5. Assembly and Recreation 5(a)	Cinemas, theatres, stadium Private clubs(night clubs) Casinos, public halls, Sports complex	02 01 02
5(b)	Passenger terminals, bus terminals, railway stations Airports	01 05
5(c)	Conference centers, churches Gymnasias, skating rings	01 03
6. Industrial 6(a)	Flammable/hazardous production & processing facilities	10
6(b)	Manufacturing & processing Power generation Commercial kitchen/laundries/ laboratories	03 15 05
6(c)	Mechanical & electrical plant room, IT equipment room, server room	10

7. Storage 7(a)	Container terminals, freight stations, logistic centers	30
7(b)	Place for the storage of goods or materials other than 7 (a)	20
8. Car Parks 8(a)	Underground car parks	30
8(b)	Automated car parks, enclosed car parks	-
8(c)	Garages, car parks above ground	20

Note :1. In Residential (Dwellings); 1(a) with one apartment unit is taken as 04 persons

2. In residential (Institutional); 2(a) Hotel Guest Room taken as 02 persons.

TABLE 4
Type of Occupancy

Type of Occupancy	Maximum Occupant Load with one Door
Go downs, storage buildings	50
Shops, departmental stores and business premises	75
Factory buildings without high hazards	50
Place of public resort	75
Educational buildings	75
Hospitals	15
Hotels, boarding houses, hostels, residential block etc.	20
Patient accommodation area	50
Class rooms	50
Assembly	50

Any area falling under high hazard occupancy shall have a minimum of two doors irrespective of the occupancy.

TABLE 5 - Occupancy Characteristics and Means of Escape Requirements**Table 5 (1) -Occupancy characteristics**

Occupancy Characteris-tic	Description	Examples
A	Occupants who are awake and familiar with building	Office and industrial premises
B	Occupants who are awake and unfamiliar with the building	Shops, exhibitions, museums, leisure centers, other assembly buildings, etc
C	Occupants who are likely to be asleep:	
Ci ^{*(1)}	Long-term individual occupancy	Individual flats without 24 h maintenance and management control on site
Cii ^{*(2)}	• Long-term managed occupancy	Serviced flats, halls of residence, sleeping areas or boarding schools
Ciii	• Short-term occupancy	Hotels
D ^{*(2)}	Occupants receiving medical care	Hospitals, residential care facilities ^{*(3)}

^{*(1)} Categories Ci and Cii are included for completeness within this table but are covered in more depth in BS 9991.

^{*(2)} Currently occupancy characteristics **D** medical care is dealt with in other documentation and is outside the scope of this British Standard.

^{*(3)} Under some circumstances residential care facilities may be classified as occupancy characteristics Cii.

Table 5 (H) -Risk Profile

Occupancy characteristic	Fire growth rate	Risk profile
A (Occupants who are awake and familiar with the building)	1 Slow	A1
	2 Medium	A2
	3 Fast	A3
	4 Ultra-fast	A4 ^{*(1)}
B (Occupants who are awake and unfamiliar with the building)	1 Slow	B1
	2 Medium	B2
	3 Fast	B3
	4 Ultra-fast	B4 ^{*(1)}
C (Occupants who are likely to be asleep)	1 Slow	C1
	2 Medium	C2 ^{*(2)}
	3 Fast	C3 ^{*(2), *(3)}
	4 Ultra-fast	C4 ^{*(1), *(2)}
^{*(1)} These categories are unacceptable within the scope of BS9999. Addition of an effective localized suppression system or sprinklers will reduce the fire growth rate and consequently change the category. ^{*(2)} Risk profile C may be divided into sub-categories. ^{*(3)} Risk profile C3 is unacceptable under many circumstances unless special precautions are taken.		

Table 5 (III) - Fire Growth Rate

Category	Fire growth rate	Fire growth Parameter kJ/s ³	Description	Typical examples
1	Slow	0.003	Evenly distributed low level fire load small discrete packets of fuel or material of limited combustibility	Reception areas concourses (without concession outlets) and halls with limited fire load such as sports stadia and foyers
2	Medium	0.012	Evenly distributed low to mid-level fire load comprising a mix of combustible materials	Offices, lounges, classrooms, auditoria, seating areas, galleries and car parks
3	Fast	0.047	Stacked combustibles (on or off racking and shelving but excluding high rack storage), some small quantities of materials other than materials of limited combustibility (or where larger quantities are stored in separate fire resisting enclosures), process, manufacturing or storage of combustible materials	Shop sales areas, workshops, factories and small storage buildings
4 G	Ultra-fast	0.188	Medium to large quantities of materials other than materials of limited combustibility, high racked storage, flammable liquids and gases or where	Warehousing, processing plants and car parks, utilizing a car stacker or similar method where there is no fire separation between stacked cars.

Table 5 (IV) - Maximum Travel Distance When Minimum Fire Protection Measures are Provided

Occupancy Characteristics	Risk Profile	Fire Growth Rate	Travel Distance in metres (m)	
			Two-way Travel	One-way Travel
A Occupants who are awake and familiar with the building	A1	Slow	65	26
	A2	Medium	55	22
	A3	Fast	45	18
	A4	Ultra-fast	Not applicable (t)	Not applicable (*)
B Occupants who are awake and unfamiliar with the building	B1	Slow	60	24
	B2	Medium	50	20
	B3	Fast	40	16
	B4	Ultra-fast	Not applicable (*)	Not applicable (*)
C Occupants who are likely to be asleep	C1	Slow	27	13
	C2	Medium	18	9
	C3	Fast	14	7
	C4	Ultra-fast	Not applicable (*)	Not applicable (*)
Where exact travel distances are not known, direct distances should be taken as two thirds of the travel distance.				
<i>Note : Where premises contain provisions for the consumption of alcoholic beverages then a reduction in the travel distances of 25% might be advisable for those particular parts of the premises.</i>				
<i>ω These categories are unacceptable within the scope of this regulation. Addition of an effective localized suppression system or sprinkler system will reduce the fire growth rate and consequently change the category.</i>				

Table 5 (V) - Absolute Minimum Width of Stairs

Occupancy characteristic	Dimensions in millimeters	
	Width of stair for downward travel	Width of stair for upward travel
A	1 000	1 200
B (except assembly)	1 000	1 200
B (assembly only)	1 100	1 200
C	1 000	1 200

Table 5 (VI) -Exit widths when minimum fire protection measures are provided

Refer this table in conjunction with Reg. 2(9)

Risk profile	Minimum exit width per person mm
A 1	3.3
A2	3.6
A3	4.6
A4	Not applicable
B 1	3.6
B2	4.1
B 3	6.0
B4	Not applicable
C1	3.6
C2	4.1
C3	6.0
C4	Not applicable

Table 5 (VII) - Minimum width of escape stairs for simultaneous evacuation

Risk Profile	Minimum width of stair per person served over total number of floors served									
	i floor	2 floors	3 floors	4 floors	5 floors	6 floors	7 floors	8 floors	9 floors	10+floors
A1	3.90	3.40	2.95	2.45	2.15	2.00	1.80	1.70	1.50	1.40
A2	4.50	3.80	3.25	2.75	2.45	2.20	2.00	1.90	1.70	1.60
A3	5.40	4.60	4.00	3.50	3.10	2.80	2.60	2.30	2.10	2.00
A4	-	-	-	-	-	-	-	-	-	-
B1	4.20	3.60	3.10	2.60	2.30	2.10	1.90	1.80	1.60	1.50
B2	4.80	4.00	3.40	2.90	2.60	2.30	2.10	2.00	1.80	1.70
B3	7.00	6.00	5.30	4.60	4.20	3.70	3.40	3.10	2.80	2.60
B4	-	-	-	-	-	-	-	-	-	-
C1	4.20	3.60	3.10	2.60	2.30	2.10	1.90	1.80	1.60	1.50
C2	4.80	4.00	3.40	2.90	2.60	2.30	2.10	2.00	1.80	1.70
C3	7.00	6.00	5.30	4.60	4.20	3.70	3.40	3.10	2.80	2.60
C4	-	-	-	-	-	-	-	-	-	-

Note : The widths of stairs have been calculated on the assumption that all floors are evacuating simultaneously.

This is conservative, as the occupants on the fire floor are likely to move more quickly than on the other floors.

Table 5 (VIII) - Dead-end Travel Distance

Purpose Group	Types of Premises	Protected Corridor / Passage	Balcony
1 (a), (b)	Residential - Apartment or Flat	15 m	
2 (b), (c)	Hotel, Hostel, School, Dormitory, Vocational Institution, Military	15 m	24 m
2 (a)	Hospital, Healthcare and Child care facility, Elder's home,	12 m	24 m
4, 5, 6, 7	Office, Commercial, Public Assembly, Industrial, Storage	18 m	24 m

TABLE 6**Extent of Compartmentation**

Compartments	Maximum Floor Area (m ²)	Maximum Cubic Extent (m ³)
Below ground level - Compartment consisting of not more than one story	2000	7500
If sprinklered	4000	11000
Between ground and a height of 30m. - Compartment consisting of not more than 3 storeys.	4000	15000
If sprinklered	8000	22000
More than 30 m from ground level - Compartment consisting of not more than one story	2000	7500
If sprinklered	4000	11000

TABLE 7

Multi-storey Industrial and Storage Buildings (Purpose Group 6 and 7)

Compartments	Floor Area of Compartment (m ²)
Height above ground level	
Not Sprinklered	
Not more than 18 m	7000
More than 18 m	2000
Sprinklered	
Not more than 18 m	14000
More than 18 m	4000
Basement - Sprinklered	3000

Note :

Compartment volumes in excess of those given above shall be considered based on individual merits giving full consideration to the performance of equivalent safety standards available in the provision of enhanced means of escape, means of access for fire fighting, fire service installations and improved fire resistance of the structure.

TABLE8**Minimum Period of Fire Resistance**

Minimum Period of Fire Resistance						
	Minimum Periods (minutes) for Elements of Structure in a:					
	Basement storey (refer Note 1, below) including floor over		Ground or upper storey			
	Depth (m) of a Lower basement		Height (m) of top floor above ground, in building or separated part of building			
Purpose Group	More than 9	Not more than 9	Not more than 18	Not more than 30	Not more than 60	More than 60
Residential						
1 (a)	120	90	60	90 (ref.3)	120(ref 3)	120
1(b) & (c)	NR	90(ref.2)	60	NR	NR	NR
2 (a)	120	90	60	90	120(ref 5)	120
2 (b)	120	90	60	90	120(ref 5)	120
2 (c)	NR	NR	60	NR	NR	NR
3 Office	120	90	60	90	120(ref.5)	120
4 Shop/Commercial						
4(a)	120	90	NR	NR	NR	NR
4(b)	120	90	60	90	120	Note 7
4(c)	120	90	60	60	NP	NP
5 Place of public Assembly and Recreation						
5(a)	120	90	NR	NR	NR	NR
5(b)	120	90	60	90	120	120
5(c)	120	90	60	NR	NR	NR
5(d)	120	90	60	90	120	NR
5(e)	120	90	60	90	120	NR

6 . Industrial / Factories	120	90	60	90	120	NP
7 Storage						
Any building or part not described elsewhere	120	90	60	90	120	NP
8. Car Parks						
8(a)	120	90	NA	NA	NA	NA
8(b)	120	90	60	90	120	NA
8(c)	NA	NA	60	90	120	NA

Notes

NR — not relevant

NP — not permitted

NA — not applicable

Sprinklered means that the building is fitted throughout with an automatic sprinkler system meeting the relevant recommendations of BS 5306: Part 2; ie the relevant occupancy rating together with the additional requirements for life safety.

Notes

- (1) *The floor over a basement (or if there is more than one basement, the floor over the topmost basement) should meet the provisions for the ground and upper storeys that period is higher.*
- (2) *Increased to a minimum of 60 minutes for compartment walls separating buildings.*
- (3) *Reduced to 60 minutes for any floor within a maisonette, but not if the floor contributes to the support of the building.*
- (4) *Multi-storey hospitals should have a minimum 60 minutes standard*
- (5) *Reduced to 90 minutes for elements not forming part of the structural frame.*
- (6) *Increased to 30 minutes for elements protecting the means of escape.*
- (7) *Special Approval shall be obtained from*
- (8) *30 minutes in the case of three storey dwelling houses, increased to 60 minutes' minimum for compartment walls separating buildings*
- (9) *The car park should comply with the requirements for open sided car parks specified in approved Document B Section 11.*

TABLE 9**Table to Regulation 3(40) and 3 (41)**

Provision as to method of test and minimum period of fire resistance				
Part of Building	Method of Test	Minimum period as to		
		BS 476		
		Stability Integrity Insulation.		
		BS 476		
(1)	(2)	Collapse (3)	Passage (4)	Insulation (5)
1. External wall situated exposure of 1m more from relevant boundary (excluding any part of such a wall which is described in item 2.	exposure of inside of wall to test by fire.	*	*	15 min
2. External wall situated less than 1m from any point on relevant boundary. 3. Separating wall. 4. Compartment wall. 5. Structure (other than an external wall) enclosing protected shaft. 6. Structure referred to in Regulation 21 (2) (a).	exposure of each side of structure separately to test by fire.	*	*	*
7. Compartment floor.	exposure of underside of floor to test by fire.	*		*
8. Floor of upper storey in building of purpose group 1 which has 2 storeys	exposure of underside of floor to test by fire	30 min	15 min	15 min
9. Door other than a door described in item 10 or 11.	exposure to test by fire when fitted in its frame.	*	*	No requirement
10. Door referred to in both regulation 19(5) and regulation 17(1) (a) (i) or 21 (2) (b)	exposure to test by fire when fitted in its frame.	30 min	20 min	No requirement
11. Door referred to in both regulation 19 (6) and regulation 18(7)(a) or 18(7)(b).	exposure to test by fire when fitted in any rebated frame	30 min	30 min	No requirement

* Denotes 'period of fire resistance specified'

TABLE 10**Unprotected Areas in any Side of a Building**

Minimum Distance between side of building and relevant boundary in meters	Maximum Total Percentage of Unprotected Area
1	4
2	8
5	20
10	40
15	60
20	80
25	100%

TABLE 11

Classification of Fire Resisting Doors

Specifications: Construction and Material		Class of Door
.	Solid hardwood timber not less than 45 mm finished thickness.	A
2.	Laminated timber core solid throughout covered both sides with ply facing and with all edges protected with a solid timber capping covering the full thickness of the core and facings, the total thickness of door not less than 45 mm.	A
3.	Timber stiles and top and bottom rails not less than 96 mm wide and a middle rail not less than 165 mm wide rebated to receive 9.5 mm plasterboard in filling on both sides strengthened by 45 mm wide intermediate rails, the whole covered both sides with plywood facings, the total thickness of door not less than 45 mm.	A
4.	Doors tested in accordance with the provisions of Section 7 of British Standard "Fire Test on Building Materials and Structures" BS 476 and certified as being capable of resisting the action of fire for 1/2 hour. Class A doors shall be hung to open in one direction only in solid timber or metal frames with rebates or door stops not less than 13mm deep, centre opening doors may be in two leaves each hung to the frame with buti-jointed meeting stiles where only minimum clearance is allowed at the meeting edges. Provided that Class A doors which open into a corridor or lobby with enclosures capable of resisting the action of fire for a period not less than 1/2 hour, may open in 2 directions without rebated frames provided minimum clearance only is allowed at the meeting edge. Glazing fixed shut may be incorporated in a Class A door if it is capable of resisting the action of fire for a period of 1/2 hour, or if it comprises a single vision panel of 6.35 mm thick clear glass not exceeding 650 cm² in area in a solid timber frame at least 45 mm and 45 mm clear of rebates.	A
5.	Doors in accordance with paragraph (1), (2) or (3) of this table, but covered completely on both sides either externally or beneath plywood facings with suitable fire resisting insulating board not less than 4.75 mm in thickness.	B
6.	Doors tested in accordance with the provision of Section 7 of the British Standard BS 476 and certified as being capable of resisting the action of fire for 1 hour. Class B doors shall be hung to open in one direction only, shall not exceed 910 mm width or 2,130 mm in height and shall be hung in solid timber or metal frames (not aluminium) with rebates or door stops not less than 25 mm deep and having a width at right angles to the face of the door or not less than 35mm.	B
	A panel of glazing not exceeding 152 mm wide by 300 mm high may be incorporated in a Class B door if it is filled with glass reinforced with wire and capable of resisting the action of fire for a period of 1 hour.	B

Notes : (a) 1/2 hr fire rating shall be class A

(b) 1 hr fire rating shall be class B

(c) The word "timber" (wearing in this Table shall mean "hardwood" as defined in Chapter I.

(d) BS EN 1634-1 ; 2000 Fire Resistance tests -fire doors and shutter assemblies

TABLE 12

**Classification of Surfaces of Walls and Ceilings Relating to
Flame Spread**

		Class of surface for both walls and ceilings (except where separately specified).		
Purpose group of building of compartment (1)	Maximum floor area of small room (in m ²) (2)	Small rooms (see column) (2) (3)	Rooms other than small rooms. (4)	Circulation spaces and protected shafts. (5)
(i) Small residential- (House having not more than two storeys.)	4	3	(Wall) 1 (Ceiling) 3	(Wall) 1 (Ceiling) 3
Any other house	4	3	1	0
(ii) Other residential	4	3	1	0
(iii) Institutional	4	1	(Wall) 0 (Ceiling) 1	0
(iv) Office	30	3	1	0
(v) Shop	30	3	1	0
(vi) Factory	30	3	1	0
(vii) Place of public resort	30	3	1	0
(viii) Storage and general	30	3	1	0

TABLE 13

Materials for Roofing and Roof Framing Permitted for use with Ceilings of Flame Spreads as in Table 12	
Materials for Roofing	
(1)	Cement Tiles
(2)	Clay Tiles
(3)	Metal
(4)	Concrete
Materials for Roof Framing	
(1)	Concrete
(2)	Metal
(3)	Timber

TABLE 14

Provision of Manual Call Points, Automatic Fire Detection, Voice Evacuation and Fireman Communication systems According to Purpose Group, Height and Size of the building.

Purpose Group	Height (Ground - Highest Habitable Floor Level)	Manual Call Points	Fire Detection	Voice Evacuation	
1. Residential (Dwellings) 1(a)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	NR R R R	NR NR R R	NR NR NR R	NR NR NR NR
1 b)	Height up to 18m	NR	NR	NR	
1(c)		NR	NR	NR	
2. Residential 2(a)(Institutional)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	NR R R R	NR R R R	NR NR NR R	
2(b) (Hotels)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	NR R R R	NR R R R	NR NR R R	NR NR NR NR
2,(c) (Hostels)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	NR R R R	NR R R R	NR NR NR R	NR NR NR NR
2(d) (Detention)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ²	NR R	NR R	NR NR	NR NR
3. Office	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	NR R R R	NR R R R	NR NR NR R	NR NR NR NR
4. Commercial 4(a) (Underground)	> 800m ²	R	R	R	NR
4(h) (Shops)	< 800m ² > 800m ² Height between 18m - 30m > 800m ² Height more than 30m	NR R R	NR R R	NR NR R	NR NR NR
4(c) (Shops)	< 800m ²	NR	NR	NR	NR
5. Assembly and Recreation 5(a)(Underground)	Floor area less than 800 m ² Floor area more than 800 m ²	NR R	NR R	NR R	NR NR
5(b) (Cinemas)	Floor area less than 800 m ² Floor area more than 800 m ²	NR R	NR R	NR R	NR NR

5(c) (Passenger Terminal)	Open Buildings	R	NR	NR	NR
	Enclosed above ground more than 800 m ²	R	R	R	NR
	Enclosed above ground more than 800 m ² (airports)	R	R	R	R
	Underground more than 400 m ²	R	R	R	R
5(d) Colleges	Naturally ventilated buildings less than 800m ²	NR	NR	NR	NR
	Naturally ventilated buildings more than 800m ² or more than 18m height	R	R	R	NR
	Enclosed Buildings less than 800 m ²	NR	NR	NR	NR
	Enclosed buildings more than 800m ² or more than 18m height	R	R	R	NR
5(e) Conf. Halls	Floor area less than 800 m ²	R	NR	NR	NR
	Floor area more than 800m ² or more than 18m height	R	R	R	NR *
6. Industrial 6(a) (Flammable)	Floor area less than 800 m ²	R	NR	NR	NR
	Floor area more than 800 m ²	R	R	R	NR*
6(b) (Power Generation)		R	R	R	NR*
6(c) (M&E Plant rooms)		R	R	R	NR*
6(d) (IT facility)		R	R	NR	NR
7. Storage 7(a), 7(b) ,7(c)	Basements	R	R	NR	NR
	Floor area less than 800 m ²	NR	NR	NR	NR
	Floor area more than 800 m ²	R	R	NR	NR*
8. Car Parks 8(a)	Underground	R	R	R	R
	Automated mechanized car park	NR	R	NR	NR
8(c)	Parking structures above ground/Naturally ventilated	R	R	NR	NR

Notes :

- (1) If an automatic detection system exceed 100 or more detectors addressable system shall be applicable.
- (2) The above are guidelines only. These shall be enhanced after Fire Risk Assessment is carried out on the building. AHJ reserves the right to revise the above requirements based on the risk assessment
- * - Designer should take into account the Potential Fire risk and fire spread and decide on the type of detection system .

TABLE 15**Minimum Sizes of Fire Extinguishers for Class A Fire Hazards**

Minimum Extinguisher Rating	Maximum Travel Distance to Extinguishers	Area to be protected per Extinguisher		
		Low Hazard Occupancy	Ordinary Hazard Occupancy	High Hazard Occupancy
3A	30m	30m ²	15m ²	8m ²
5A	30m	50m ²	25m ²	15m ²
8A	30m	80m ²	40m ²	20m ²
13A	30m	130m ²	65m ²	35m ²
21A	30m	210m ²	105m ²	55m ²
27A	30m	270m ²	135m ²	70m ²
34A	30m	340m ²	170m ²	85m ²
43A	30m	430m ²	215m ²	110m ²
55A	30m	550m ²	275m ²	140m ²

TABLE 16**Minimum Sizes of Fire Extinguishers for Class B Fire Hazards**

Type of Hazard	Basic Minimum Extinguisher Rating	Maximum Travel Distance to Extinguishers
Light	34B	15m
	144B	25m
Ordinary	144B	10m
	183B	15m
High	183B	7m
	233B	10m

TABLE 17

Provision of Portable Fire Extinguishers, Hose Reel, Landing Valve, Sprinkler according to Purpose Group, Height and Size of the Building

Purpose Group	Height (Ground - Highest Habitable Floor Level)	Portable Fire Extinguishers	Hose Reels	Landing valves	Sprinklers
I. Residential (Dwellings)					
1(a)	Height up to 18 m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	R R R R	NR R R R	NR NR R R	NR NR NR R
1(b)	Height up to 18m	R	NR	NR	NR
1(c)		NR	NR	NR	NR
2. Residential					
2(a)(Institutional)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	R R R R	NR R R R	NR NR R R	NR NR NR R
2(b) (Hotels)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	R R R R	NR R R R	NR NR R R	NR NR NR* R
2(c) (Hostels)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	R R R R	NR R R R	NR NR R R	NR NR NR R
2(d) (Detention)	Height up to 18m & floor area less than 800 m ² Height up to 18m & floor area more than 800 m ²	R R	NR R	NR R	NR NR
3. Office	Height up to 18m & floor area less than Height up to 18m & floor area more than 800 m ² Height between 18m - 30m Height more than 30m	R R R R	NR R R R	NR NR R R	NR NR NR R
4. Commercial					
4(a) (Underground)	> 800m	R	R	R	R
4(b) (Shops)	< 800m ² > 800m ² Height between 18m - 30m > 800m ² Height more than 30m	R R R	NR R R	NR R R	NR NR R
4(c) (Shops)	< 800m ²	R	NR	NR	NR
5. Assembly and Recreation					
5(a)(Underground)	Floor area less than 800 m ² Floor area more than 800 m ²	R R	R R	NR R	NR NR
5(b) (Cinemas)	Floor area less than 800 m ² Floor area more than 800 m ²	R R	NR R	NR R	NR R
5(c) (Passenger Terminal)	Open Buildings Enclosed above ground more than 800 m ² Enclosed above ground more than 800 m ² (airports) Underground more than 400 m ²	R R R R	NR R R R	NR R R R	NR NR R R

5(d) Colleges	Naturally ventilated buildings less than 800m ²	R	NR	NR	NR
	Naturally ventilated buildings more than 800m ² or more than 18m height	R	R	R	NR *
	Enclosed Buildings less than 800 m ²	R	NR	NR	NR
	Enclosed buildings more than 800m ² or more than 18m height	R	R	R	NR*
5(e) Conf. Halls	Floor area less than 800 m	R	NR	NR	NR
	Floor area more than 800m ² or more than 18m height		R	R	R*
6. Industrial 6(a) (Flammable)	Floor area less than 800 m ²	R	NR*	NR	NR*
	Floor area more than 800 m ²	R	R*	R*	R*
6(b) (Power Generation)		R			NR*
6(c) (M&E Plant rooms)		R	R*	R*	
6(d) (IT facility)	Refer note 2 below	R			
7. Storage 7(a), 7(b) ,7(c)	Basements	R	R	R	R*
	Floor area less than 800 m ²	R	NR	NR	NR
	Floor area more than 800 m ²	R	R	R	R*
8. Car Parks 8(a)	Underground	R	R	R	R*
8(b)	Automated mechanized car park	R	NR	NR	R**
8(c)	Parking structures above ground/Naturally ventilated	R	R	R	

Notes :

⁽¹⁾ The above are guidelines only. These shall be enhanced after Fire Risk Assessment is carried out on the building. AHJ reserves the right to revise the above requirements based on the risk assessment.

Consider use of automatic gaseous fire suppression systems

Designer should take into account the Potential Fire risk and fire spread and decide on the type of protection system

*** Intermediate sprinklers may be required

TABLE 18
Classification of Typical hazards For Sprinkler Installations

Tables A1, A.2 and A.3 contain lists of minimum hazard classification. They shall also be used as guidance for occupancies not specifically mentioned.

Table 18.1
Light Hazard occupancies

Schools and other Educational Institutions, Offices, Churches, Clubs, Conference Rooms, Prisons, Data processing, Gyms, Health Clubs, Hospitals, Meeting Rooms

(Above referred table is as per BSEN12845. Similar references in NFPA are also acceptable)

Table 18.2
Ordinary Hazard occupancies

Occupancy	0111	OH2	OH3	0114
Glass and ceramics	Glass Manufacturing		Glass factories	
Chemicals	Cement works	Photographic laboratories photographic film factories	Dyers works soap factories	Candle wax factories Match factories paint application shops
Engineering	Sheet metal product Factories, Maintenance Workshops	Car workshops engineering factories, Emergency Generator Rooms, Lift M/C Rooms, Machine Shops, Mechanical Plant Rooms, Metal Workshops, Services Rooms	Electronics factories Radio equipment factories refrigerator factories washing machine factories	
Food and beverages	Abattoirs Dairy processing, Dairy Stores, Pantry, Food Courts and Restaurant seating Areas	Bakeries Biscuit factories breweries chocolate factories confectionery factories, Distilleries, Kitchens	Animal fodder factories corn mills Dehydrated vegetable and soup factories Sugar factories	alcohol distilleries

Table 18.2
Ordinary Hazard occupancies (Contd...)

Occupancy	OH1	OH2	OH3	OH4
Miscellaneous	Hospitals hotels libraries (excluding book stores) Restaurants schools offices, Laboratories (Class C &	Laboratories (Class A & B) Laundries Car parks museums, Resin Application Areas, Solvent Extraction Areas, Performing Stages, Wood Machining, Wood Product Assembly	Broadcasting studios railway stations Plant room	Cinemas and theatres concert halls tobacco factories
Paper		Paper & Pulp Mills, Paper processing Plants, Piers & Wharfs, Stationery Stores	Book binding factories cardboard factories paper factories printing works	Waste paper processing
Rubber and plastics			Cable factories Injection moulding (plastics) plastics factories and plastic goods (excluding foam plastics) Rubber goods factories synthetic fiber factories (excluding acrylic) vulcanization works	Rope factories
Shops and offices	data processing (computer room, excluding tape storage) offices, Fire Control Rooms, Canneries,	Libraries > 900 m , Mercantile Establishments, Post Offices, Printing & Publishing Houses, Retail Areas, Show Rooms	Department stores Shopping centres	Exhibition halls

Table 18.2
Ordinary Hazard occupancies (Contd...)

Occupancy	OH I	OH2	OH3	01-14
Textiles and clothing	Laundries	Leather goods factories, Textile Manufacturing	Carpet factories (excluding rubber and foam plastics) cloth and clothing factories fibre board factories footwear factories, Knitting factories, linen factories, mattress factories (excluding foam plastics) sewing factories weaving mills, woolen and worsted mills	Cotton mills Flax preparation plants Hemp preparation plants
Timber and wood			Woodworking factories furniture factories (without foam plastics) Furniture showrooms upholstery (without foam plastics) factories	Saw mills chipboard factories plywood factories
ether	Electronic Plants, Non Hazardous General Stores	Cereal Mills, Chemical Plants (ordinary), Confectioneries, Garbage Rooms, Horse Stables, Tire Manufacturing, Tobacco Product Manufacturing		
Note : Where there is painting or other similar high fire load areas in OH1 or OH2 occupancy. they shall be treated as OM.				

(Above referred table is as per BSEN12845. Similar references in NFPA are also acceptable)

Table 18.3
High Hazard Process Occupancies

HHP 1	HHP2	HHP3	HHP4
Floor cloth and linoleum manufacture Paint, colour and varnish manufacture Resin, lamp black and turpentine manufacture Rubber substitute manufacture Wood Wool manufacture	Fire lighter manufacture Manufacture of foam plastics, foam rubber and foam rubber goods Tar distilling Depots for buses, unladen lorries and railway carriages	Cellulose nitrate manufacture	Fire work manufacture

----- (Above referred table is as per BSEN12845. Similar references in NFPA arc also acceptable)